

PHYSOLY

Geometrical Optics Handout

For Physics Olympiads

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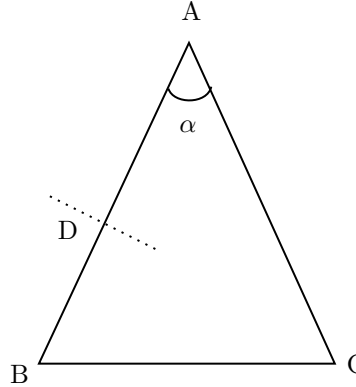
In this handout we will cover all the Geometrical Optics which you will need for Olympiads. We will present example problems and ideas about how they can be solved. These will be followed by Exercises. The Solutions to the Exercises are given at the end but you are requested to spend considerable time thinking before moving on to them.

June 11, 2026

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Problem 1

As shown in the figure, the vertex angle of a prism $\angle ABC$ is α ($\alpha < 90^\circ$). Suppose that light is incident at point D on the side AB at an arbitrary angle of incidence, and after being refracted once, it is incident on the side AC . The light ray got totally reflected at AC .



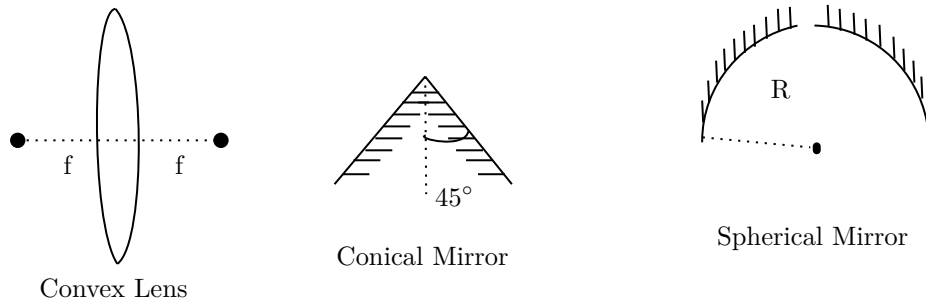
It is known that the critical angle for total internal reflection of light on the side AC is θ , $\theta < 45^\circ$. Assume that AC is sufficiently long. Find the range of the apex angle α of the triangular prism in the following two situations:

1. If light is incident only from below the normal on the side AB .
2. If light is incident only from above the normal on the side AB .

Problem 2

A **Hollow Laser Beam** is a cylindrical beam with zero intensity at its center along the propagation direction. Because of this feature, it can confine certain tiny particles on its axis, and act as laser catheters, laser tweezers, optical wrenches, etc. to achieve precise control of nano-particles, biological cells and other tiny particles. Hollow laser technology has been widely used in biology, laser processing, atomic cooling, etc., and is gradually becoming a new branch of Physics.

The basic method of generating a hollow laser beam is to use an optical system to convert a solid cylindrical laser beam into a hollow laser beam. Given the following optical components: a convex lens with focal length f , a conical mirror with semi-vertex angle 45° , and a spherical mirror with a radius R (with a circular hole in the middle), as shown in the following figure:



Use the above mentioned optical devices to design an optical system that transforms a very thin solid cylindrical laser beam incident on it into a hollow outgoing laser beam with the radius of the cylindrical cavity being r . Answer the following questions:

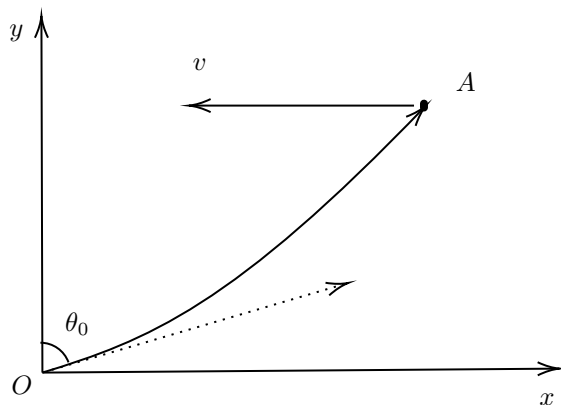
1. Draw the optical path diagram of this optical system.
2. Find the distance from the vertex of the conical mirror to the center of the spherical mirror, x in this optical system.

Problem 3. Laser Aiming Systems

The design of a laser aiming system needs to take into account the change of refractive index in air. Due to the influence of various factors such as surface conditions, altitude, temperature, humidity, and air density, the refractive index of air is unevenly distributed in the atmosphere. So the propagation path of laser light is not a straight line. For the sake of simplicity, suppose that the refractive index of air at a certain place varies with height as such -

$$n^2 = n_0^2 + \alpha^2 y$$

where n_0 is the refractive index of air at ground level ($y = 0$), and both n_0, α are positive constants. The propagation time of the laser itself is assumed to be negligible. The laser transmitter is located at the origin of coordinates O , as shown in the figure.



- If the angle between the direction of incident laser and the vertical axis is θ_0 (where $0 < \theta_0, \pi/2$), find the equation describing the propagation path of the laser.
- Assume that a target A is placed in the first quadrant. When its height is y_a , find the maximum x -coordinate $x_{a,max}$ where it can be placed for the laser to hit it.
- For damage to be caused to the target by the laser, a certain irradiation time is required. Suppose that the target is within the attackable range of the laser transmitter with initial position (x_0, y_0) . It moves towards the y -axis with a constant speed along negative x -axis. In order for the laser to always illuminate the target, how should the laser exit angle θ_0 change over time t ?
- Attack using a laser usually follows the principle of safe destruction, that is, both the target aircraft must be destroyed, and no bomb thrown by the target aircraft should hit the laser launcher (the bomb is stationary with respect to the aircraft when it is thrown). Assume that once the object enters the attackable range of the laser transmitter, the laser transmitter immediately irradiates it with laser light. It is known that the height of the horizontal flying target is y_a , the duration of continuous laser irradiation required to destroy it is t_a , and the laser located at the origin of the coordinates can safely destroy it; find the speed range of the aircraft over which this aiming system can work. Neglect air resistance. Acceleration due to gravity is given by g .

Problem 4. NIMs

The refractive index n of an optical medium can either be greater than or less than zero. Media in which $n < 0$ are called negative-index meta-materials (NIMs). When light propagates in a NIM, its optical path length is negative. If we say that the angle of refraction is negative when the incident ray and refracted ray are on the same side of the normal, Snell's law

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

would still hold, even if the refractive index on either side were negative. Here, n_1 and n_2 can either be positive or negative, and θ_2 is the angle of refraction. We use the convention where $\theta_1 \geq 0$ always.

- Suppose that a beam of light is incident upon an interface between two materials with different refractive indices. For each possible case (positive and negative n_2), show the path of the rays entering material 2 and the corresponding wavelets on the figures. Hence, show that the generalised Snell's law holds.

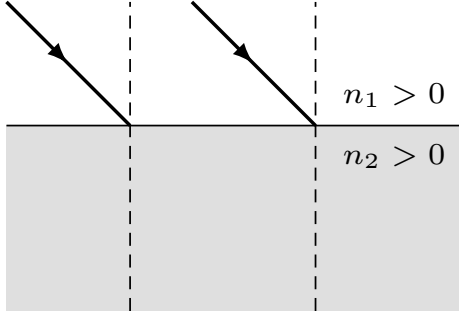


Figure 1

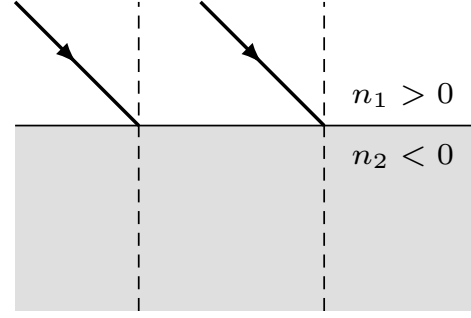


Figure 2

2. Refer to Figure 3. A spherical surface of radius R and centre C partitions three-dimensional space into an outer region and an inner region, with refractive indices $n_1 > 0$ and $n_2 < 0$ respectively. Considering any optical axis passing through C , we set the origin at O , the intersection of the axis with the interface. Light ray x is incident upon the interface at M and refracts to give ray y , as shown. Let s_1 and s_2 be the object distance and image distance respectively. Under the paraxial approximation, derive the equivalent lens equation (a relationship between s_1 , s_2 , and the given parameters of the system) and an expression for the magnification of an image. Note down the sign of each quantity in the final results *explicitly*.

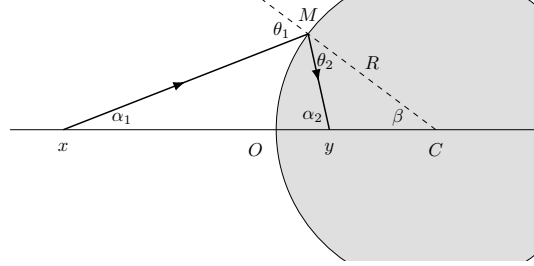


Figure 3: A spherical NIM.

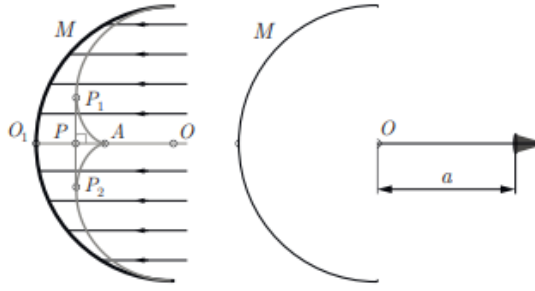
3. Suppose that medium 1 is air, i.e. $n_1 \approx 1$, and n_2 can either take positive or negative values. We place a thin convex lens of focal length $f = 1.5R$ in front of the spherical interface such that its optical axis passes through C , one focal point is inside the NIM, and the distance between O and the centre of the lens O' is d . A beam of light rays all parallel to the axis is incident upon the lens. For each set of parameters in Table , obtain the distance between O and the point at which all light rays converge. Also draw a figure illustrating the path of the light rays in case 4.

Case	n_2	d
1	1.5	$0.35R$
2	1.5	$0.85R$
3	-1.5	$0.35R$
4	-1.5	$0.85R$

Problem 5. Caustics

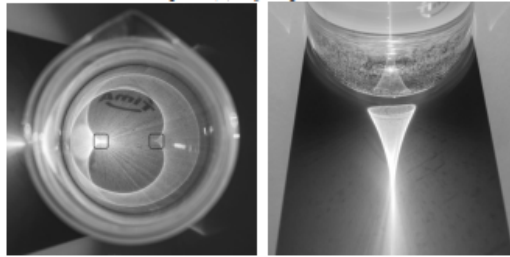
A **Caustic** is defined as the envelope of rays that do not intersect at one point. In this case, the caustic is a curve that is tangent to all rays reflected from one surface or refracted at some interface. The light intensity near caustics increases, so the caustic curves are clearly visible to the naked eye and in photographs.

1. The gray curve in the figure is a caustic formed after a beam of parallel rays is reflected by a cylindrical surface M ; the radius of surface being R . Determine the distance from axis of the cylinder, i.e. point O , to the vertex of the caustic, point A , namely distance OA , as well as the distance PA . OO_1 is the axis of symmetry and P_1P_2 is the tangent line to the caustic.



2. A point source illuminating a cylindrical surface of radius R is located at a distance $a = 4R$ from point O on the axis of symmetry of the OO_1 system. Determine the distance from point O to the vertex of the caustic formed by the reflected rays.

The glass with water in it is illuminated using a lantern. In each picture the light beam makes a different angle α with the horizontal surface of table as shown in Diagrams 1 and 2. In Diagram 1, the glass is photographed from above (the optical axis of the objective is perpendicular to the bottom of the glass), and it is illuminated from the right. In Diagram 2, the lens axis deviates slightly from the perpendicular.



Picture 1

Picture 2

3. This part is about the situation shown in picture 1. The angle α is about 45° . Near the bottom of the glass, there is a curve with two peaks, which corresponds to the situation in the first part. In the figure, the peaks are marked using squares. Explain the observed pattern.
4. This part is about situation shown in Diagram 2. The radius of the base of the glass is $R = 8$ cm. The lantern is located at a distance $a \approx 10R$ from the axis of the glass. The angle α can be considered small. Outside the glass, the caustic is visible, formed by the rays passing through the glass. Determine the approximate distance from the glass to the top of the caustic (to the top of the "cone"). The refractive index of water is $n \approx 1.33$.

Problem 6. Sound Propagation

The speed of propagation of sound in the ocean varies with depth, temperature and salinity. Figure below shows the variation of sound speed c with depth z for a case where a minimum speed value c_0 occurs midway between the ocean surface and the sea bed. Note that for convenience $z = 0$ at the depth of this sound speed minimum, $z = z_S$ at the surface and $z = -z_b$ at the sea bed. Above $z = 0$, c is given by

$$c = c_0 + bz$$

Below $z = 0$, c is given by

$$c = c_0 - bz$$

In each case $b = \left| \frac{dc}{dz} \right|$, that is, b is the magnitude of the sound speed gradient with depth; b is assumed constant

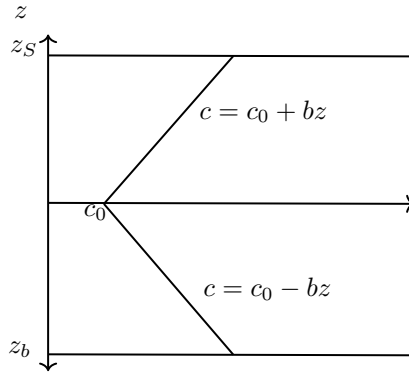


Figure 4: Speed of sound due to depth

Figure below shows a section of the $z - x$ plane through the ocean, where x is a horizontal direction. The variation of c with respect to z is shown in Figure 4. At the position $z = 0, x = 0$, a sound source S is located. A “sound ray” is emitted from S at an angle θ_0 as shown. Because of the variation of c with z , the ray will be refracted.

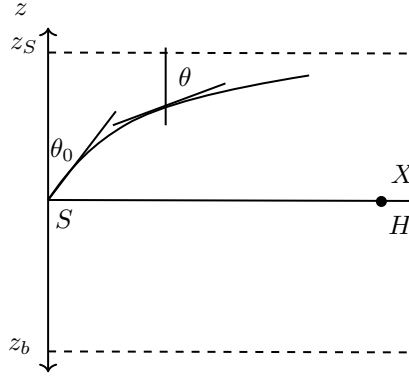


Figure 5: Trajectory of a sound ray

- (a) Show that the trajectory of the ray, leaving the source S and constrained to the $z - x$ plane forms an arc of a circle with radius R where

$$R = \frac{c_0}{b \sin \theta_0} \text{ for } 0 \leq \theta_0 \leq \frac{\pi}{2}$$

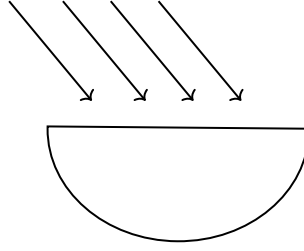
- (b) Derive an expression involving z_S, c_0 and b to give the smallest value of the angle θ_0 for upwardly directed rays which can be transmitted without the sound wave reflecting from the sea surface.
- (c) Figure 5 shows the position of a sound receiver H which is located at the position $z = 0, x = X$. Derive an expression involving b, X and c_0 to give the series of angles θ_0 required for the sound ray emerging from S to reach the receiver H . Assume that z_S and z_b are sufficiently large to remove the possibility of reflection from sea surface or sea bed.
- (d) Derive an expression to give the time taken for sound to travel from S to H following the ray path associated with the smallest value of angle θ_0 , as determined in part (c). Find the value of this transit time. The following result may be of assistance:

$$\int \frac{dx}{\sin x} = \ln \tan\left(\frac{x}{2}\right)$$

Find the time taken for the direct ray to travel from S to H along $z = 0$. Which of the two rays will arrive first, the ray for which $\theta_0 = \pi/2$, or the ray with the smallest value of θ_0 as found for part (c)

Problem 7

Parallel light rays are falling on the plane surface of a semi-cylinder made of glass, at an angle of 45° , in such a plane which is perpendicular to the axis of the semi-cylinder (see figure), index of refraction is $\sqrt{2}$. Where are the rays emerging out of the cylindrical surface?



Problem 8. Spherical Aquarium

A spherical aquarium, filled with water, is placed in front of a flat vertical mirror. The radius of the aquarium is R , and the distance between its center and the mirror is $3R$. A small fish, which is initially at the point nearest to the mirror, starts to move with velocity v along the wall. An observer looks at the fish from a very large distance along a horizontal line passing through the center of the aquarium.

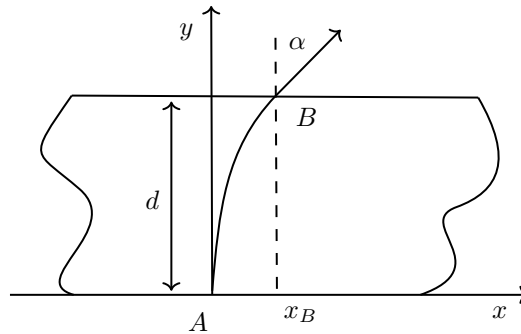
What is the relative velocity v_{rel} at which the two images of the fish seen by the observer will move apart? Express your answer in terms of v . Assume that:

- The wall of the aquarium is made of a very thin glass.
- The index of refraction of water is $n = 4/3$.

Problem 9

Consider a parallel, transparent plate of thickness d – see figure. Its refraction index varies as

$$n = \frac{n_0}{1 - \frac{x}{R}}$$



A light beam enters from the air perpendicularly to the plate at the point A ($x_A = 0$) and emerges from it at the point B at an angle α .

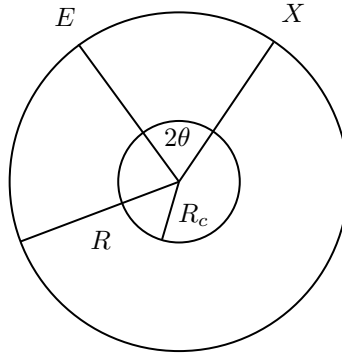
1. Find the refraction index n_B at the point B .
2. Find x_B (i.e. value of x at the point B)
3. Find the thickness d of the plate.

Take into account:

$$n_0 = 1.2 \quad R = 13 \text{ cm} \quad \alpha = 30^\circ$$

Problem 10. Seismic waves through the Earth

Early 20th century a model of the earth was proposed in which it was assumed to be a sphere of radius R consisting of a homogeneous isotropic solid mantle down to radius R_c . The core region within radius R_c contained a liquid.



The velocities of longitudinal and transverse seismic waves P and S waves respectively, are constant, V_P , and V_S within the mantle. In the core, longitudinal waves have a constant velocity $V_{CP} < V_P$ and transverse waves are not propagated.

An earthquake at E on the surface of the Earth produces seismic waves that travel through the Earth and are observed by a surface observer who can set up his seismometer at any point X on the Earth's surface. The angular separation between E and X , 2θ is given.

- (i) Show that the seismic waves that travel through the mantle in a straight line will arrive at X at a time t (the travel time after the earthquake), is given by

$$t = \frac{2R \sin \theta}{v} \quad \text{for } \theta > \arccos\left(\frac{R_c}{R}\right)$$

where $v = v_P$ for P waves and $v = v_S$ for S waves

- (ii) For some of the positions of X such that the seismic P waves arrive at the observer after two refractions at the mantle-core interface. Draw the path of such a seismic P wave. Obtain a relation between θ and i , the angle of incidence of the seismic P wave at the mantle-core interface, for P waves
- (iii) Using the data

$$\begin{aligned} R &= 6370 \text{ km} \\ R_c &= 3470 \text{ km} \\ v_{CP} &= 10.58 \text{ km s}^{-1} \\ v_s &= 6.31 \text{ km s}^{-1} \\ v_P &= 9.02 \text{ km s}^{-1} \end{aligned}$$

and the result obtained in (ii), draw a graph of θ against i . Comment on the physical consequences of the form of this graph for observers stationed at different points on the Earth's surface.

Sketch the variation of the travel time taken by the P and S waves as a function of θ for $0 \leq \theta \leq 90$ degrees.

- (iv) After an earthquake an observer measures the time delay between the arrival of the S wave, following the P wave, as 2 minutes 11 seconds. Deduce the angular separation of the earthquake from the observer using the data given in Section (iii).
- (v) The observer in the previous measurement notices that some time after the arrival of the P and S waves there are two further recordings on the seismometer separated by a time interval of 6 minutes 37 seconds. Explain this result and verify that it is indeed associated with the angular separation determined in the previous section.

Problem 11

- a) Consider a plane-parallel transparent plate, where the refractive index, n , varies with distance, z , from the lower surface (see figure). Show that $n_A \sin \alpha = n_B \sin \beta$. The notation is that of the figure

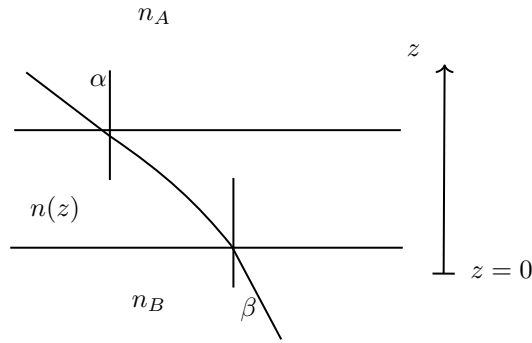


Figure 6: Light's path on varying refractive index

- b) Assume that you are standing in a large flat desert. At some distance you see what appears to be a water surface. When you approach the “water” it seems to move away such that the distance to the “water” is always constant. Explain the phenomenon.
- c) Compute the temperature of the air close to the ground in b) assuming that your eyes are located 1.60 m above the ground and that the distance to the “water” is 250 m. The refractive index of the air at 15°C and at normal air pressure (101.3 kPa) is 1.000276. The temperature of the air more than 1 m above the ground is assumed to be constant and equal to 30°C . The atmospheric pressure is assumed to be normal. The refractive index, n , is such that $n - 1$ is proportional to the density of the air. Discuss the accuracy of your result.

Problem 12. Extremum Principle in Optics

A light ray travels from medium I to medium II with refractive indices n_1 and n_2 respectively. The two media are separated by a line parallel to the x -axis. The light ray makes an angle i_1 with the y -axis in medium I and i_2 in medium II (see Fig. 7). To obtain the trajectory of the ray, we make use of another extremum (minimum or maximum) principle known as Fermat's principle of least time.

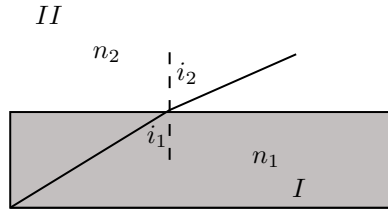


Figure 7: Light travels between two mediums

1. The principle states that between two fixed points, a light ray moves along a path such that time taken between the two points is an extremum. Derive the relation between $\sin i_1$ and $\sin i_2$ on the basis of Fermat's principle.

Shown in Fig. 8 is a schematic sketch of the path of a laser beam incident horizontally on a solution of sugar in which the concentration of sugar decreases with height. As a consequence, the refractive index of the solution also decreases with height.

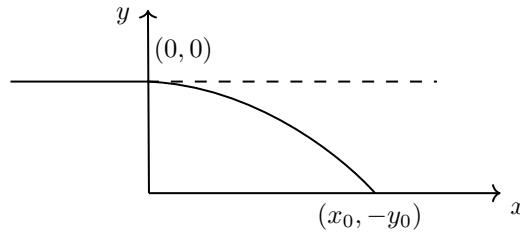


Figure 8

2. Assume that the refractive index $n(y)$ depends only on y . Use the equation obtained in part 1 to obtain the expression for the slope $\frac{dy}{dx}$ of the beam's path in terms of refractive index n_0 at $y = 0$ and $n(y)$.
3. The laser beam is directed horizontally from the origin $(0,0)$ into the sugar solution at a height y_0 from the

bottom of the tank as shown in figure 3. Take $n(y) = n_0 - ky$ where n_0 and k are positive constants. Obtain an expression for x in terms of y and related quantities for the actual trajectory of the laser beam.

You may use

$$\int \sec \theta d\theta = \ln(\sec \theta + \tan \theta) + C$$

$$\int \frac{dx}{\sqrt{x^2 - 1}} = \ln(x + \sqrt{x^2 - 1}) + C$$

4. Obtain the value of x_0 , the point where the beam meets the bottom of the tank. Take $y_0 = 10$ cm, $n_0 = 1.5$, $k = 0.05$ cm

Problem 13. Exoplanet Microlensing

With the discovery of planets orbiting around stars in recent years, the observation of exoplanets from astronomical distances became a challenge to scientists. Gravitational microlensing is one of the detection methods. It makes use of Einstein's discovery in general relativity that when a light ray passing near a spherically symmetric body of mass M , its direction will be deflected towards the body by a small angle given by

$$\alpha = \frac{4GM}{Rc^2}$$

where G is the gravitational constant, c is the speed of light, and R is the distance of closest approach of the light ray to the body. In this problem, we will study the principle of detecting exoplanets by microlensing.

- Consider a distant star S located at a distance D_s from Earth E , acting as the light source. Another star L of mass M and located at distance $D_l (< D_s)$ from Earth acts as the lens. The lines EL and ES make a small angle β between them. Construct the following sketches: (a1) the line EL , (a2) the line ES , (a3) the distances D_l and D_s , (a4) the angle β (remark: although this angle is small in practice, it should not be drawn too small for the purpose of clarity), (a5) a line perpendicular to EL through L , acting as the gravitational lens, (a6) the light ray from S to E , assuming that each of the segments between S and the lens and that between the lens and E are straight lines, (a7) the deflection angle α , (a8) the apparent angle θ of the star S as observed on Earth (relative to line EL).
- Derive an equation for the angle θ in terms of the parameters D_s, D_l, G, M, c and β , assuming that all angles are small.
- Consider the case that the lens is exactly aligned with the source ($\beta = 0$). The image of S appears to be a ring known as an Einstein ring. Derive the expression for the angular radius θ_E of the Einstein ring.
- Calculate the Einstein radius for the following typical values: $M = 0.3$ solar mass, $D_s = 10$ kpc, $D_l = 3$ kpc. Give your answer in milli-arc-seconds. You may use the following constants: $G = 6.67 \cdot 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$, 1 solar mass = $1.99 \cdot 10^{30} \text{ kg}$, $c = 3 \cdot 10^8 \text{ m/s}$, 1 kpc = $3.09 \cdot 10^{19} \text{ m}$, 1 radian = 206265 arc seconds.
- When the lens and the source are not exactly aligned, there will be two images of S . It is convenient to express the angles β and θ in multiples of the Einstein radius θ_E . Hence we define $u = \frac{\beta}{\theta_E}$ and $y = \frac{\theta}{\theta_E}$. Derive the expressions for the angular positions y of the two images in terms of u .
- To study the effect of the finite size of star S , we introduce Cartesian coordinates on the plane normal to ES and through S , with the y axis lying in the plane containing E , L and S . Consider the corners $(0, u + \delta)$ and (δ, u) of a square on the surface of star S ($\delta \ll u$). Calculate the coordinates of the two corners of the two images when viewed from Earth.
- Calculate the areal magnifications of the two images of star S in terms of u . Following the practice in astronomical observations, give your answer in absolute values.
- In practice, since the images cannot be resolved, astronomers measure the sum of the magnifications of the two images. Derive the expression for the total magnification. Describe its behavior when star S is remote (u approaches infinity) and when S approaches perfect alignment with L and E (u approaches 0).

- (i) A planet P of star L has mass m and is located in the plane of E , L and S at the same distance D_l from Earth. EP and EL makes an angle θ_p . Derive an equation for the angle θ taking into account the gravitational lensing effects of both star L and planet P . Expressions in the equation should be written in terms of the parameters $D_s, D_l, G, M, c, \beta, m$ and θ_p , assuming that all angles are small. Simplify the equation by introducing the mass ratio $q = \frac{m}{M}$ and the rescaled positions $u_p = \frac{\theta_p}{\theta_E}, u = \frac{\beta}{\theta_E}, y = \frac{\theta}{\theta_E}$.
- (j) In typical exoplanet detections, there is a motion of star S relative to star L . As star S approaches the closest distance to star L and moves away, u decreases with time to a minimum value u_0 and increases again. By plotting the magnification of the image of star S versus time, one observes a smooth and relatively broad peak in the magnification curve due to gravitational lensing by star L . In addition, one can observe a side peak due to the presence of the planet. For $q \ll 1$, estimate the width of this side peak, that is, the range of u in which the side peak is significant.
- (k) For $q \ll 1$, consider the situation that light rays pass very near to planet P , so that the gravitational lensing by star L becomes relatively insignificant. Calculate the position of star S where the total magnification of its image diverges, and the behavior of the total magnification in the neighborhood of this location

Problem 14. Gravitational Lens Simulator

The gravitational field of a massive body exercises a lens-like condensing action upon radiation passing through it. A simulated gravitational lens was constructed of Plexiglas for use in demonstrating the lens phenomenon. According to general relativity, a light ray passing at a distance of closest approach to the center of a spherically symmetric body of mass M , will be deflected toward the body through an angle which, for small deflections, is given by

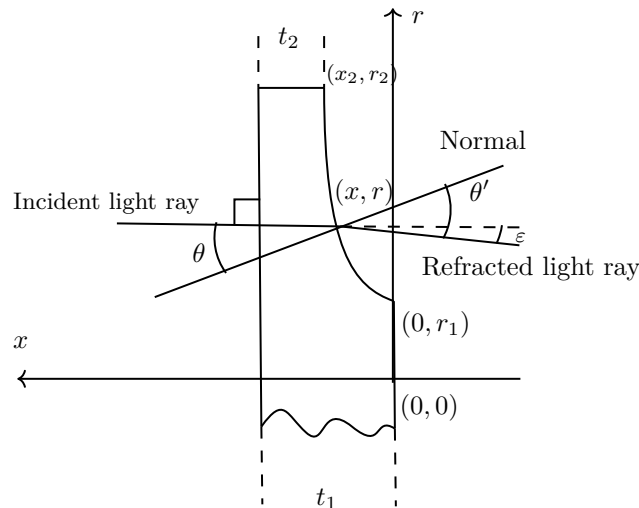
$$\alpha = \frac{4GM}{rc^2}$$

where G is the gravitational constant and c is the speed of light. We, therefore, require of our simulator that it deflects transmitted light through an angle

$$\epsilon = \frac{R}{r}$$

where R is a constant.

The lens, illustrated in cross section in Figure, is designed to be hand held within the range between roughly one foot and arm's length from the observer. The object for viewing is assumed to be at a distance much larger than one meter on the left hand side of the lens. This implies that one can assume the incident light ray to be normal to the plane front surface of the lens and refraction is thus assumed to take place entirely at the back surface. The angle of incidence of the light ray at the back surface is designated by θ , the angle of refraction by θ' , and the angle of deflection by ϵ , which are all assumed to be small angles. The refractive index of the lens is n .



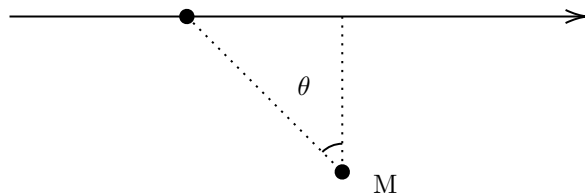
- (a) Derive the expression $\frac{dx}{dr}$ for the slope of the back refracting surface of the lens in terms of ϵ and n .

- (b) Derive the expression $x(r)$ of the back refracting surface of the lens. Express your answer in terms of n , R and r_1 .
- (c) A light ray is incident at impact parameter r equal to 2 cm. If we require the ray to cross the lens axis at a horizontal distance of 30 cm from the point at which it departs from the lens, find R .
- (d) Find the effective gravitational mass of the lens.

Problem 15. Bending Light with Gravity

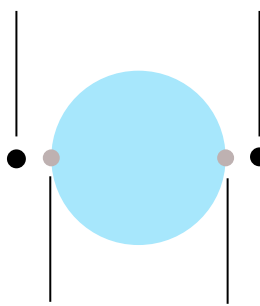
- (a) The path of light bent by gravitational field is an effect of General Relativity. However, one can analyse such phenomenon and obtain an approximate answer using Newtonian mechanics by assuming that a light beam is made of photons traveling at speed c , with the effective mass of each photon being equal to $m = E/c^2$, where E is the energy of the photon. Let us first consider the case of a classical particle flying by a fixed point mass M at high initial speed v . The trajectory of the particle will be bent due to the gravity of the point mass. Because of the high speed of the particle, the bending angle is very small and can be approximately calculated in the following way.

- (i) Assume that the trajectory of the particle remains straight, as if the point mass was not there. The distance from the trajectory to the fixed point mass is b , as shown in the figure. Calculate the total impulse the particle receives from the point mass as it travels from the far left to the far right in terms of M , b , G the universal gravity constant, and m the mass of the particle. You may find it easier to use the angle θ as the integration variable.



- (ii) Determine the bending angle of the particle.
- (b) The angular distance of two stars in the sky happens to be the same as the sun's angular diameter.

Apparent position



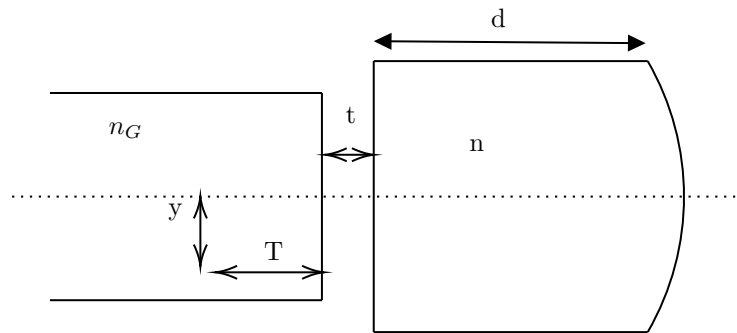
Original position

- (i) The sun-earth distance is $1.5 \cdot 10^{11}$ m, and the sun's diameter is $1.4 \cdot 10^9$ m. Calculate the sun's angular diameter in the sky.
- (ii) When the sun happens to be right between the two stars in the sky, the stars appear to be 'pushed outwards' by the sun, as shown in the figure. Draw a diagram to briefly explain the phenomenon.
- (iii) Calculate the apparent angular distance between the two stars in (ii). The mass of the sun is $2.0 \cdot 10^{30}$ kg. The speed of light in vacuum is c . The universal gravity constant is G . Given, $c = 3.0 \cdot 10^8$ m/s and $G = 6.67 \cdot 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$.
- (iv) With the present technology, why the phenomenon in (ii) can only be observed during a total solar eclipse?
- (v) If the moon's orbit size were doubled, could we still observe total solar eclipse?

Problem 16

As shown below, a point light source is inside a glass block of refractive index n_G at distance y from the optical axis and distance T from the surface. A plano-convex lens made of transparent plastic material of refractive index n is placed near the source. The thickness of the lens is d , and the radius of its convex surface is R . The width of the air gap between the glass block and the lens is t .

1. Determine the width of the air gap t so that the light from the point source forms a parallel beam after passing through the lens.
2. Find the maximum thickness of the lens d_{max} such that it is still possible to adjust the width of the air gap to produce a parallel beam.
3. Find the angle θ between the parallel beam and the optical axis.
4. If d is allowed to change by a small amount δd , and R is allowed to change by δR , find the angular change $\delta\theta$.

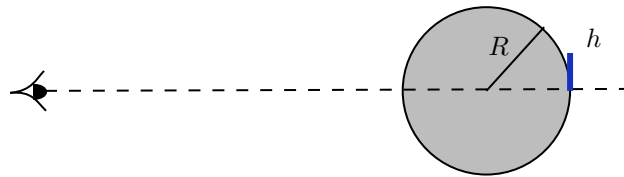


Problem 17

A cylinder with radius R is made from transparent material with refractive index n . A thin beam of light is going through the cylinder near its axis. Find distance L behind the cylinder where the beam is focused

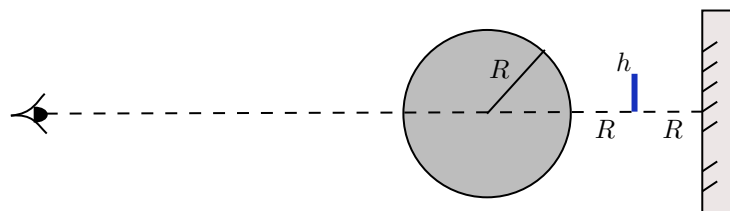
Problem 18

A small object with a height h is placed behind a glass sphere with radius R ($R \gg h$) and index of refraction n . What is apparent height of object h' seen by observer who is located at the distance much larger than the size of the sphere? Assume that observer's eyes are near going through the axis of sphere and bottom part of object



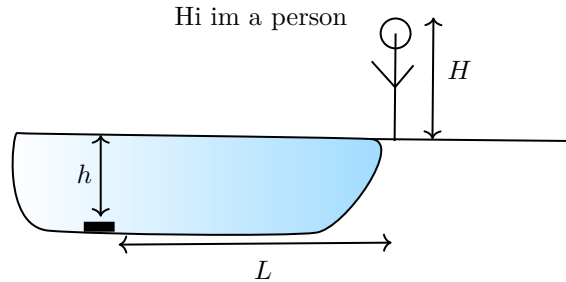
Problem 19

A small object with a height is placed between a glass sphere and a mirror as shown at the picture. What is apparent height of the object seen by an observer, which is located far away from sphere. Radius of sphere is R , ($R \gg h$), refractive index of glass is n



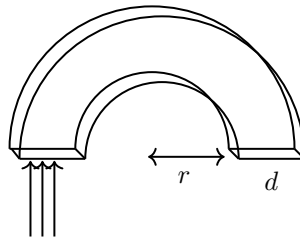
Problem 20

A person with a height $H = 1.7$ m is standing at the bank of a shallow river with depth $h = 10$ cm. Due to refraction actual and apparent depths of objects are different. What is apparent depth h' of the coin that is located at the bottom of flat river, if horizontal distance from the observer to coin is $L = 3$ m? Water's refractive index is $n = 1.33$



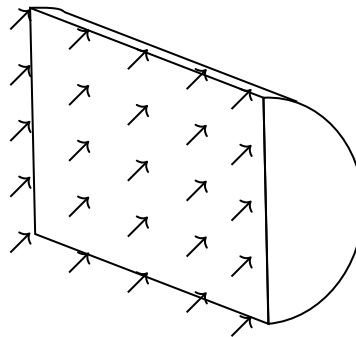
Problem 21

A parallel beam of lights falls perpendicularly at the one side of the glass toroid with a rectangular cross section. Inner radius of the glass is $r = 10$ cm. What is maximum glass thickness d , when all the beams of light entering cross section A will emerge from other toroid surface B ? Assume that all surfaces of the glass are smooth and refractive index of the glass is $n = 1.5$



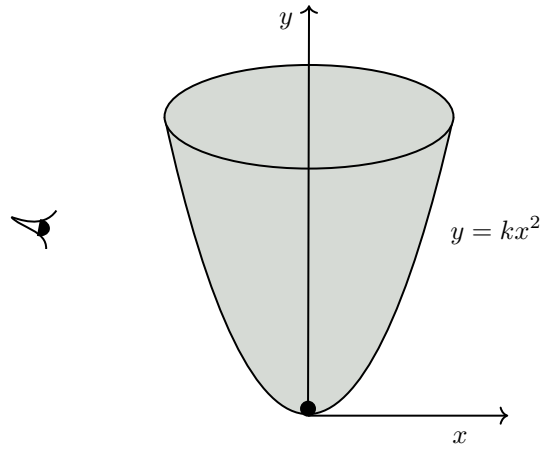
Problem 22. Light through a semi-cylinder

Glass semi cylinder with index of refraction $n = 1.5$ is illuminated by a parallel beam of light through its plain surface, as shown in the diagram. In such situation the light will go through a limited portion of arced surface of semi cylinder. Find N – fraction of area, where rays will emerge, to total area of the cylinder.



Problem 23. Hiding the impurity

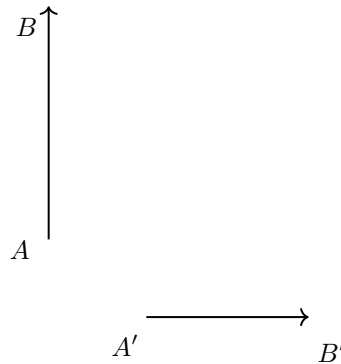
An observer is looking from the side at the paraboloid made from uniform transparent material. At what minimum index of refraction n , the observer will not see impurity at the bottom of the paraboloid with a shape $y = kx^2$?



Problem 24. Finding the Lens - 1

In the given figure, you can see the object AB and its image $A'B'$ in a thin lens. Using constructions, find:

- the optical center of the lens ;
- the lens plane ;
- the main focuses of the lens.
- Is the lens concave (diverging) or convex (collecting)? Explain the reason for this answer.



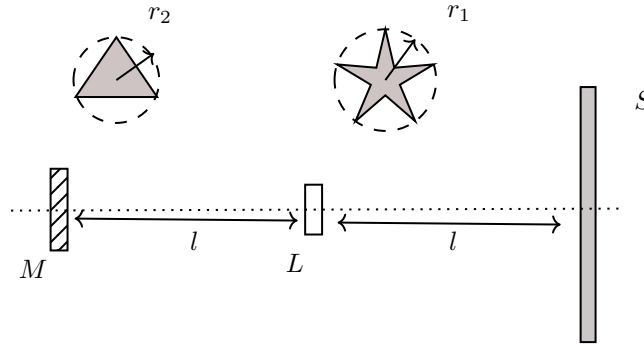
Problem 25. Finding the Lens - 2

In figure below, positions are shown of the point light source S and its image S' created by a thin lens. In the same figure, OO' the main optical axis of the lens. Plot an image of the point source S_1 created by the same thin lens.



Problem 26

At equal distances $l = 50$ cm in between the screen S and the flat mirror M , a flat matte source L emitting white light only towards the mirror is placed. The planes of the screen, mirror and source are all parallel to one other. The source has the shape of a five-pointed star inscribed in a circle of radius r_1 whereas the mirror has a regular triangle shape inscribed in a circle of radius r_2 . The source and mirror centers lie on the same axis perpendicular to the screen plane. Draw a schematic image of the source on the screen, keeping its orientation in accordance with the figure below. Estimate the sizes of all elements of the image.

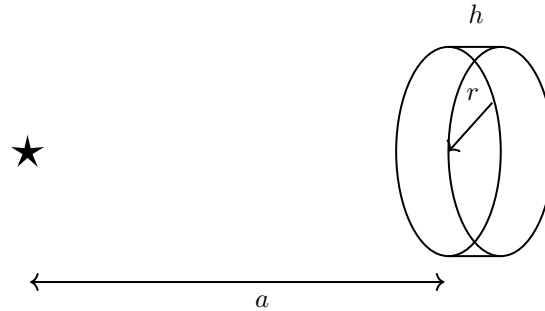


Problem 27. Flat Lens

Round transparent plane-parallel plate of the thickness h is made from the material which optical refractive index depends on the distance r to the central axis of the plate according to the law

$$n(r) = n_0(1 - \beta r^2)$$

where n_0, β are known positive constants. The plate is in the air whose refractive index $n = 1$. On the axis of the plate at a distance a ($a \gg h$) the point light source S is located. Show that the plate is a kind of lens, that is, it forms an image of the source. Determine at what distance from the plate the image of the source occurs. What is the focal length of this lens?



Problem 28. Rectangular Source

Rectangular light source of size $c \times b$ is fixed on the ceiling of the room of height $L = 3.0$ m.

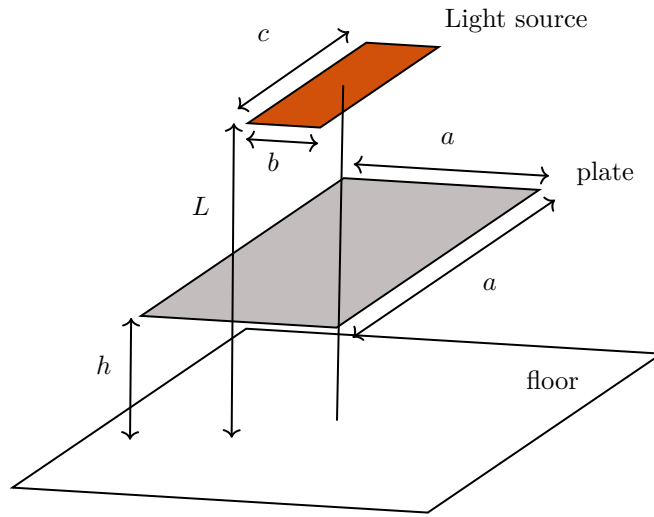


Figure 9

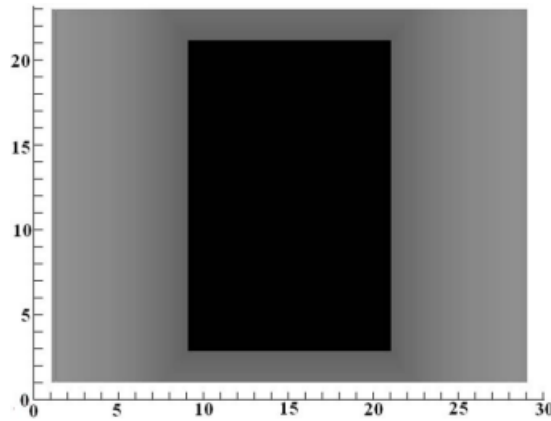


Figure 10

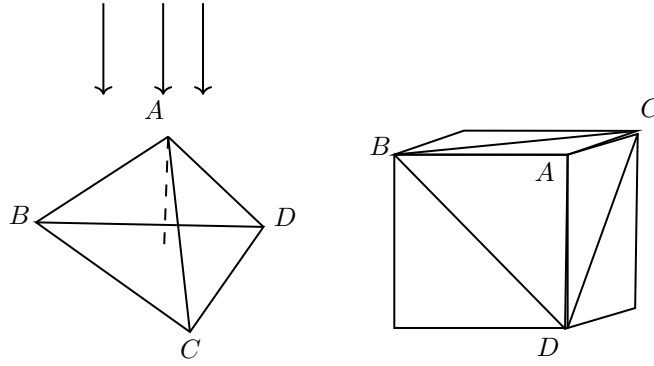
Opaque square plate with side $a = 10$ cm is placed horizontally under the lamp at a height h above the floor (Fig. 10 schematically shows the shape of the shadow casted by the plate on the room floor). The shadow consists of a dark rectangle surrounded by a lighter rectangle, and when approaching the edges of the outer rectangle the shade turns lighter. The value of the scale division in Fig. 10 is 1.0 cm. Find the sizes of the light source b and c , and the height h of the plate position. Specify the exact orientation of the light source with respect to the plate shadow.

Problem 29

Thin lens gives an image of the object, located perpendicular to its optical axis. The image size is 1 cm. If the distance from the object to the lens is increased by 5 cm, the image size remains 1 cm. Find the image size if the distance from the object to the lens is increased by another 5 cm.

Problem 30

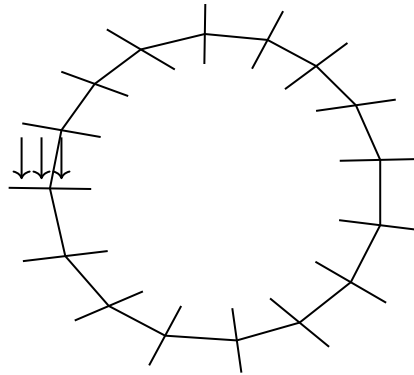
The pyramid $ABCD$ has been made with a glass with a refractive index $n = 1.6$. The top angles of pyramid are 90 degrees and its base is a equilateral triangle with side length $a = 2.0$ mm. We can think of that pyramid as a slice of a cube. Pyramid has been lightened perpendicular to its base.



1. Draw the image of the rays that passes through ABC
2. We put a screen that is parallel to pyramid's base and $L = 10$ cm away from it. Draw the image of rays on the screen

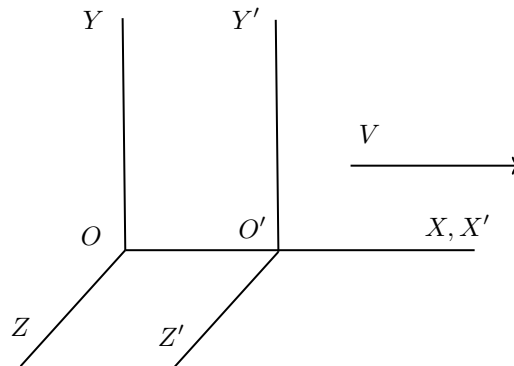
Problem 31. So Many Lenses

At the vertices of the regular 17-gon, there are 17 identical lenses. The optical centers of the lenses are located exactly at the vertices of the polygon, the planes of all lenses are perpendicular to one of the sides adjacent to the lens. The focal lengths of the lenses are all equal to $F = 10$ cm and coincide with the length of the side of the 17-gon. One of the lenses is illuminated by a parallel light flux directed along its optical axis. It turns out that one of the rays has a closed trajectory. Determine the radius of the circle inscribed in this trajectory. Consider two cases: all of the lenses are collecting; all of the lenses are diverging. Consider all angles small such that $\sin \alpha \approx \tan \alpha \approx \alpha$



Problem 32. Light in Moving Medium

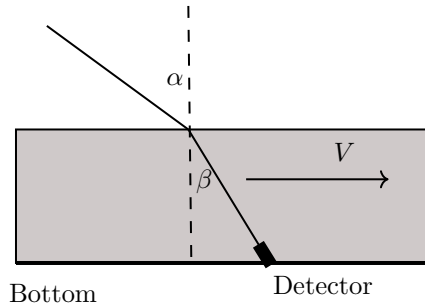
Consider two reference frames as shown below.



Let an object move in the plane $X'Y'$ of the reference frame S' such that the projections of its velocity on the axis $O'X'$ and $O'Y'$ are equal u_x and u_y , respectively.

1. Find the projections of the object velocity u_x on the axis OX and u_y on the axis OY in the reference frame S

Consider a water flow moving at the speed of V relative to the bottom of the vessel. A plane EMW falls onto the water surface to make the angle α to the normal in the laboratory reference frame. A detector is fixed at the bottom of the vessel. The refractive index of water is known to be n .



If the water velocity $V \ll c$ the expression for the sine of the angle β at which the detector registers the direction of EMW propagation, takes the form:

$$\sin \beta = A_1 + B_1 V$$

2. Find A_1 and B_1 in terms of α and n

If the water velocity $V \ll c$ the expression for velocity v_m of the light propagation in the laboratory reference frame is found as

$$v_m = A_2 + B_2 V$$

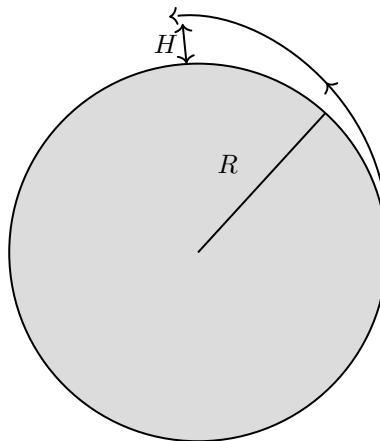
3. Find A_2 , B_2 in terms of β , n , c

Problem 33. Projectile Ray

A spherical planet with radius $R = 1000$ km has a layered atmosphere, so refractive index is varies with height h above its surface as

$$n(h) = n_0(1 - kh)$$

, where $n_0 = 1.0$ and $k = 9.8 \cdot 10^{-7} \text{ m}^{-1}$ are some known constants. A beam of light is emitted near the surface of planet in the horizontal direction. Find maximum height H above the surface of the planet that ray can go up to.



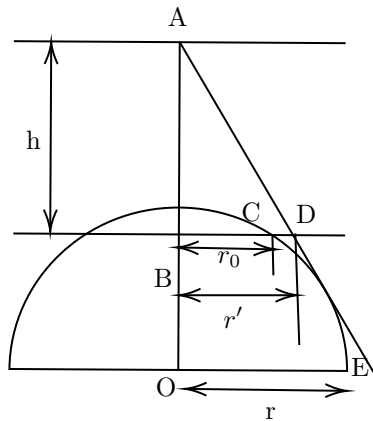
Problem 34. Estimation of the Density of the Watermelon in the Photo

The present problem proposes that only based on the picture in Figure A, we have to make some estimations for the density of a melon floating in a bucket of water and analyze the small vertical oscillations of the melon in the water bucket. In all the estimations you will make, you will consider that the watermelon is homogeneous and can be

approximated by a sphere. In the different estimations you will make, you will neglect or consider refraction and / or three-dimensional effects.



- (A) Task A proposes to analyze the factors that influence the accuracy of the density determination of watermelon and to make a first estimation of its density without considering the refraction of light on the air-water surface and the three-dimensional effects.
- Deduce the expression of V_0 , the volume of the watermelon portion which is immersed in water. Express the result as function of the radius of the melon r and of the ratio $q = \frac{r_0}{r}$ where r_0 represents the radius of the intersection circle of the melon with the surface of the water.
 - Determine the expression of the density ρ of melon as function on water density ρ_0 and ratio q .
 - Plot the graph of the ratio $\frac{\rho}{\rho_0}$ as a function of q .
 - Using the plot drawn in task A(iii) briefly describe how the value of the ratio q influences the accuracy of the density determination ρ .
 - Estimate the density of the melon from the photograph shown in Figure A, without considering the refraction of the light at the air-water surface and the separation space between the horizontal planes containing the intersection circle of the water surface with the melon and horizontal equatorial circle of the melon.
- (B) The estimation achieved in Task A can be improved by considering the spatial separation between the horizontal planes containing the intersection circle of the water surface with the melon and horizontal equatorial circle of the melon. In this task, neglect the phenomenon of refraction of light on the surface air-water. Consider that the camera with which the photograph was taken is placed in position A, at distance h away from the surface of the water in the bucket, and note the "apparent" radius of the melon $BD = r'$ (Figure B). Assume that the camera is placed high enough relative to the surface of the water in the bucket, so it can record latitudes close to the equatorial circle of the melon. In Figure B the water surface is represented by the line passing through the points B, C and D. The sketch in Figure B is not made on scale.



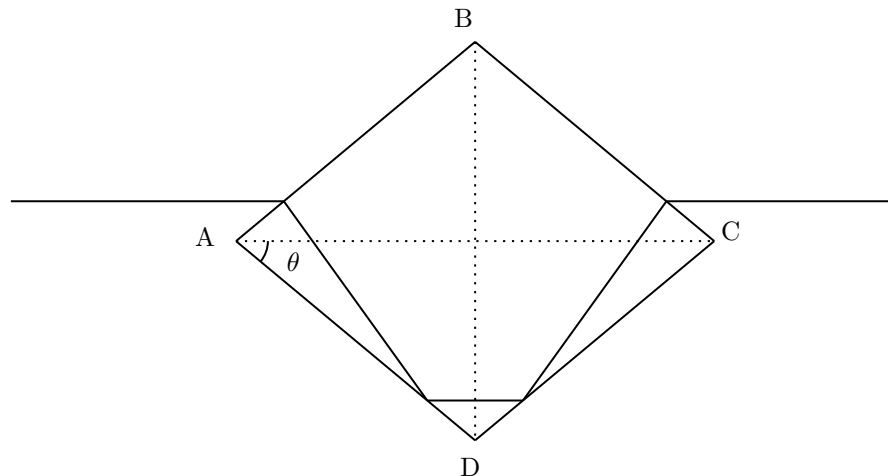
- Determine the expression of the radius r of the melon, in the conditions mentioned in this task. Express the result as function of h, r_0, r' .

- (ii) Estimate the density of the melon in the photograph shown in Figure A under the conditions specified in Task B. Consider that the photograph was made by a young man with the height of 1.70 m , standing next to the bucket having the diameter 0.30 m . The height of the bucket is 0.30 m.
- (C) In the Task C you are asked to consider whether the refraction of light significantly influences the result of the melon density estimation - if you take into account the spatial separation between the horizontal planes containing the intersection circle of the water surface with the melon and the melon's equatorial horizontal circle.
- Determine - under the conditions specified for the Task C - an expression containing the melon radius r , the refraction index for water n and the quantities h, r_0, r' whose significance was mentioned in the Task B.
 - . Estimate the density of watermelon in the photograph shown in Figure A, under the conditions specified in the Task C. The water refraction index is $n = \frac{4}{3}$.
 - Specify whether the refractive phenomena considerably influenced the result of the melon density estimation. Briefly motivate your answer.
- (D) In the Task D, it is required to analyze the small vertical oscillations of the melon in the picture shown in Figure A, using a simple modeling in which the friction forces and the moving of the water in the bucket are considered negligible.
- Determine the height $h_{c,IV}$ of the spherical cap of melon located in the air - under the conditions set out in the Task C(ii).
 - Determine the angular frequency expression ω of the small vertical oscillations of the melon around the equilibrium position shown in the picture in Figure A. Express the result as function of ρ_0, ρ, r , of the height of the spherical cap of the melon h_c and the gravitational acceleration g .
 - Write the equation of the small vertical oscillations of the melon, if their damping is negligible. Express the result as function of elongation Δh_c and pulsation ω .
 - Calculate the amount of time required to perform ten small, vertical, undamped oscillations of melon under the conditions of estimation set out in the task C(ii). Consider $g = 9.81 \text{ m/s}^2$.

Problem 35. Reflection and Refraction of Light

1. An Interesting Prism

The main section of a glass prism, situated in air ($n' = 1.00$), has the form of a rhombus with $\angle BAD = \angle BCD = \theta$. A thin yellow beam of monochromatic light, propagating towards the prism, parallel with the diagonal AC of the rhombus, is incident on the face AB (Fig. 1). The beam is totally reflected on the faces AD and DC , then emerges through its face BC . For the yellow radiation, the refraction index of the glass is $n = 1.60$.



- (a) Derive the mathematical expression for the angle θ as a function of the refraction index n of the prism, such that the total deviation of the beam that exits the prism to be zero. Under the above condition, calculate

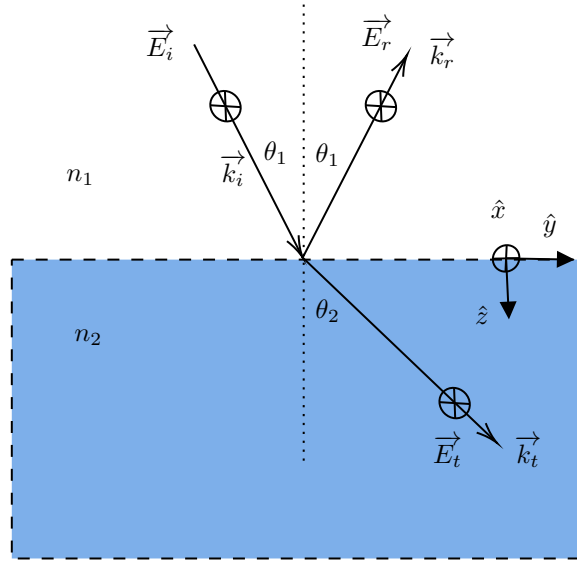
the numerical value of θ in degrees and minutes, if $n = 1.60$.

The prism with θ determined above and the direction of the incident beam remain fixed, but the nature of the light radiation changes, being formed now of the yellow doublet of the mercury. The two wavelengths have the values 579.1 nm and 577 nm respectively. The refraction indices of the glass for these wavelengths are $n = 1.60$ and $n + \Delta n$ respectively where $\Delta n = 1.3 \cdot 10^{-4}$. The light rays that exit the prism enter longitudinally into an astronomical telescope adjusted for infinite distance.

- (b) Derive the mathematical expression for ‘the angular distance’ ϵ between the two images seen through the telescope (first as a function of θ and Δn , then as a function of n and Δn) and calculate its numerical value.
- (c) If the focal distance of the telescope’s objective is $f_{ob} = 0.40$ m , derive the linear distance y between the two images, seen in the focal plane of the objective and calculate its numerical value.

2. Refraction, but... mostly reflection

2A. Total reflection in geometrical optics



Total reflection occurs when light travels from a medium with refractive index n_1 to another one, with the refractive index $n_2 < n_1$, at an incidence angle $\theta_1 \geq l$, where l is the critical value of the incidence angle, called limit angle, beyond which there will be no refracted light. At total reflection, the entire energy of the incident light beam goes to the reflected beam.

- (a) Derive the mathematical expression for the limit angle.

2B. Total reflection in electromagnetic optics

Electromagnetic optics proves that, besides being totally reflected, the incident light beam also penetrates the less refringent medium as an evanescent wave. The characteristics of the reflected and the refracted light beams depend on the angle of incidence, as well as on the orientation of the electric field of the light wave (called polarization). For simplicity, let us consider that the electric field intensity is perpendicular on the incidence plane, as represented in Fig. B. The indices i , r , and t refer to the incident, reflected and transmitted properties of light waves while $\vec{k}$ is the wave vector, giving the light propagation orientation. Moreover, $\hat{x}$, $\hat{y}$, $\hat{z}$ are the unit vectors of the chosen Cartesian reference frame.

Physical note: The perturbation produced by a plane, monochromatic wave in a point in space at a certain moment of time can be written as $\vec{E}(\vec{r}, t) = \vec{E}_0 \cos(\omega t - \vec{k} \cdot \vec{r})$, or, to simplify calculations, in the complex form $\vec{e}(\vec{r}, t) = \vec{E}_0 e^{\iota(\omega t - \vec{k} \cdot \vec{r})}$, where $\iota = \sqrt{-1}$, and then taking only the real part of the result.

Mathematical note: For the complex number $z = a + \iota b$, a is the real part and b is the imaginary part. It can be written as $z = \sqrt{a^2 + b^2} \left(\frac{a}{\sqrt{a^2 + b^2}} + \iota \frac{b}{\sqrt{a^2 + b^2}} \right) = |z| e^{\iota \phi}$ where $|z|$ is the modulus of the complex

number, the first term in the bracket denotes $\cos \phi$ and the second term in the bracket (except the ι) denotes $\sin \phi$.

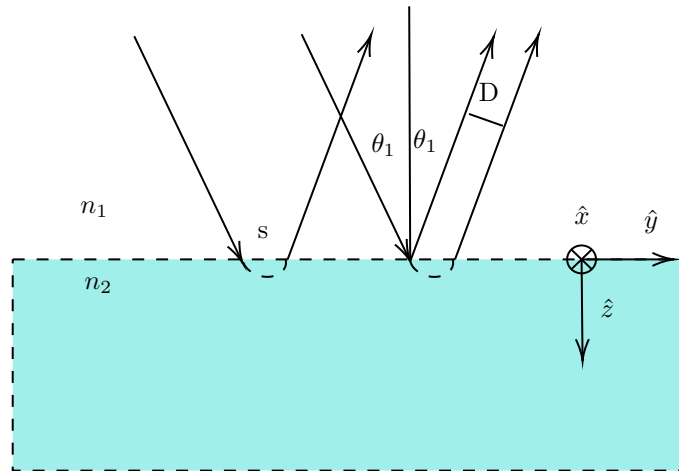
- (a) **Evanescent Wave** - Knowing that the incident wave is a plane and monochromatic one, characterized by the equation $\vec{e}_i(\vec{r}, t) = \vec{E}_{0i} e^{\iota(\omega t - \vec{k}_i \cdot \vec{r})}$, prove that the mathematical expression for the evanescent wave is $\vec{e}_t(\vec{r}, t) = \vec{E}_{0t} e^{-\alpha z} e^{\iota \phi}$ and derive the exact expression for the attenuation coefficient α as a function of the incidence angle θ_1 , the limit angle l , and the wavelength λ of the incident wave. Also, derive the exact expression for the phase ϕ of the evanescent wave.
- (b) **Penetration Depth** - Derive the mathematical expression of the distance Δz from the interface at which the amplitude of the evanescent wave is e times smaller than at the interface, as a function of the incident wavelength λ and calculate its numerical value. The first medium is glass ($n_1 = 1.6$), the second is air ($n_2 = 1.0$) and the incidence angle of light is $\theta_1 = 40^\circ$.
- (c) **The phase speed of the evanescent wave** - Derive the mathematical expression for the ratio $\frac{v_e}{v_1}$, where v_e is the phase speed of the evanescent wave and v_1 - the phase speed of the incident wave, and compute its numerical value for the case of the incidence angle of light of $\theta_1 = 40^\circ$.

The energy transferred from the incident wave to the totally reflected wave - For any value of the incidence angle, the relationship between the amplitude of the field of the reflected wave and that of the incident wave was derived by the French physicist Augustin Fresnel (1788-1829):

$$E_{0r} = \frac{n_1 \cos \theta_1 - n_2 \cos \theta_2}{n_1 \cos \theta_1 + n_2 \cos \theta_2} E_{0i}$$

Physical note: If the perturbation produced by a wave in a point in space at a given moment is expressed using complex numbers, then the wave intensity has the mathematical expression $I = \frac{1}{2} \epsilon_0 c E_0 * E_0 = \frac{1}{2} \epsilon_0 c |E_0|^2$ where $E_0 * = a - \iota b$ is the complex conjugate of the complex number $E_0 = a + \iota b$. Here ϵ_0 is the vacuum permittivity and c is the speed of light in vacuum.

- (d) Prove that the totally reflected wave has the same intensity as the incident wave.
- (e) **The Goos – Hänchen effect** - When an incident wave beam with a finite cross section undergoes total reflection at an interface between two media, the totally reflected wave beam is laterally displaced, on a distance D (see Fig. 3), that was measured for the first time by Goos and Hänchen in 1947. In Fig. 3, the displacement along the surface is s , and the Goos – Hänchen shift is the lateral shift D indicated in the diagram. This is the Goos – Hänchen effect. The explanation of this lateral shift is based on the appearance of the evanescent wave at the interface and its propagation parallel to the interface.



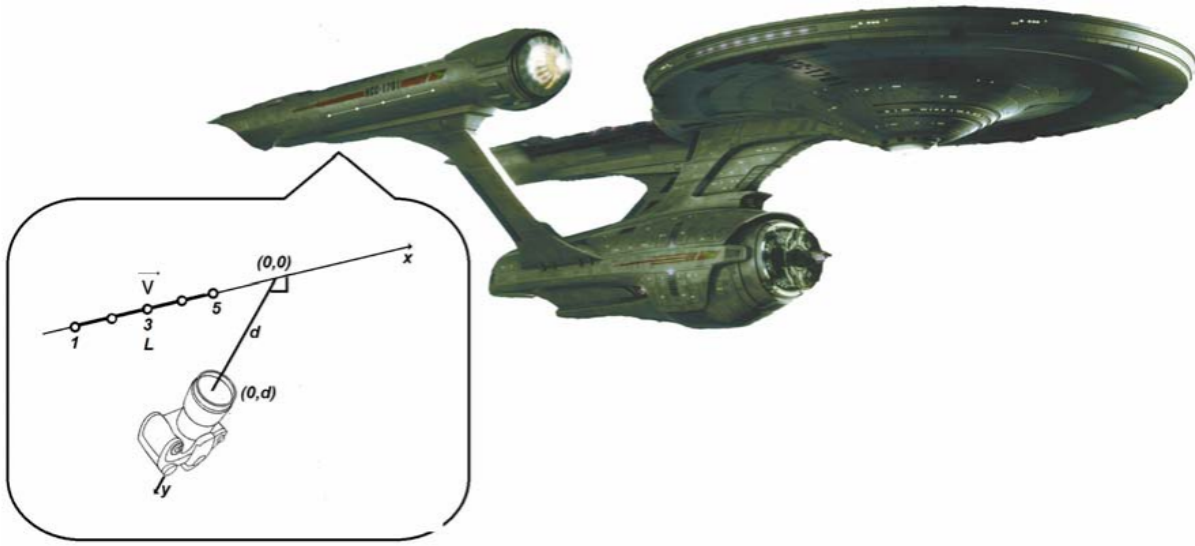
- i. **The Lateral Shift** - Derive the mathematical expression for the Goos – Hänchen lateral shift D , admitting that the phase difference between the totally reflected wave and the incident one is zero at the interface. Consequently, compute the numerical value of the displacement s along the interface as a function of the wavelength λ of the incident light, if the first medium is glass ($n_1 = 1.6$), the second is air ($n_2 = 1.0$), and the incidence angle of light is $\theta_1 = 40^\circ$.

An alternative explanation of the Goos – Hänchen shift can be given in terms of the time delay associated with the scattering of a radiation pulse at the interface. The incident radiation pulse is not scattered instantaneously by the surface, but reemerges into medium 1 after a time delay τ , during which the pulse propagates parallel to the surface and is displaced by the distance s .

- ii. **Time needed for the total reflection** - Derive the mathematical expression for the time delay τ and calculate its value if the first medium is glass ($n_1 = 1.6$), the second is air ($n_2 = 1.0$), and the incidence angle of light is $\theta_1 = 40^\circ$, and the monochromatic radiation has the wavelength $\lambda = 579.1\text{nm}$. The light speed in vacuum is $c = 3.0 \cdot 10^8 \text{ m/s}$.

Problem 36

Description of the situation



Star Ship Enterprise has on each of the two warp engines a succession of five collinear, equidistant, numbered, beacons as shown in the figure (for the propeller on right side). When traveling, SS Enterprise moves with constant speed $\vec{v}$ so that, for the questions that follow, beacons located on a propeller are always on the same line that is Ox axis of the coordinate system. In the situation room of another spacecraft which is at rest and which oversees the SS Enterprise, are analyzed the images of line of beacons. The images are taken by a camera with the aperture located at the point $(0, d)$ - in the orthogonal coordinate system shown in the inset in figure above. An image of line of beacons is produced by light rays that arrive simultaneously at camera aperture that opens for a short time. On a picture of SS Enterprise at rest, the length of line of beacons is L . In the problem, the beacons can be considered as points. The speed of light is $c = 3 \cdot 10^8 \text{ m/s}$. Employ the quantities $\beta = \frac{v}{c}$ and $\gamma = \frac{1}{\sqrt{1 - \beta^2}}$, if they help to simplify your result.

Task No. 1 - The link between image and the actual position

The actual position is the position in the frame in which the camera is at rest.

Next, the camera remains stationary in the point $(0, d)$ and the SS Enterprise is moving at velocity $\vec{v}$ on Ox direction. In an image provided by the camera described above is observed that one of the bright beacons is placed in the position x_i .

- Determine the expression for the actual position of the bright beacon when the image is formed in the camera.
- Find also the corresponding inverse relation, that is the expression of x_i in terms of x, d, L, v and c .

Task No. 2 - Apparent length of the beacons line

The camera takes a picture of the instant when the actual position of the center of beacons line is at some x_0 point.

- (A) Determine the expression for the apparent length of the beacons line on this picture.
- (B) Indicate – using a mathematical expression – how the apparent length of beacons line changes with time, when SS Enterprise moves along the Ox axis. The space ship comes from far away, passes through the origin of the system and then receding far away.

Task No. 3 - Symmetrical image

One of the pictures taken by the camera shows both ends of beacons line at the same distance from the aperture of camera.

1. Determine the expression for the apparent length of beacons line on this picture.
2. Determine the expression for the actual position of the middle beacons at the time when this picture is taken.
3. Determine where will be situated on the picture the image of the middle beacons.

Task No. 4 - Pictures of SS Enterprise, being far away and approaching or being far away and receding

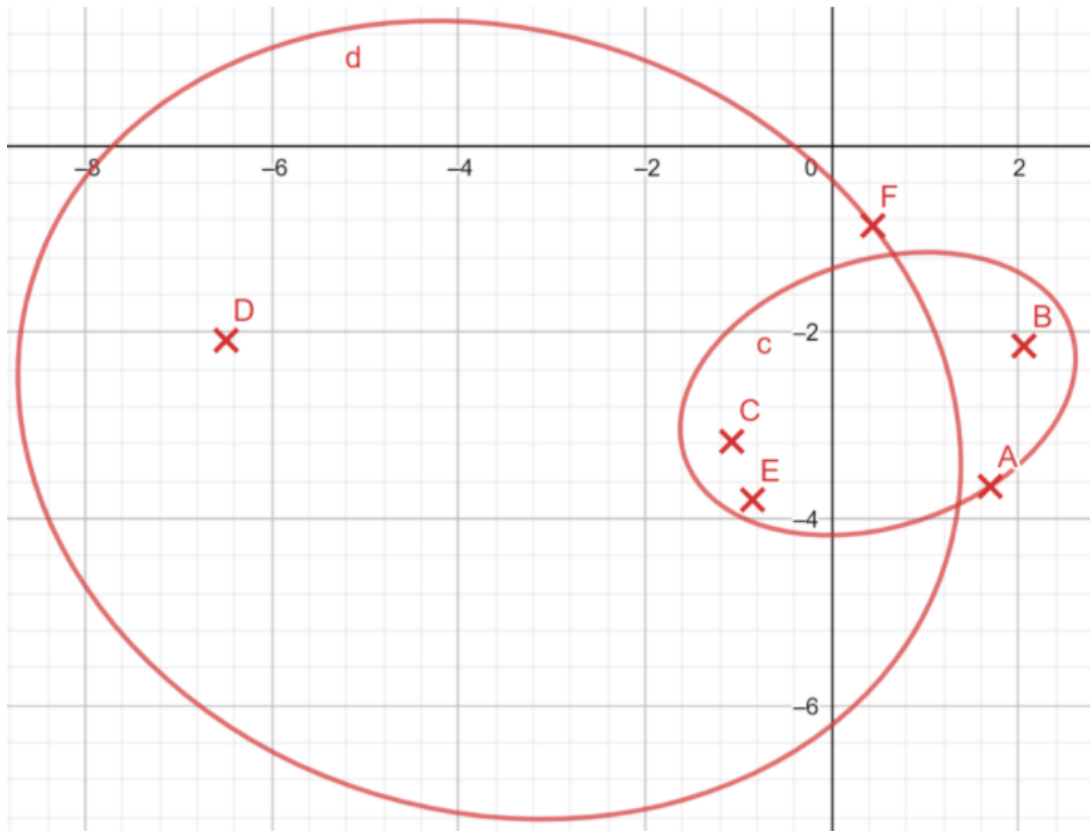
Camera takes a picture when SS Enterprise is approaching from far away and another image when the ship (and the beacons line) is far away and receding. On one of these images the apparent length of the line of beacons is 160 m and the other is 600 m.

1. Decide which of the following statements is correct.
 - i. Apparent length is 200 m on the image of 'approaching' ship and 600 m on the image of 'receding' ship.
 - ii. Apparent length is 600 m on the image of 'approaching' ship and 200 m on the image of 'receding' ship.

Write in the appropriate box in the answer sheet the letter corresponding to the answer you think is correct. Briefly justify your choice.
2. Determine the speed v of the ship.
3. Calculate the value of length of the beacons line at rest.
4. Calculate the value of length of the beacons line on the symmetric picture, described at Task no. 3.a

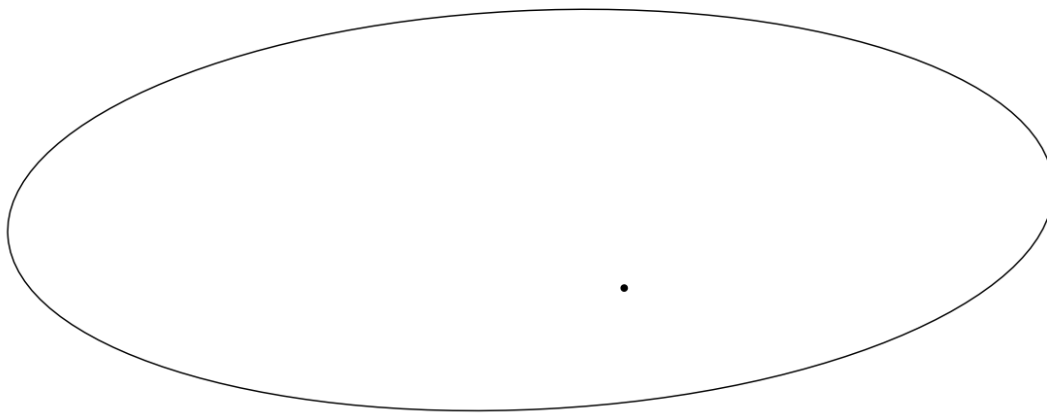
Problem 37

The two ellipses depicted in the figure below, available as a [Geogebra file](https://www.geogebra.org/), represent real images of circles, created by an ideal thin lens. Both the ellipses and the main optical axis of the lens lay in the plane of the figure. Reconstruct the position of the lens (i.e. the position of the centre and the orientation). Remark: it is recommended to use Geogebra, <https://www.geogebra.org/>, either as an online service, or as a downloaded application. In you construction steps, you may use everything what Geogebra has to offer: drawing a line through two given points, drawing parallel and perpendicular lines to a given line through a given point, drawing tangents through a given point to a given shape (i.e. a given curve), finding intersection points of shapes (i.e. of curves or lines), drawing circles with a given centre through a given point, or through three given points, or through two diametrically opposite points on it, drawing ellipse with given foci through a point laying on it, or through five given points. Trial-and-error approach (for instance, displacing a certain shape until three certain intersection points lay on a straight line) is not accepted. Alternatively, you may print the ellipses onto a sheet of paper and make constructions by hand using a ruler and a compass (be aware that there may be no solution if you start with arbitrarily positioned two ellipses of arbitrary shape, so make sure to either print the drawing, or take the initial shapes close enough to the provided one).



Problem 38

Ellipse depicted in the figure below represents a real image of a circle, created by an ideal thin lens. The dot inside it depicts the image of the center of the circle. Both the ellipse and the main optical axis of the lens lay in the plane of the figure. Reconstruct the position of the lens (i.e. the position of the centre and the orientation). Remark: you need to copy the ellipse onto a sheet of paper and find there the position of the lens, eg. by using geometrical constructions. Alternatively, you can use Geogebra, <https://www.geogebra.org/>.



Problem 39

What is the diameter of the lens which was used to make the photo below?

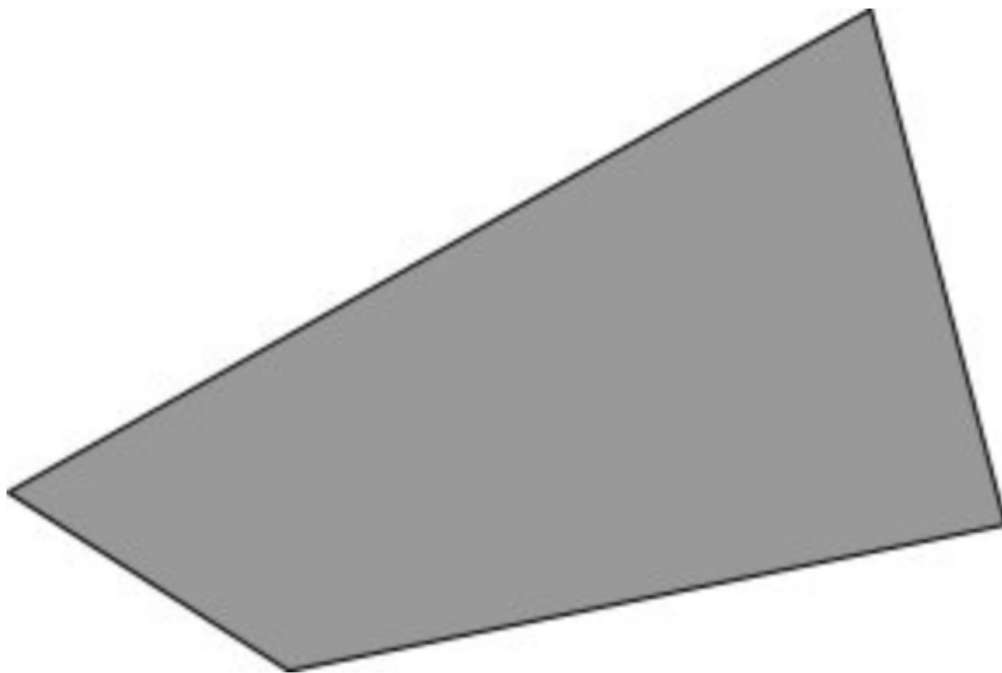
Remark: although the photographic lenses are made of several optical components, for many practical calculations – including this problem – they can be considered as ideal thin lenses.



Problem 40

Quadrilateral depicted in Figure represents a real image of a square, created by an ideal thin lens. Both the quadrilateral and the main optical axis of the lens lay in the plane of the figure. Reconstruct the position of the lens (ie. the position of the centre and the orientation).

Remark: you need to copy the quadrilateral onto a sheet of paper and find there the position of the lens, eg. using geometrical constructions.

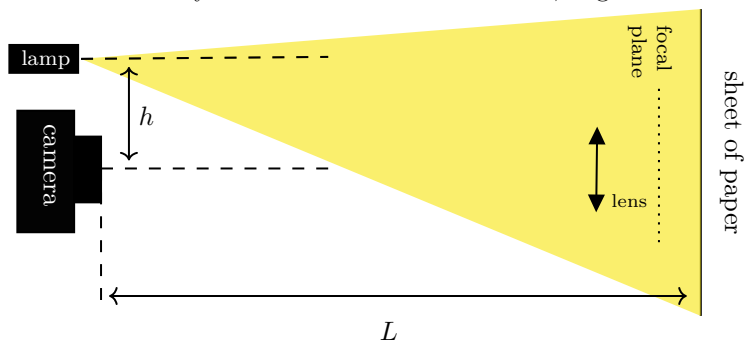


Problem 41. Cat Eyes

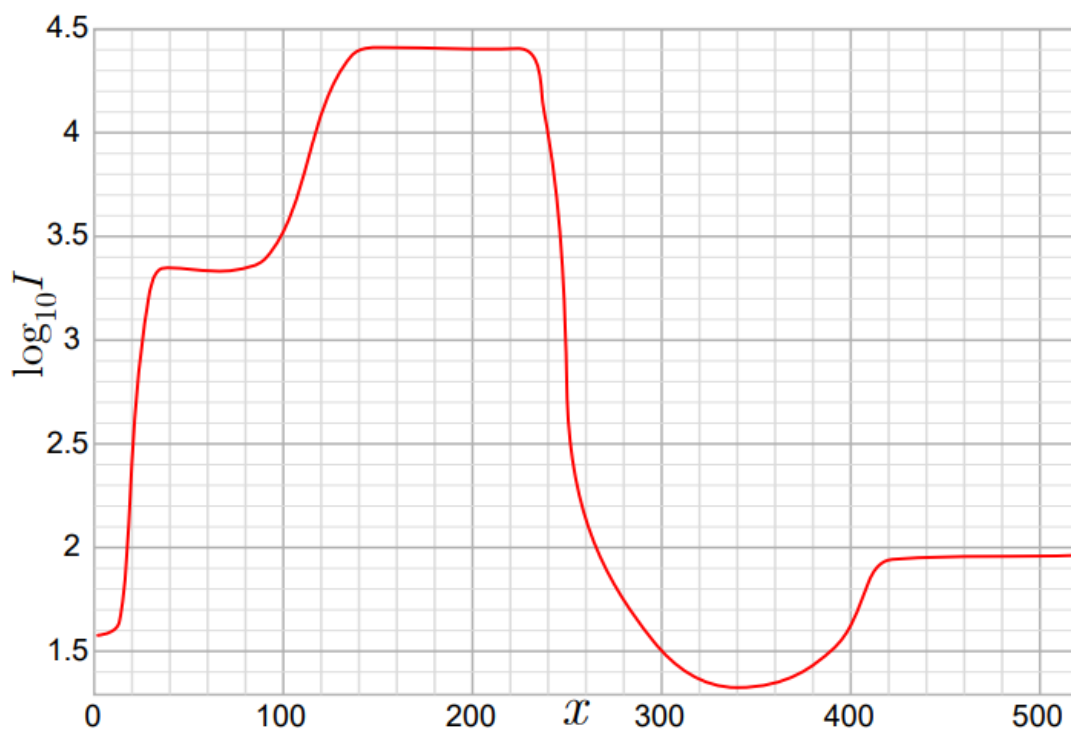
You may have noticed that in darkness, when a cat is within the light beam of a headlamp, its eyes appear very bright, see the photo below (left). This phenomenon can be modelled by a lens setup, see the photo on right, and the diagram beneath the photos.



Figure 11: Left : The eyes of a Cat when seen in dark, Right : The lens setup



The photo on right was taken by a digital single-lens reflex camera. The light intensity at the camera sensor pixels marked by a red line (in the photo) is shown in the graph below: the log base 10 of the light intensity (measured as the number of photons caught by each pixel) is plotted against the x -coordinate, with the pixels' side length serving as the unit length.



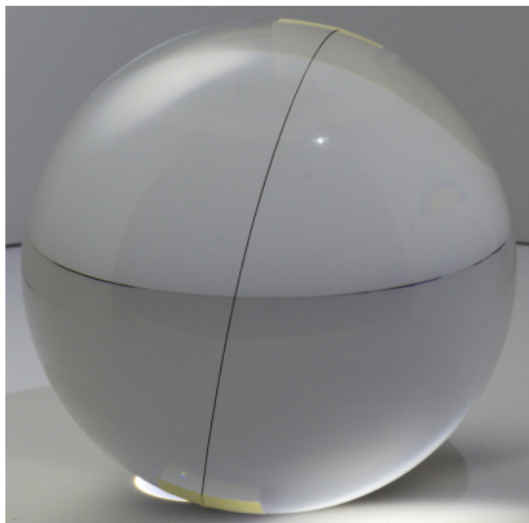
The lens modelling cat eyes can be treated as an ideal thin lens of focal length $f = 55$ mm and diameter $D = 39$ mm; however, you should keep in mind that the given graph shows real measurement data, and the lens has certain non-ideal features. Most importantly, partial reflections of brightly lit areas from the lens surfaces may decrease the contrast: dark areas seen through the lens appear less dark than they actually are; this effect can be neglected for the camera lens, but not so for the lens serving as a model of a cat's eye. Based on the given data, estimate (with the accuracy of ca 20%) the distance h between the axis of the camera and the axis of the lamp (which can be considered as a point source) if the distance of the camera from the paper sheet was $L = 4.8$ m.

Problem 42. Glass Ball

The first photo here is taken with a digital camera and shows a glass ball, backlit with diffuse dichromatic light which has only two narrow spectral lines (red 630 nm and violet 400 nm). This diffuse light comes from the white floor (marked with '1' in the figure) and the white walls (marked with '2'), both illuminated with violet and red LED lamps. The sensor of the camera has only red, blue and green sensors so that violet light appears in the photo as blue. The photo is taken from a distance much greater than the radius of the ball. On the back side of the ball, a very narrow opaque thread is glued to the glass surface, forming an arc of a great circle on the ball. In the photo, the thread is obstructed by the ball and cannot be seen directly. However, hugely deformed images of a very short segment of the thread are seen as blue (marked with 'b') and red ('r') ellipses. The letter 'p' indicates purple-coloured areas in the photo.

In the first photo, the centre of the ball is marked with a cross, and the perimeter of the ball is traced with a dashed line. You can find a larger version of the first photo on a separate sheet. You may take distance measurements there. In the larger photo, the boundary between the red and purple regions is also traced with a dashed line.

The second photo is taken while illuminating with a white LED with the ball turned so that the thread can be seen directly



- Explain qualitatively using a ray diagram why a segment of the thread is seen as a closed loop in the first photo.
- Determine the coefficient of refraction n_R for the red light.
- Determine the difference of the coefficients of refraction $\Delta n \equiv n_V - n_R$ for the red and violet light (with n_V denoting the coefficient of refraction for the violet light).



Problem 43. 2022 OPhO Open Exam

Consider two points S and S' randomly placed inside a D -dimensional [hyper-rectangular](#) room with walls that are perfect-reflecting $(D - 1)$ -dimensional hyper-plane mirrors. How many different light-rays that start from S , reflect N times on one of the walls and $N - 1$ times on each of the rest, then go to S' ? Use $D = 7$ and $N = 3$.

Problem 44

Two thin rods are placed in water so that they are parallel and the distance between them is equal to a . One of the rods was hit hard. How long does it take for the sound from the impact to reach the point on the second rod, remote from the impact site at a distance $\sqrt{a^2 + \ell^2}$ if the speeds of sound in the water and in the rod are equal to c and v , respectively?

Problem 45

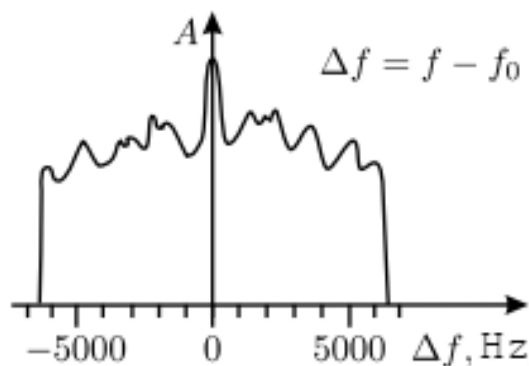
There are two ships in the ocean at a distance of $L = 3$ km from each other. The depth of ocean at their location is $H = 1$ km. A cannon shot was fired on one of the ships. How long after the shot the second ship will get first, second and third sound signals? The speed of sound in the water is $v_1 = 1.5$ km/s. The ocean floor is flat and consists of rocks, in which the speed of sound propagation is $v_2 = 4.5$ km/s. The speed of sound in the air during firing is $v_3 = 333$ m/s. There is no waving on the ocean surface.

Problem 46

An accordion player is riding on a rotating carousel with a radius of $R = 5$ meters. At what maximum angular speed ω of rotation of the carousel does the music performed by the accordion player sound out of tune for listeners on the ground, if good hearing allows one to discern a quarter-tone pitch? Two sounds differ by a quarter tone when their frequency ratio is $\sqrt[4]{2} \approx 1.0293$. The speed of sound in air under experimental conditions is considered equal to $c = 346$ m/s.

Problem 47

It is difficult to determine the wind speed in a tornado with conventional meteorological instruments (since the tornado is small in size and moves) and unsafe. It is proposed to measure it from a distance using a portable radar, since there is a lot of dust and small objects reflecting radio waves inside the tornado. The radar emits radio waves at a frequency of $f_0 = 10^{10}$ Hz. The spectrum of the signal reflected from the tornado is shown in the figure. Find the maximum wind speed in a tornado.



Problem 48

Recent studies have shown that in the ocean, the properties of water vary greatly with depth. For example, in northern latitudes, the speed of sound increases with depth according to the law $c(z) = c_0(1 + az)$, where c_0 is the speed of sound at the water surface, z is the depth, and a is a constant. If a sound wave is sent to ocean with angle α with the vertical, to what maximum depth it can go? The law of refraction of sound waves is completely analogous to the law of refraction of light.

Problem 49

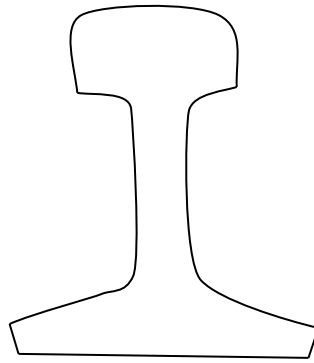
Without waiting for the bus, the pedestrian walked to the next bus stop, the pavilion of which was visible in the distance. After a while, he discovered that the apparent height of this pavilion is $k = 1.5$ times smaller than the apparent height of the pavilion behind him. After passing another $L = 100$ meters, the pedestrian noticed that, on the contrary, the pavilion in front seemed to him $k = 1.5$ times bigger than the pavilion behind him. Find the distance between stops. Consider the apparent size of an object to be inversely proportional to the distance to it. The stop pavilions are the same, a pedestrian walks along a straight line connecting them.

Problem 50

A passenger on a bus traveling along a straight canal with water observes a glare of light that is thrown to the calm surface of the water by a lantern on the opposite bank of the canal. Find the speed of the glare movement along the water surface relative to the channel banks, if the height of the lantern above the water surface is H , the height of the passenger's eyes above the water surface is h , and the bus speed is v .

Problem 51

How far is the train from the entrance to the metro station when a passenger standing on the edge of the platform near the end of the tunnel begins to see the glint of the headlights on the rails? Before entering the station, there is a fairly long turn with a radius of curvature R . Consider that the tunnel is horizontal and its cross-section is a rectangle of width $l \ll R$, the distance between the rails is $h \ll R$, the headlights of the train are located exactly above the rails, the rail profile is shown in the figure



Problem 52

If you look closely at your reflection, visible in a flat glass mirror with a silvered back surface, then in addition to the main image, you can see two more additional images of lower brightness. How will they be positioned relative to the main image? The thickness of is d and the refractive index of it is n .

Problem 53

If you look at the snow on a sunny winter day, you can see that the snow "sparkles". Assuming that the snow surface consists of randomly located flat crystals with an average face size $d = 1$ mm, estimate the average distance D between adjacent sparkles. How does D change when you close your one eye? The angular diameter of the Sun is $\varphi = 0.5^\circ$.

Problem 54

You are looking from a distance of $L = 2$ m at your reflection in a Christmas ball with a diameter of $D = 10$ cm. At what distance should your twin stand from you so that you see him as small as your image in the ball?

Problem 55

Two flat mirrors form an 80° dihedral angle, in which a point light source is located between them. How many different source images can you see in total?

Problem 56

Two flat mirrors form a dihedral angle. The point light source is located equidistant from the mirrors. At what values of the angle α between the mirrors will the source have exactly $N = 100$ different images?

Problem 57

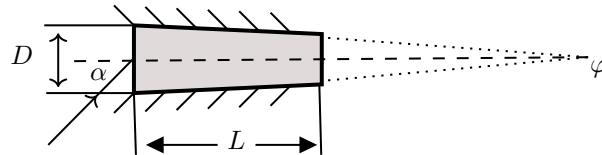
On the horizontal table there is a transparent cylinder with a base radius R and a height H_1 , made of glass with a refractive index $n = 1.5$. A point light source is located on the cylinder axis at a height H_2 above the upper base of the cylinder. Find the area of the shadow cast by the cylinder on the table surface.

Problem 58

An observer, who is in a room at a distance of $L = 3$ m from a window, covered from the outside with many small water drops, sees on it a bright spot with a radius of $r = 10$ cm from a very distant lantern located on the same level with the observer. Determine the maximum angle between droplet's surface and the surface of glass. Refractive index of water $n = 4/3$. Do not take into account the diffraction of light by droplets

Problem 59

The glass plate has the shape of an isosceles trapezoid in cross section (see figure). The base of the trapezoid is D , the height is L , and the angle between the lateral sides is $\varphi \ll 1$. The side surfaces of the plate are silvered, the refractive index of the glass is n . At what angles of incidence α will the light beam incident on the base pass through the plate?

**Problem 60**

Acetone and benzene mix with each other in any proportions, forming a clear solution. The volume of the mixture is equal to the total volume of the components before mixing. The refractive index n of the mixture depends on the concentrations of the acetone N_A and benzene N_B molecules as follows: $n_2 = 1 + K_A N_A + K_B N_B$, where K_A and K_B are some constants (polarizability of acetone and benzene molecules). The flask contains $V = 200$ ml of a mixture of acetone and benzene at a temperature of $T_1 = 50^\circ\text{C}$. A glass rod, lowered into a flask, is illuminated with light with a wavelength of $\lambda = 546$ nm and is not visible in this solution at a given temperature. How much and what liquid – acetone or benzene – should be added to the flask after it has cooled to a temperature of $T_2 = 20^\circ\text{C}$, so that after stirring the solution the glass rod is not visible under the same light? The refractive indices of light with a given wavelength for these liquids at a temperature of T_2 are $n_A = 1.36$ and $n_B = 1.50$, respectively, and for glass $n_g = 1.47$. The volumetric expansion coefficients of both fluids in the temperature range from T_2 to T_1 are the same and equal to $\mu = 0.00124\text{K}^{-1}$. Neglect thermal expansion of glass and evaporation of liquids.

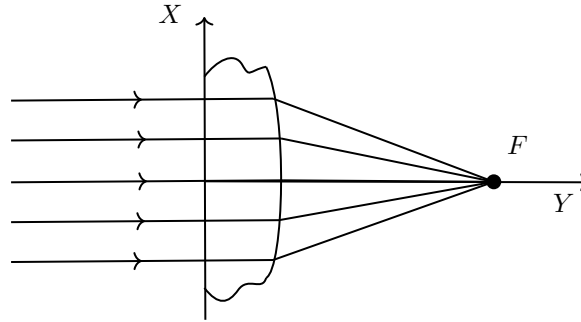
Problem 61

In a substance, the refractive index of which monotonously depends on one of the Cartesian coordinates, a light ray

can propagate along an arc of a circle. Find the type of dependence of the refractive index on this coordinate.

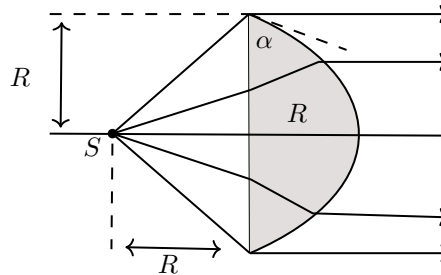
Problem 62

A wide, parallel beam of light falls perpendicularly onto a flat surface of a glass plate with a refractive index n . Find what condition the function $y = y(x)$ must satisfy (see the figure), which determines the shape of the right surface of the plate, so that all rays, passing through the plate, would be collected at point F , and show that for small deviations x from the axis of symmetry, the right surface is a sphere



Problem 63

There is a thick plano-convex uniform axisymmetric lens (see figure). The radius R of its flat base is equal to its thickness. The angle α between the surfaces bounding it at the point of their intersection is less than 90° . A point light source is placed on its axis of symmetry from the side of the flat base. The distance from it to the lens is R . The convex surface of the lens is smooth, and its shape is such that all rays passing through the lens without reflections form a strictly parallel beam with a flat front, the diameter of which is equal to the diameter of the lens. Find the angle α .

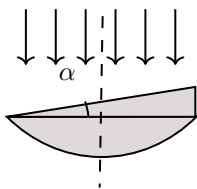


Problem 64

A yellow light filter made of glass with a thickness of $d = 1$ mm with a refractive index $n = 1.5$ is put between the camera lens with a focal length of $F = 16$ mm and the film. The camera is focused to infinity, shooting is taken, after which the filter is removed without changing the position of the lens. How far will the machine be focused now?

Problem 65

Someone made a strange flat-convex lens. The radius of the spherical surface R , the angle α is small. The lens thickness anywhere is much less than its radius r . What will a lens do with a parallel beam of light falling on it as shown? Will it have a focus, and if so, where? The refractive index of the lens glass is n



Problem 66

A point light source is located at a distance L from the screen. A thin converging lens with a focal length $F > L/4$, parallel to the screen, is moved between the source and the screen. At what position of the lens will the diameter of spot in the screen will be minimal?

Problem 67

The oscillations of the mathematical pendulum are filmed with a movie camera at a speed of $f = 24$ frames per second. One full swing takes $N = 48$ frames. The length of the pendulum on the film is $l = 10$ mm, the focal length of the objective is $F = 70$ mm. From what distance L was the video of the pendulum shot?

Problem 68

In case of lens opacity, people undergo an operation to replace the glasses with an artificial lens. The artificial lens for the eye is made so that the wearer can clearly see distant objects without glasses. Unlike the natural lens, the curvature of the surfaces of which can change (in this case, the eye focuses on selected objects - this is called accommodation), the artificial lens is rigid and cannot be “rebuilt”. Estimate the optical power of the lenses, which make it possible to easily read a book located at a distance of $d = 0.3$ m from the eye.

Problem 69

A severely myopic person wearing glasses with an optical power of $D = -10$ diopters can clearly see distant objects if the glasses are worn normally. To what maximum distance will he be able to see clearly if his glasses slide down onto his nose and are $l = 1$ cm further from his eyes than usual?

Problem 70

At a distance of $a = 20$ cm from the thin convex lens, a thin short rod is located along its main optical axis. The length of its actual image, given by the lens, is $k = 9$ times the length of the rod. How many times will the length of the image change if you move the stick along the axis by $\Delta a = 5$ cm further from the lens?

Note: For $|x| \ll 1$ you may assume that $\frac{1}{1+x} \approx 1-x$

Problem 71

It is known that when a beam of light strikes a flat glass plate perpendicular to its surface, 8% of the light energy is reflected, and 92% is transmitted. In other words, the reflection coefficient is $R = 0.08$, the transmittance is $T = 0.92$ (reflection from both surfaces of the plate is already taken into account here). Find the transmittance of a stack of n plates.

Problem 72

Find the reflectance, which is the fraction of the energy that is reflected from a system consisting of a thick, plane-parallel glass plate and a mirror behind it. The light falls perpendicular to the surface of the plate. The mirror is parallel to the plate and reflects all the light falling on it. The reflection, absorption and transmission coefficients of the glass plate are equal to R , A and T , respectively, with $R + A + T = 1$. Disregard the scattering and absorption of light in air.

Problem 73

The X-ray apparatus consists of a point source S and a receiver R , rigidly fixed on the frame. A cylindrical thick-walled balloon is moved between the source and the receiver (see figure). In this case, the intensity A of X-ray radiation recorded by the receiver depends on the x coordinate as shown in the figure. Is there any X-ray absorbing content inside the balloon?

Problem 74

The glass sphere, silver-plated from the inside (mirrored), has a round hole with an opening angle of 2α , into which a homogeneous parallel beam of rays falls perpendicular to the plane of the hole (see figure). Part of the rays, having undergone one reflection, will leave the sphere back through the hole. What fraction of the input beam power do they make up? Angle α is arbitrary (not necessarily small).

Problem 75

The inner surface of the pipe, the length of which is much longer than the diameter, is mirrored by half, and the remaining half is blackened (see figure). The pipe is placed on a black table with the mirror half down and the solar cell located on the table and on the pipe axis. In this case, the illumination of the photocell is equal to E_0 . What will it become if the pipe is turned over? The tube table is illuminated with uniformly diffused (isotropic) light.

Problem 76

A viscous liquid with a refractive index $n = 1.5$ is poured into a vertical cylindrical glass. A parallel beam of light of constant intensity falls vertically into the glass from above. The glass with liquid was spun around its axis to an angular velocity $\omega = 1$ rad/s, and the height of the liquid column on the axis of the glass became equal to $h = 30$ cm. Free fall acceleration $g = 10$ m/s², light absorption in liquid and its reflection inside the glass should be neglected. What is the ratio of initial intensity of the light at the bottom center of the cylinder to the intensity after rotating?

Problem 77

The traveler, returning home to his village on a dark night, on the road leading directly to his house, from a distance of $r = 5$ km, saw a candle light in one of the windows. Inside the house, near the next window, there is a Christmas tree decorated for the New Year with mirror balls. Estimate how far from the house will the traveler see the reflection of a candle in a Christmas ball with a diameter of $D = 10$ cm, if it perfectly reflects the light and is at a distance of $a = 1.8$ m from the candle on a line perpendicular to the road? The windows are the same, the candle burns evenly.

Problem 78. Little Mermaid

In the open sea, the water is completely blue, like the petals of the most beautiful cornflowers, and transparent, like clear glass - but deep down there! Not a single anchor will reach the bottom; on the bottom of the sea many, many bell towers would have to be placed one on top of the other, only then would they be able to stick out of the water. Mermaids live at that very bottom.



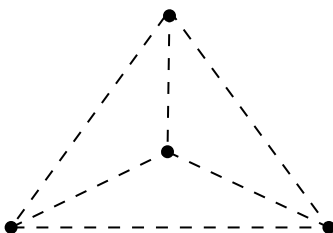
A prince $H = 1.8$ m high was standing at a coast of a quiet Atlantic Ocean and watched the Moon road, which starts at a distance $D_P = 5$ m from him horizontally and has a length $L_P = 50$ m. At the same time near the coast under the water, H meters below the water level, lied the Little Mermaid.

1. At what distance D_M from herself (also horizontally) will the Mermaid watch the Moon road?
2. What length L_M will this road have?

There are small waves over ocean surface. The refractive index of water is $n = 1.35$. Neglect Moon's angular size.

Problem 79. Lens Restoration

In Snell's archive, people found an optical scheme on which an ideal lens, an object, and its image were shown. The object is a rod with length ℓ and has two point sources at its ends. The rod and the main optical axis were drawn to be in the plane of the figure. The rod did not cross the lens. From time to time, the ink faded and only the sources themselves and their images remained visible in the drawing, and it is not known which of the four points corresponds to which. Interestingly, these are located at the vertices of and at the center of an equilateral triangle (see Figure).



1. Determine whether a source or its image is located at the center of the triangle.
2. Restore the optical scheme (object, image, lens, optical axis, foci) up to rotation by 120° and reflection.
3. Find the focal length of the lens.

Note: The lens is ideal i.e. a parallel beam gets focused in the focal plane precisely.

Problem 80. Lunar Eclipse

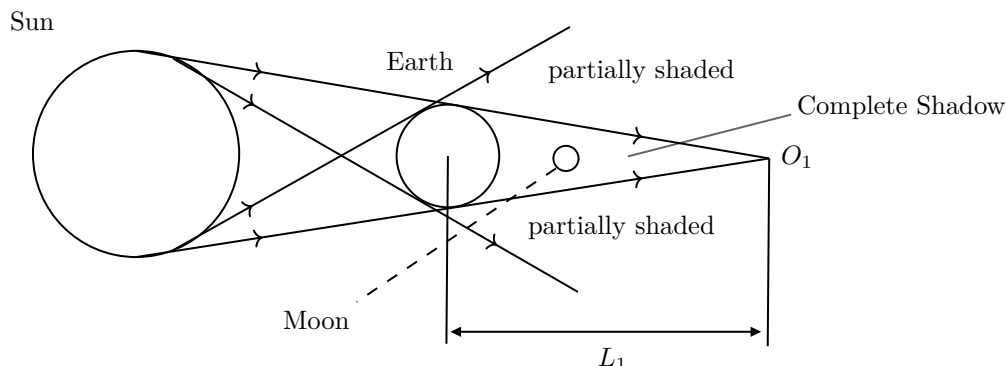
The sun is not a point source of light and has a small angular diameter as seen from Earth, $2\delta = 0.52^\circ$. Thus the area of the complete shadow of earth behind itself is finite.

1. Let the refraction of sunlight in Earth's atmosphere be absent. At what maximum distance L_1 , from the Earth will the complete shadow still be observed? Find the duration of total lunar eclipse in this case.
2. Refraction has a significant effect on the size of the complete shadow area. Let the Earth's atmosphere have an equivalent height of $h = 8$ km with average refractive index $n = 1.00028$.

Assuming that the border of the shadow is formed by rays travelling tangent to the Earth's surface determine

the maximum distance L_2 until which the complete shadow will be observed? How much of the lunar disk area will be shaded in this case?

The radius of Earth is $R = 6400$ km, acceleration due to gravity on Earth's surface is $g = 9.8$ m/s², the angular diameter of the moon is equal to the angular diameter of the Sun ($= 2\delta$) and the period of rotation of the Moon around Earth is $T_0 = 27.3$ days.



Problem 81. Lenses in a Circle

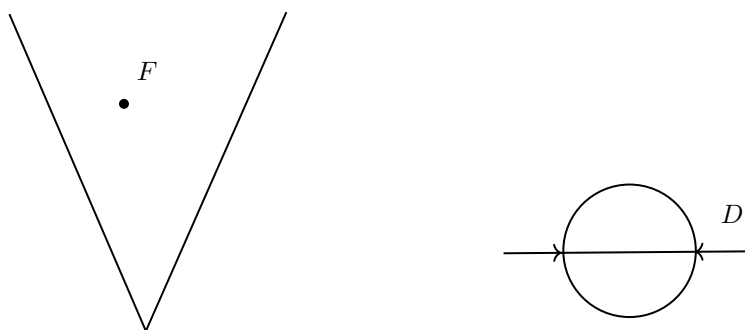
A manuscript found in Snell's archives discussed how a ray can go through a system of N identical lenses, the optical centers of which lie on a circle, with their planes being perpendicular to the circle which passes through the centers of the lenses. Over time, the ink faded and only traces from the planes of two adjacent lenses and the focus of one of them remained visible on the diagram (see Figure). It followed from the text that a light ray, being refracted by each of the lenses, goes along the sides of a regular N -gon. The type of lens and its diameter D are shown in the second Figure.

1. What kind of lenses could they be - convex (converging) or concave (diverging)?

By construction (using a compass and a ruler without divisions), restore:

2. The position of two more lenses (to the left and right of those shown in the figure);
3. Possible positions of the optical centers of the four resulting lenses;
4. The possible path of the beam through these lenses.

Justify your answer.

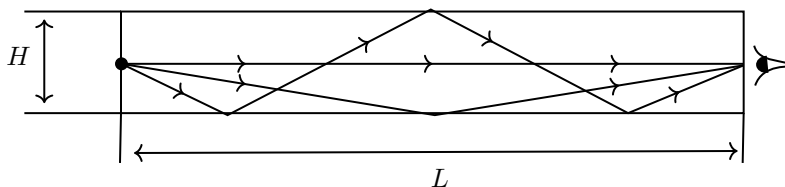


Problem 82. Flat Plate

Near the left end of a well-polished transparent plate, the refractive index of which is n , a point light source S is located (see figure). The thickness of the plate is $H = 1$ cm, its length is $L = 100$ cm. Light from the source falls on the left end of the plate at all possible angles of incidence ($0 - 90^\circ$). Both direct rays from the source and rays that have repeatedly experienced total internal reflection on the side faces of the plates fall into the eye of the observer.

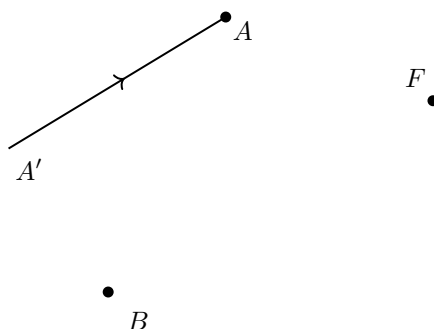
1. What is the maximum number of reflections that a ray from the source exiting at the right end of the plate can experience? Solve the problem for two values of refractive index: $n_1 = 1.73$, $n_2 = 1.3$.

2. In which of these two cases does light partially exit the plate through the side faces?



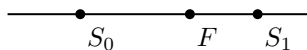
Problem 83. Refracted Ray

An optical scheme was found in Snell's archives. Over the time, ink faded and only three points can be seen now: the right focus F of thin lens, point A at which incident ray AA' is refracted, and point B which belongs to the left focal plane of the lens. Using this information, restore the position of the lens, the optical axis and the beam path behind the lens.



Problem 84. Archive of Snell

In Snell's archives, they found a drawing of an optical scheme, which showed the lens, the position of the point light source S_0 and its image S_1 . Over time, the ink faded, and only the position of the optical axis of the lens, the source S_0 , the image S_1 , and one of the foci F remained visible on the diagram (see figure). Using a compass and a ruler without divisions (i.e. a straight edge), restore the possible positions of the lens.



Problem 85. Millicar

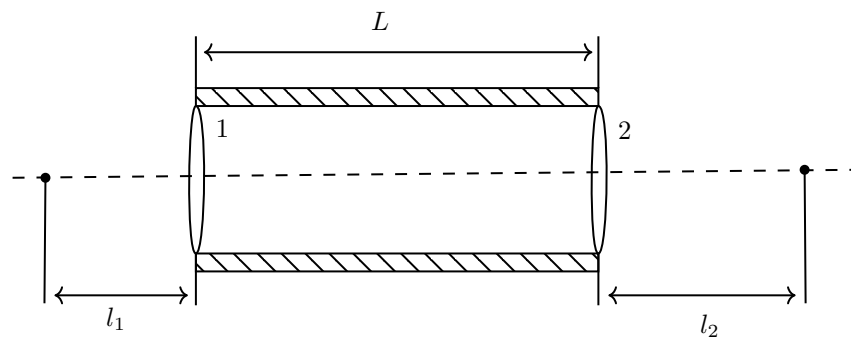
A very small car (as small as an ant) moves on a flat horizontal surface parallel the optical axis of a lens of focal length f . A point source of light is fixed to its roof such that it always moves on the optical axis of the lens. Speed of the car changes so that the speed of the image remains constant at v_0 . Determine the distance from the lens at which such a motion of the car is possible. Coefficient of friction between the wheels of the car and the road is μ .

Problem 86. The Task with 2 Lenses

In a Physics competition, participants had to determine the focal lengths of two converging lenses fixed at the ends of a hollow cylinder with length $l = 20.0$ cm. Vasya performed some experiments and found the following results -

1. The light rays emanating from a point source placed at a distance $l_1 = 5.0$ cm behind the left end of the system exit the cylinder as a parallel beam.
2. A parallel beam sent in through the left end of the cylinder converges to a point at a distance $l_2 = 10.0$ cm to the right of the cylinder's right end on its axis.

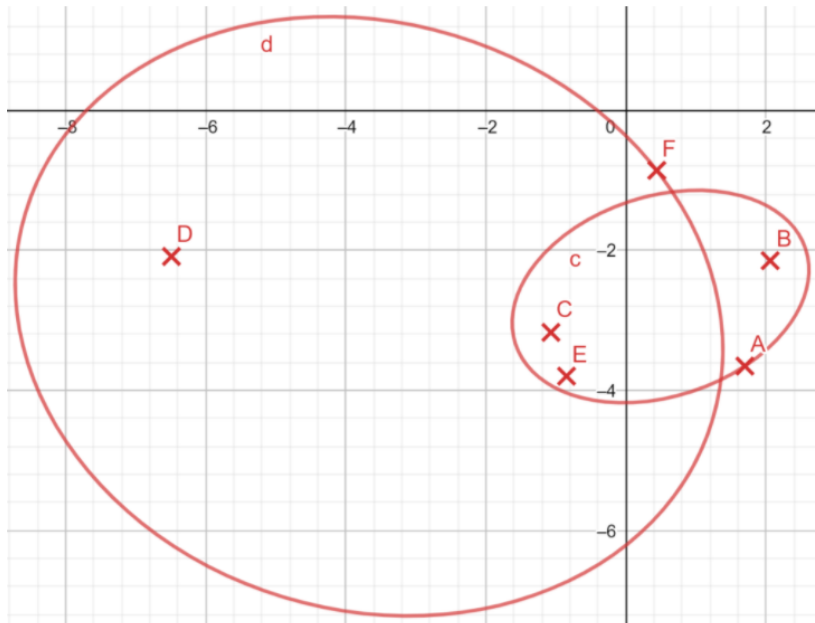
However, Vasya couldn't calculate the focal lengths F_1 and F_2 of the lenses using this set of data. Help poor Vasya.



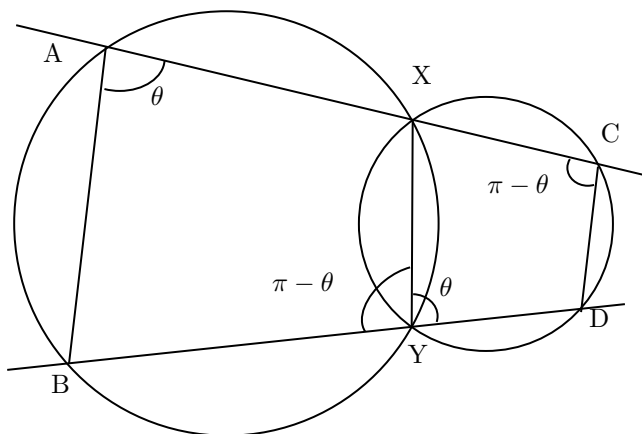
Solution 1. 37

This problem uses the hints *MENTIONA* and *MENTIONB*. As per these, the aim is to find two pairs each of parallel lines and perpendicular lines in the pre-image of the diagram whose corresponding images can be exactly identified. At this point we should look at the original diagram carefully, which is given below. We spot the points of intersection and the line through them.

Note - Point Z' refers to the image of the point Z .



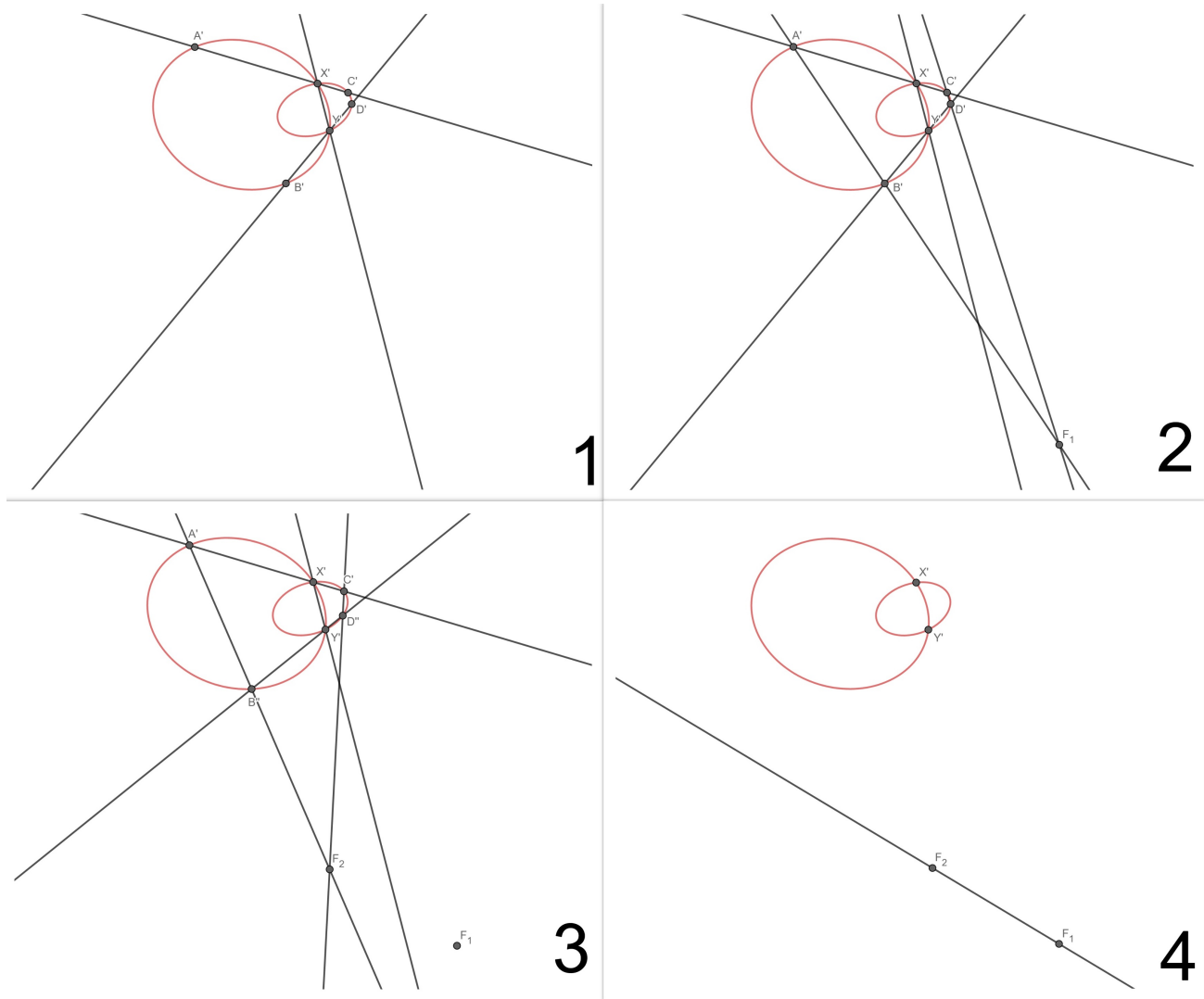
Consider two arbitrary circles S_1, S_2 intersecting at two points X, Y . First we try finding two parallel lines. We know that in a cyclic quadrilateral, the sum of two opposite angles is 180° . The only line whose image can be identified without additional constructions is XY . So we should try introducing extra elements, the first choice obviously being an arbitrary line through one of the two points X or Y . However adding just one line won't result in a parallel pair. So we draw two arbitrary lines, one each through X, Y and intersecting each circle once except at X/Y . Let the points be marked as shown in the figure.



We complete the quadrilateral $ABDC$. Let $\angle XAB = \theta$. This gives $\angle BYX = \pi - \theta$ using the property of cyclic quadrilaterals. Using the fact that B, Y, D are collinear by construction and then reusing the cyclic-quadrilateral identity, we get $\angle XCD = \pi - \theta$. However it is well known that the sum of internal angles made by an intercept between two line is $\pi = 180^\circ$ iff the two lines are parallel. This gives us the conclusion that AB, CD must be parallel in the given figure of the pre-image. Now we can use the fact that parallel lines meet at a point on the focal plane.

Construction 1 - Draw any two arbitrary lines, one each through X', Y' which intersect the image ellipses once, other than at X' or Y' (at A', B', C', D'). Now choose one of the ellipses and draw a line through its point of intersections with the two lines drawn through X', Y' (this line refers to $A'B'$ here) . Repeat this for the other ellipse (to get $C'D'$). The intersection of these lines gives a point on the focal plane.

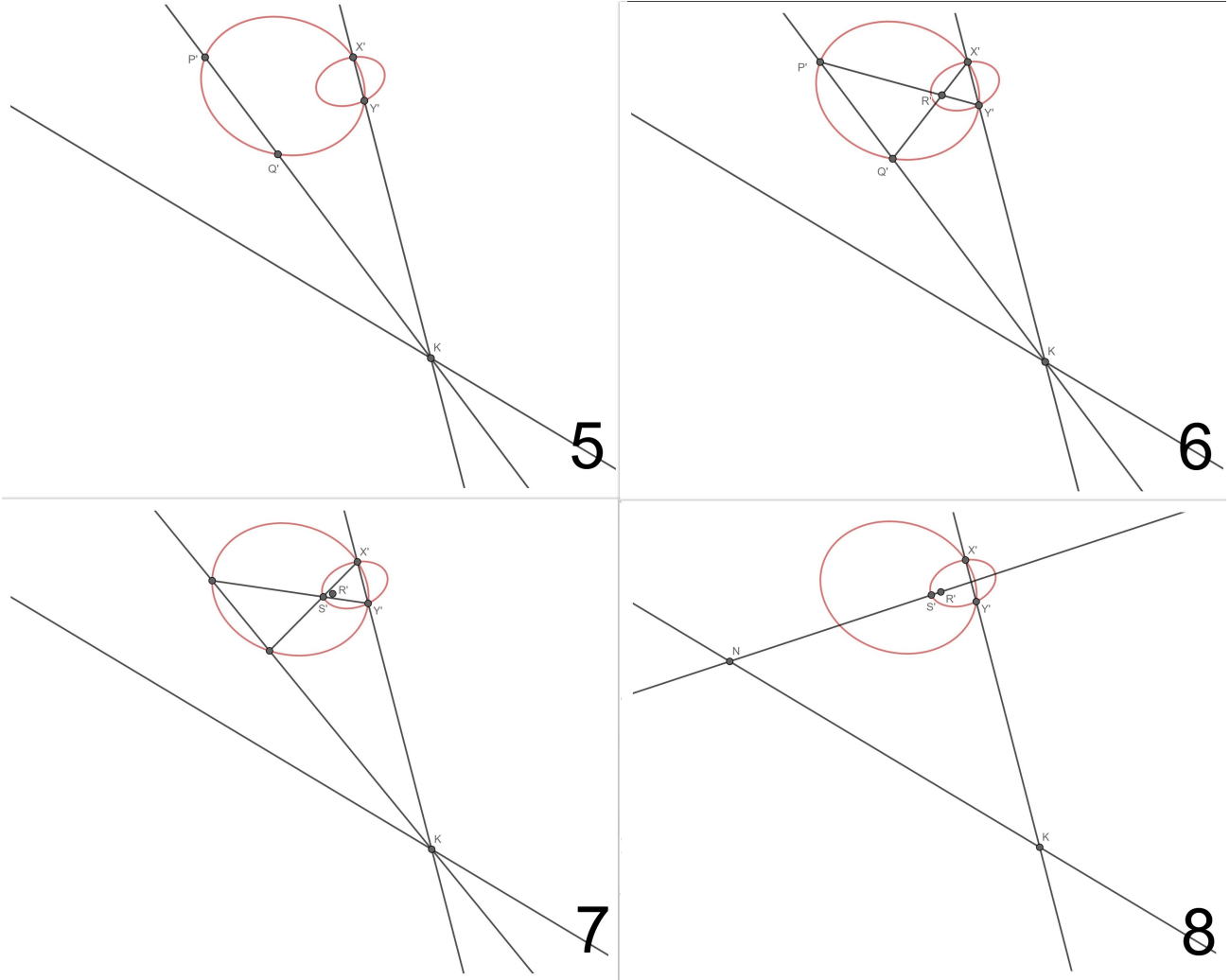
Now repeat the process for another pair of arbitrary lines through X' and Y' . Note that we can keep the diagram comparatively cleaner by introducing only one new line through either of X' or Y' while using the previously drawn line through the other point. This gives another focal point. Join the two focal points to get the focal plane.



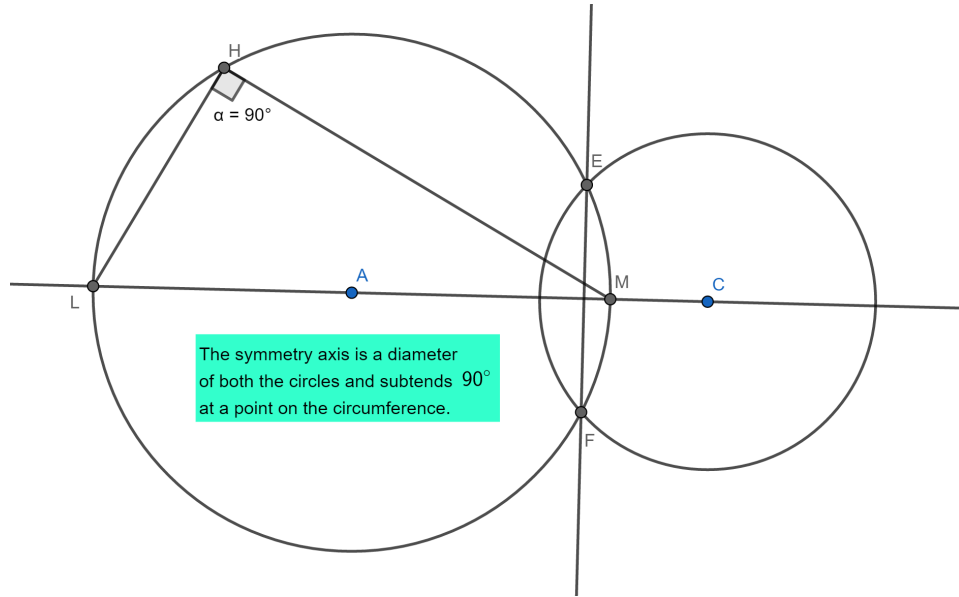
Now we try finding two pairs of perpendicular lines. An obvious pair of perpendicular lines in the pre-image containing two circles is the line XY and the symmetry axis of the system, the line through the centers of the two circles. If we are able to find any two points on the image of the symmetry axis, we can construct it. Let the point of intersection of the line $X'Y'$ with the focal plane be K . Note that all lines which were parallel to the line XY in the pre-image will have their images pass through K . Also these lines parallel to XY will be perpendicular to the symmetry axis.

Consider a line parallel to the line XY drawn to cut either of the two circles in the pre-image at two distinct points. This gives us a trapezium if we join the point of intersection of the drawn line with the circle to X and Y . Note that this trapezium will have its lateral sides equal in length by symmetry about the line joining the centers. Hence, the point of intersection of its diagonals will lie on the symmetry axis. So we can trace a point on the symmetry axis' image by constructing image of this trapezium.

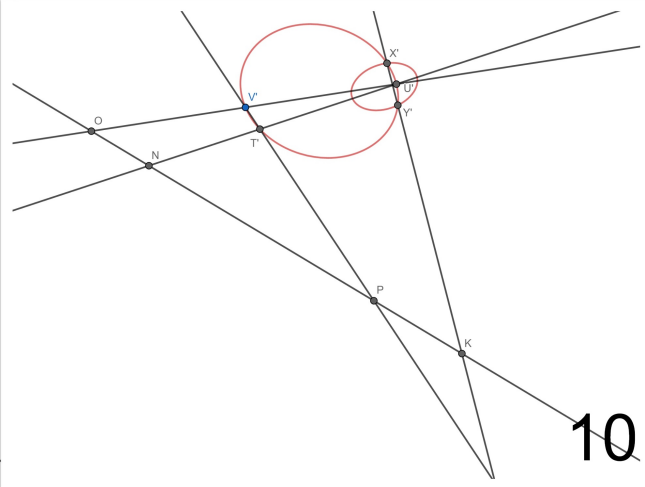
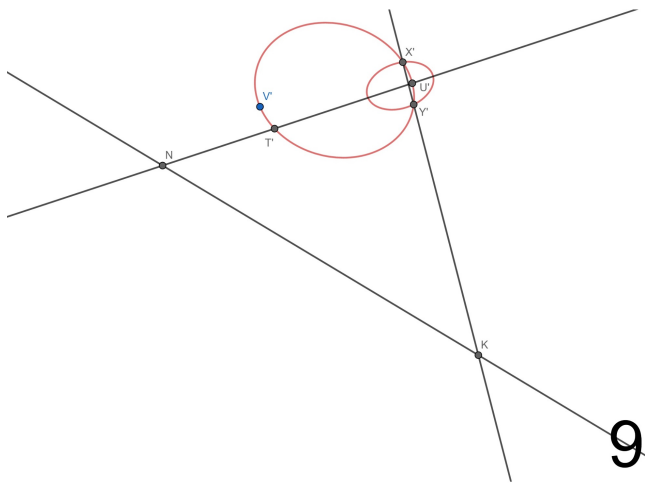
Construction 2 - Start by drawing an arbitrary line through K that cuts one of the ellipses at two points. Let this line's point of intersections with the ellipse be P', Q' . The pre-image of $P'Q'$ would have been parallel to XY . So the trapezium described in the above paragraph is $PQYX$. The image of the point of intersection of diagonals of $PQYX$ will be that of diagonals of $P'Q'Y'X'$. So join $P'Y'$ and $Q'X'$ and obtain one point on the image of the symmetry axis R' . Similarly take another arbitrary line through the point K and call the point obtained after repeating above steps as S' . Join R' and S' to get the symmetry axis. Now we have the images of a pair of perpendicular lines.



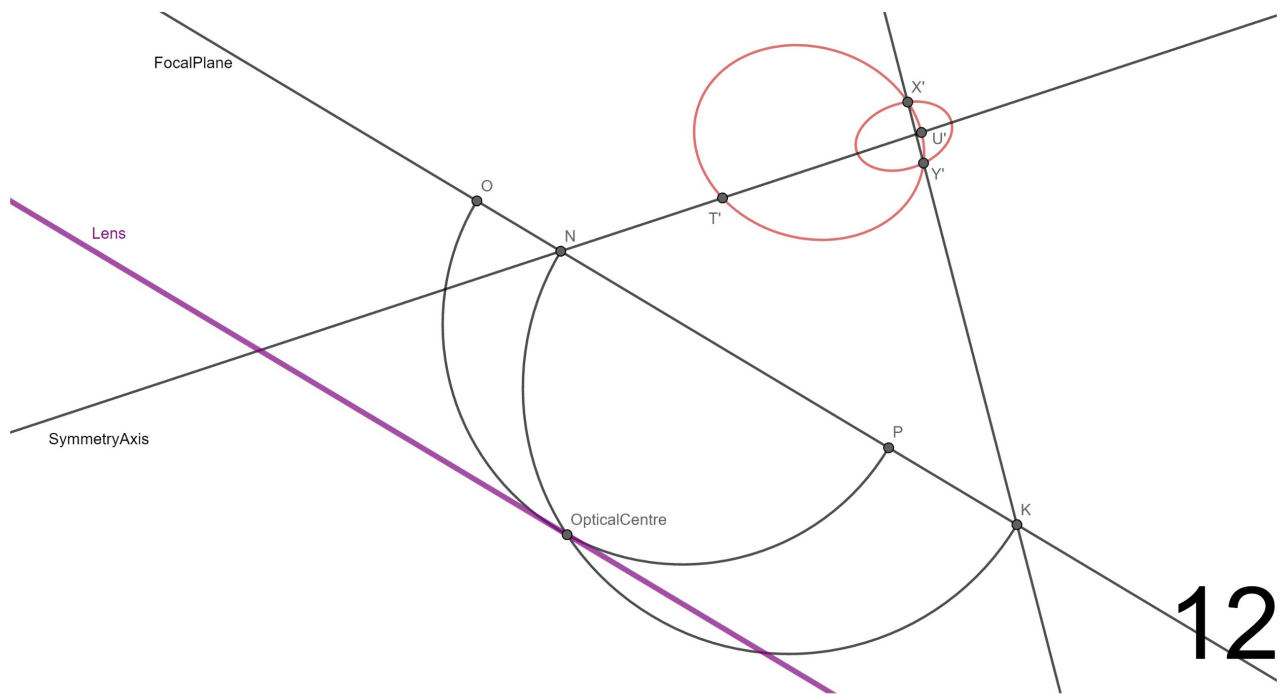
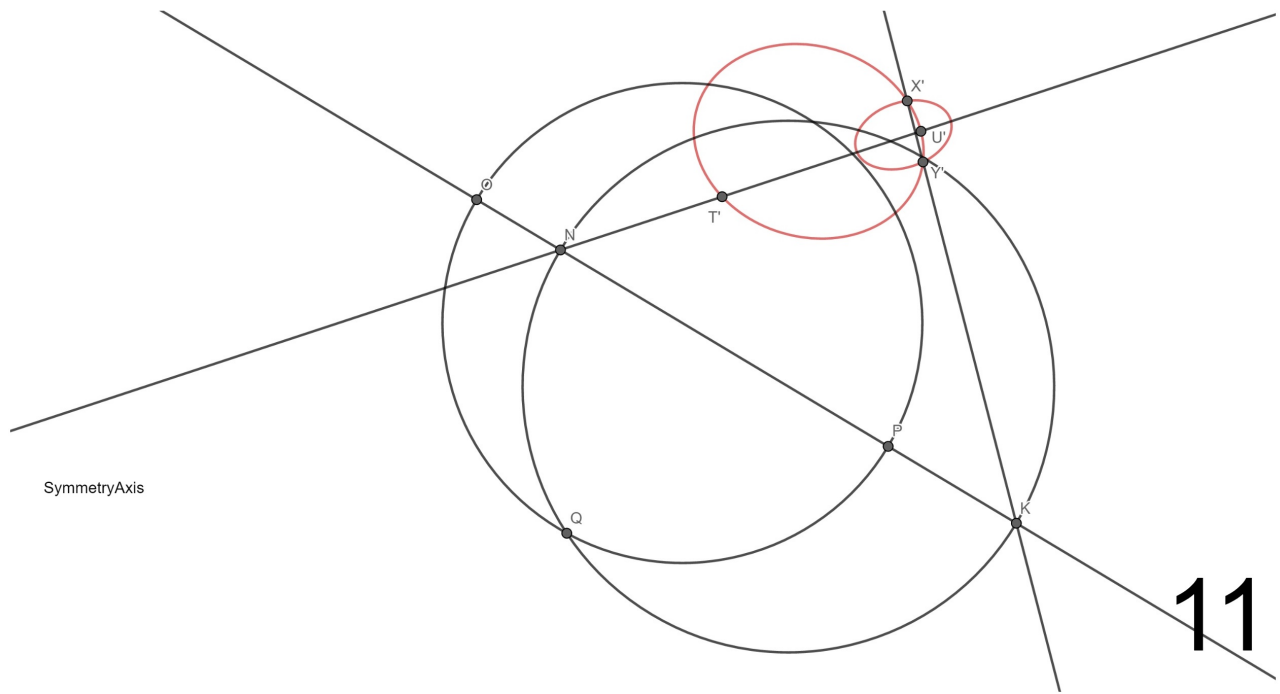
To identify another such pair, note that the angle subtended by the diameter of a circle at an arbitrary point on it is 90° . Also the symmetry axis is just a continuation of one of the diameters of the circle. We can use this property as no special objects are needed except the axis (found in Construction 2) and an arbitrary point.



Construction 3 - Call the intersection of the image of the symmetry axis with any one of the ellipses to be T', U' . Take an arbitrary point on the same ellipse V' . Then $\angle TVU = 90^\circ$ (Note that the indices without a ' are correspondences in the pre-image). Hence, we have another pair of lines whose pre-images were perpendicular - $T'V'$ and $U'V'$.



Construction 4 -With two pairs of perpendicular lines, we can now repeat the steps similar to *Problem XYZ* to get the optical centre. Take the points of intersection of $T'V'$ and $U'V'$ with the focal plane (P, O respectively) as diameter and construct a circle. Do the same with the points of intersection of the focal plane with $X'Y'$ (K) and the image of the symmetry axis (N). These two circles intersect at two points - one lies on the same side of the straight line and the ellipses and other lies in between them. However, the images are real and the corresponding focal plane of the image and the image itself must lie on the same side of the the optical centre. This specifies which point is of interest to us and we get the exact location of the optical centre, called Q here. Now we can draw a line parallel to the focal plane through the optical centre. This gives us the line representing the lens (its intersection with the plane of the image, to be precise) and we are done.



Centre of circles

Tangents to the circles (made at their interceptions) intersect at the line joining the centers because of symmetry:

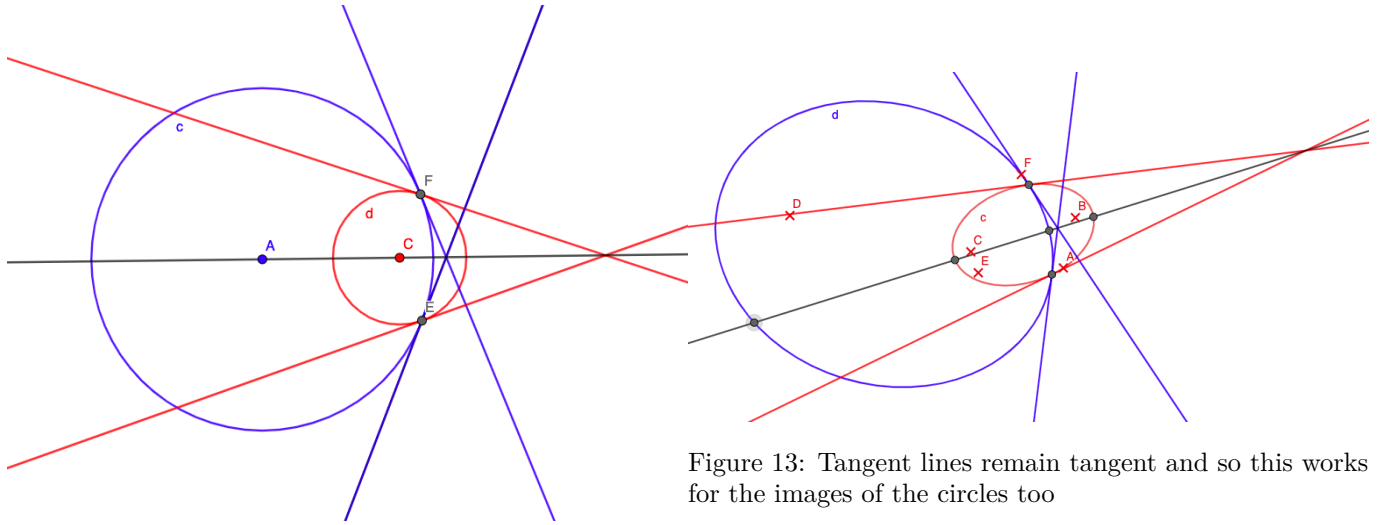


Figure 12: Original circles

Thus we find the line on which the original centers lie. Now if we consider lines GF and FH (in the following figure part (a)) they are at right angles because of Thales theorem in the smaller circle, therefore JK is a diameter of the greater circle, and hence we find its centre:

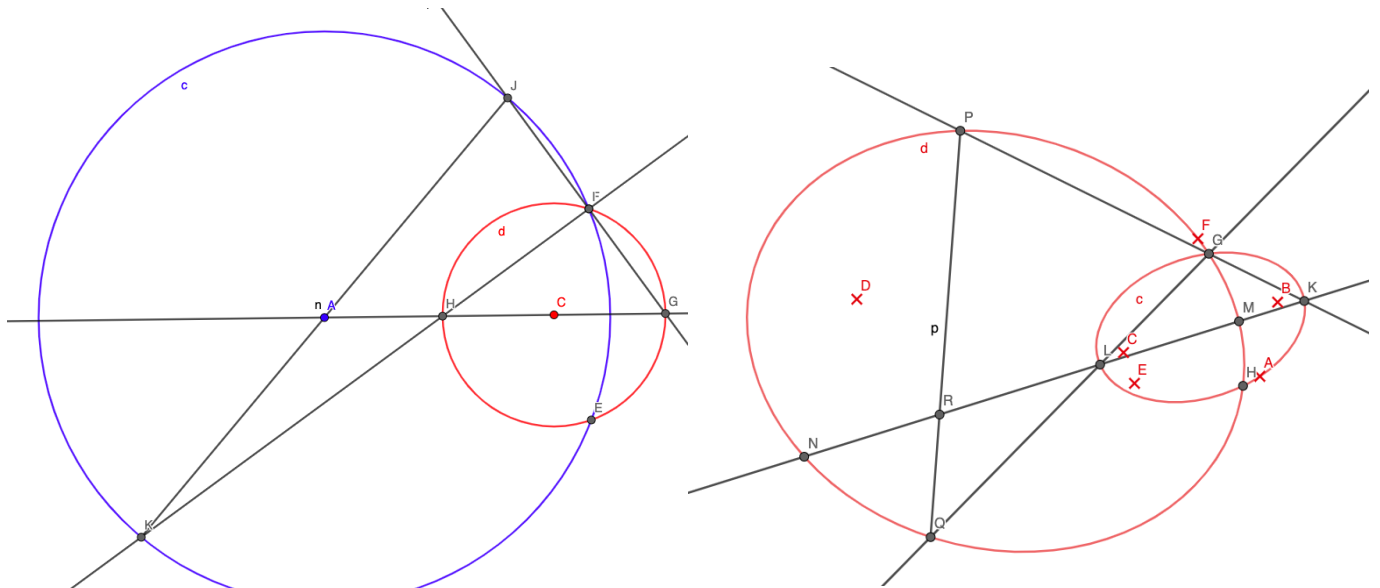


Figure 14: Finding another diameter of original bigger circle, its center is A

Figure 15: With their images. Centre of bigger circle is R

Lens

Now consider a pair of lines (GT and SH in the figure below) going through the centre of the bigger ellipse, in the original circle these lines were diagonals of a rectangle inscribed in the bigger circle. If we extend two parallel sides of the rectangle, their point of intersection (name it point A) lies at infinity, and image of this point (A') therefore lies in the image focal plane of the lens. We can thus find two points (name them A' and B' in the figure) which lie in the focal plane from one rectangle (2 pairs of such sides) and hence find the focal plane. The rays coming from A and B are at right angles, if we consider rays going through the centre of lens (C), they don't get refracted and therefore

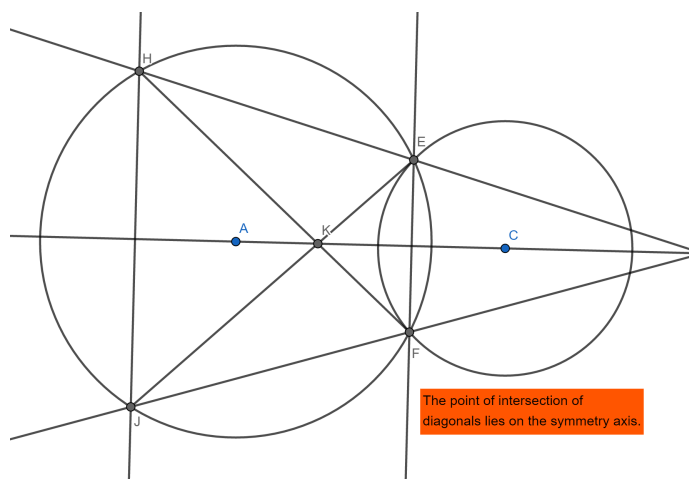
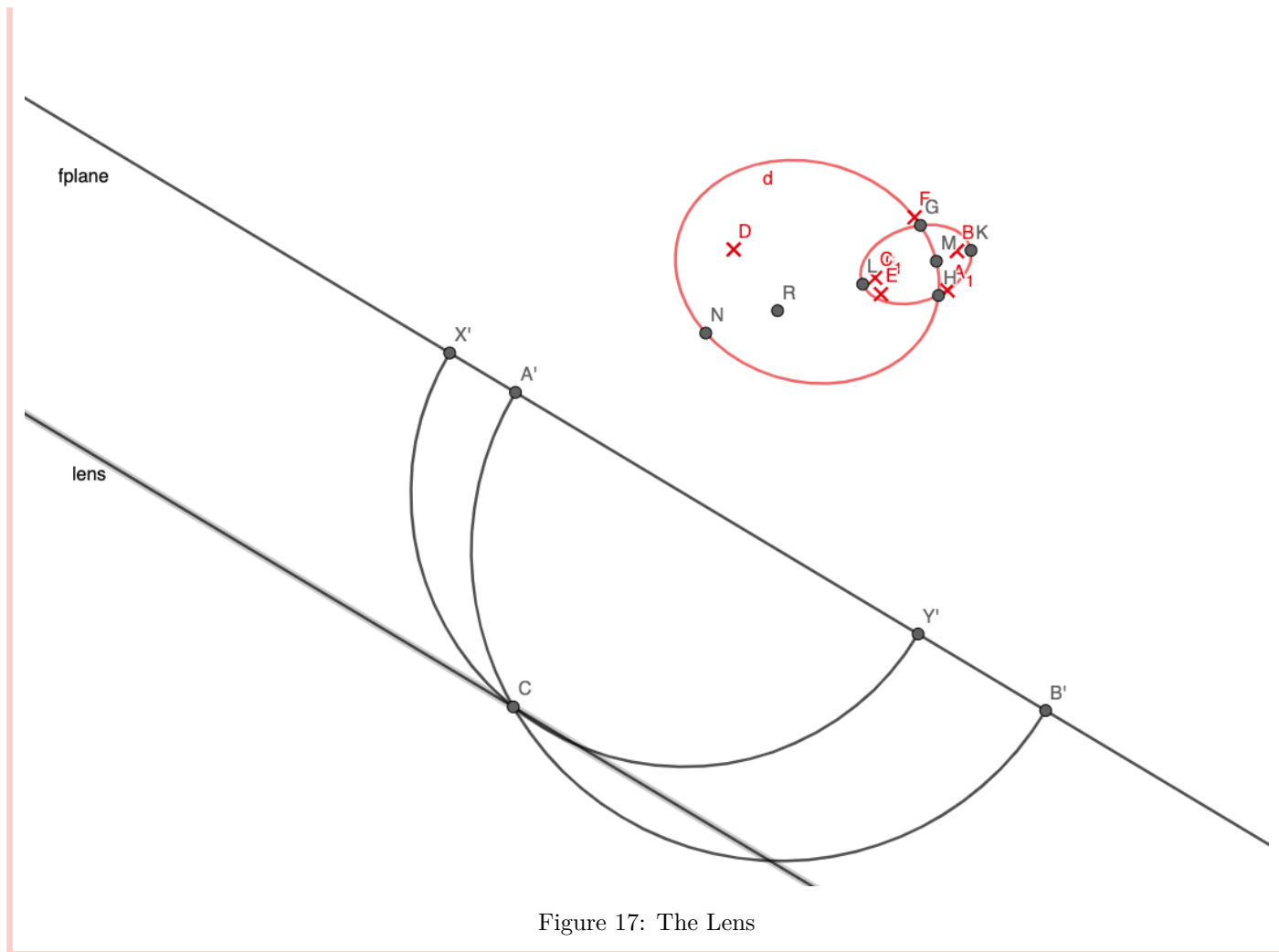
$\angle A'CB' = 90^\circ$. Hence the centre of lens C lies at a Thales circle with points A' and B' forming its diameter. The image focal plane must lie between the image and the lens plane hence we only need to consider one semicircle of the Thales circle:

Figure 16: Finding the Optical Centre

So we create two such rectangles in the greater circle, by choosing two pairs of arbitrary lines going through the centre of circle. We get hence two Thales circles on which the centre of lens C lies, and therefore we find C as the intersection of the two circles. The lens will be parallel to the focal plane:

Figure 16: Finding the Optical Centre

So we create two such rectangles in the greater circle, by choosing two pairs of arbitrary lines going through the centre of circle. We get hence two Thales circles on which the centre of lens C lies, and therefore we find C as the intersection of the two circles. The lens will be parallel to the focal plane:





Solution 3. 6

- (a) Snell's Law may be expressed as $\frac{\sin \theta}{\sin \theta_0} = \frac{c}{c_0}$ (1) where c is the speed of sound.

Consider some element of ray path ds and treat this as, locally, an arc of a circle of radius R . Note that R may take up any value between 0 and ∞ . Consider a ray component which is initially directed upward from S .

$$ds = R d\theta, \text{ or } \frac{ds}{d\theta} = R$$

From equation (1), for a small change in speed dc , $\cos \theta d\theta = \sin \theta_0 \frac{dc}{c_0}$.

For the upwardly directed ray $c = c_0 + bz$ so $dc = b dz$ and

$$\frac{\sin \theta_0}{c_0} b dz = \cos \theta d\theta, \text{ hence } dz = \frac{c_0}{\sin \theta_0} \frac{1}{b} \cos \theta d\theta$$

We may also write (here treating ds as straight) $dz = ds \cos \theta$. So

$$ds = \frac{c_0}{\sin \theta_0} \frac{1}{b} d\theta$$

Hence

$$\frac{ds}{d\theta} = R = \frac{c_0}{\sin \theta_0} \frac{1}{b}$$

This result strictly applies to the small arc segments ds . Note that from equation (1), however, it also applies for all θ , i.e. for all points along the trajectory, which therefore forms an arc of a circle with radius R until the ray enters the region $z < 0$.

- (b) Here

$$\begin{aligned} z_s &= R - R \sin \theta_0 \\ z_s &= R(1 - \sin \theta_0) = \frac{c_0}{b} \frac{1 - \sin \theta_0}{\sin \theta_0} \\ \theta_0 &= \arcsin \frac{c_0}{bz_s + c_0} \end{aligned}$$

- (c) The simplest pathway between S and H is a single arc of a circle passing through S and H . For this pathway:

$$X = 2R \cos \theta_0 = \frac{2c_0}{b} \cot \theta_0$$

Hence

$$\cot \theta_0 = \frac{bX}{2c_0}$$

The next possibility consists of two circular arcs linked as shown. For this pathway:

$$\frac{X}{2} = 2R \cos \theta_0 = \frac{2c_0}{b} \cot \theta_0$$

$$\cot \theta_0 = \frac{bX}{4c_0}$$

In general, for values of $\theta_0 < \pi$, rays emerging from S will reach H in n arcs for launch angles given by

$$\cot \theta_0 = \frac{bX}{2nc_0}$$

Note that when $n = \infty$, $\theta_0 = \pi$ as expected for the axial ray.

(d) With the values cited, the four smallest values of launch angle are

$$n = 1 \Rightarrow \theta_0 = 86.11$$

$$n = 2 \Rightarrow \theta_0 = 88.09$$

$$n = 3 \Rightarrow \theta_0 = 88.73$$

$$n = 4 \Rightarrow \theta_0 = 89.04$$

(e) The ray path associated with the smallest launch angle consists of a single arc

$$t_{13} = \int_1^3 dt = \int_1^3 \frac{ds}{c}$$

$$t_{13} = 2 \int_1^2 \frac{ds}{c} = \int_{\theta_0}^{\pi/2} \frac{Rd\theta}{c}$$

Using $R = \frac{c}{b \sin \theta}$ gives

$$t_{13} = \frac{2}{b} \int_{\theta_0}^{\pi/2} \frac{d\theta}{\sin \theta}$$

$$t_{13} = -\frac{2}{b} \ln \tan \frac{\theta_0}{2} = 6.6546s$$

The axial ray will have travel time given by

$$t = \frac{X}{c_0} = 6.6666s$$

thus this axial ray travels slower than the example cited for $n = 1$, thus the $n = 1$ ray will arrive first.

Solution 4. 7

Let us use angle φ to describe the position of the rays in the glass (Fig. 5). According to the law of refraction $\sin 45^\circ = \sqrt{2} \sin \beta$, so $\sin \beta = 0.5$, or $\beta = 30^\circ$. The refracted angle is 30° for all of the incoming rays. We have to investigate what happens if φ changes from 0 to π . It is easy to see that φ can not be less than 60° . Hence, the critical angle is given by $\sin \beta_{crit} = 1/n$ or $\beta = 45^\circ$. In the case of total internal reflection, $\varphi = 180^\circ - 60^\circ - 45^\circ = 75^\circ$. If φ is more than 75° the rays can emerge the cylinder. Increasing the angle we reach the critical angle again. Thus the rays are leaving the glass cylinder if: $75^\circ < \varphi < 165^\circ$ and the arc of the emerging rays, subtends a central angle of 90° .

Solution 5. 9

Consider a light ray passing through a system of parallel plates with different refractive indexes. From the Snell law we have

$$n_1 \sin \beta_1 = n_2 \sin \beta_2$$

Thus, in general:

$$n_i \sin \beta_i = \text{const}$$

This relation does not involve plates thickness nor their number. So, we may make use of it also in case of continuous dependence of the refractive index in one direction (in our case in the x direction)

At the point A the angle $\beta_A = 90^\circ$. The refractive index at this point is n_0 . Thus, we have

$$n_A \sin \beta_A = n_B \sin \beta_B = n_0$$

.

Additionally, from the Snell's law applied to the refraction at point B , we have

$$\frac{\sin \alpha}{\sin 90^\circ - \beta_B} = n_B$$

Therefore $\sin \alpha = n_B \cos \beta_B = \sqrt{n_B^2 - n_0^2}$

And finally

$$n_B = \sqrt{n_0^2 + \sin^2 \alpha} = 1.3$$

The value of x_B can be found from the dependence $n(x)$.

$$n_B = \frac{n_0}{1 - \frac{x_B}{R}}$$

$$x_B = R \left(1 - \frac{n_0}{n_B} \right) = 1 \text{ cm}$$

The answer to the third question requires determination of the trajectory of the light ray. According to considerations described at the beginning of the solution we may write

$$n(x) \sin \beta(x) = n_0$$

$$\sin \beta(x) = \frac{R - x}{R}$$

or equivalently

$$\tan \beta(x) = \frac{R - x}{\sqrt{R^2 - (R - x)^2}}$$

But $\tan \beta(x)$ is the derivative of $y(x)$.

$$\frac{dy}{dx} = \frac{R - x}{\sqrt{R^2 - (R - x)^2}}$$

$$y = \sqrt{R^2 - (R - x)^2} + C$$

The value of C is determined from the initial conditions $y(0) = 0$.

Therefore $y^2 + (R - x)^2 = R^2$. It means that the ray moves in the plate along a circle.

To calculate the width of the plate, we may write that

$$R^2 = d^2 + (R - x_B)^2 = 5 \text{ cm}$$

Solution 6. 11

- (a) From the Snell's Law, $n_A \sin \alpha = n_B \sin \beta$
 (b) The phenomenon is due to total reflection in a warm layer of air when $\beta = \pi/2$. This gives $n_B = n_A \sin \alpha$
 (c) As the density, ρ , of the air is inversely proportional to the absolute temperature, T , for fixed pressure we have
- $$n = 1 + k\rho = 1 + \frac{k}{T}$$

The value given at 15°C determines the value of $k = 0.0795$. In order to have total reflection we have $n_{30} \sin \alpha = n_T$ or

$$\left(1 + \frac{k}{303}\right) \frac{L}{\sqrt{h^2 + L^2}} = 1 + \frac{k}{T} \text{ with } h = 1.6 \text{ m and } L = 250 \text{ m.}$$

As $h \ll L$ we can use a power expansion in h/L :

$$T \approx 303 \left(1 + \frac{303h^2}{2kL^2}\right) = 328 \text{ K}$$

Solution 7. 12

- (a) The speed of light in medium I is $\frac{c}{n_1}$ and in medium II is $\frac{c}{n_2}$, where c is the speed of light in vacuum. Let the two media be separated by the fixed line $y = y_1$. Then time $T(\alpha)$ for light to travel from origin $(0, 0)$ to (x_0, y_0) is

$$T(\alpha) = \frac{n_1 \sqrt{y_0^2 + \alpha^2}}{c} + \frac{n_2 + 2\sqrt{(x_0 - \alpha)^2 + (y_0 - y_1)^2}}{c}$$

Differentiating w.r.t. α and setting the derivative of $T(\alpha)$ to zero

$$\frac{n_1 \alpha}{\sqrt{y_0^2 + \alpha^2}} - \frac{n_2(y_0 - \alpha)}{\sqrt{(x_0 - \alpha)^2 + (y_0 - y_1)^2}} = 0$$

Or equivalently

$$n_1 \sin i_1 = n_2 \sin i_2$$

- (b) From Snell's law $n_0 \sin i_0 = n(y) \sin i$

$$\frac{dy}{dx} = -\cot i$$

$$n_0 \sin i_0 = \frac{n(y)}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}}$$

$$\frac{dy}{dx} = -\sqrt{\left(\frac{n(y)}{n_0 \sin i_0}\right)^2 - 1}$$

- (c)

$$\int \frac{dy}{\sqrt{\left(\frac{n_0 - ky}{n_0 \sin i_0}\right)^2 - 1}} = -\int dx$$

Note $i_0 = 90^\circ$ so $\sin i_0 = 1$.

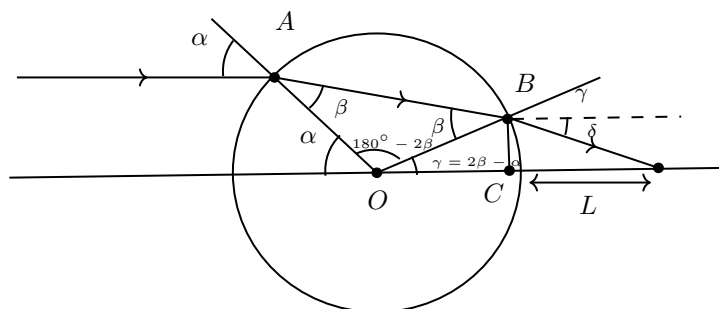
The solution, using the integral in the text problem, is

$$x = \frac{n_0}{k} \ln \left(\frac{n_0 - ky}{n_0} + \sqrt{\left(\frac{n_0 - ky}{n_0} \right)^2 - 1} \right) + C$$

From the initial condition $y(0) = 0$ we get $C = 0$

Solution 8. 17

Let's draw a large and comprehensive picture



Looking at isosceles triangle formed by two radii of the cylinder it is possible to define all the angles in terms of incident angle α and refraction angle β which are related with each other with Snell's law:

$$\sin \alpha = n \sin \beta$$

We can approximate the sine using small angle approximation since the beam is thin

$$\sin \phi \approx \tan \phi$$

Then,

$$\alpha = n\beta$$

Also we can define the length BC in terms of two different equalities so

$$BC = OB \sin \gamma = CF \tan \delta$$

Approximating

$$R\gamma = L\delta$$

And we can express angles γ and δ in terms of β :

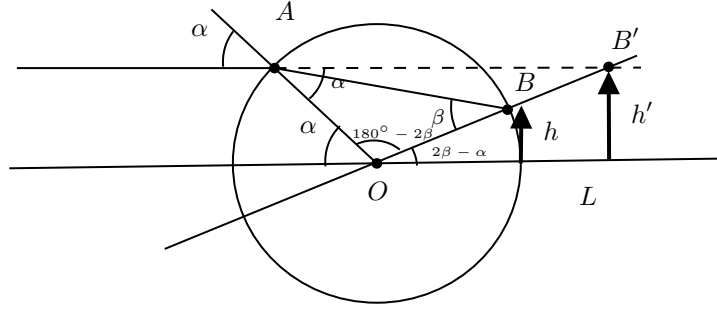
$$\begin{aligned}\gamma &= 180^\circ - ((180^\circ - 2\beta) + \alpha) = 2\beta - \alpha = (2 - n)\beta \\ \delta &= \alpha - \gamma = 2(n - 1)\beta\end{aligned}$$

Solving last 3 equations gives us

$$L = \frac{2-n}{2(n-1)}R$$

Solution 9. 18

Every point of object is formed a lot of beams which are focused in eye. So we will consider two rays emitted from “head” of the objects one is going directly to the center of the sphere other one is the the ray which goes parallel to the main axis after refractions for sake of simplicity. We should draw a comprehensive diagram in order to see what is going on



It's clearly seen from picture the eye will assume the parallel beam of light will go in straight light and make the top of the object at point B' . Using Snell's law:

$$\sin \alpha = n \sin \beta$$

So using small angle approximation (since $R \gg h$)

$$\alpha = n\beta$$

Apparent length can be estimated as

$$h' = R \sin \alpha \approx R\alpha$$

While the actual length is

$$h = R \sin(2\beta - \alpha) \approx R(2\beta - \alpha)$$

Solving these three equations results

$$h' = \frac{n}{2-n}h$$

Alternative approach

From one of the previous problems you derived (should) the focus distance of spherical lens is

$$F = \frac{2-n}{2(n-1)}R + R = \frac{n}{2n-2}R$$

So sphere can be considered as a convex lens. Using the lens equation

$$\frac{1}{R} + \frac{1}{f} = \frac{2n-2}{nR}$$

Therefore the image distance is

$$f = \frac{n}{n-2}R$$

From similar triangles we can find the apparent height:

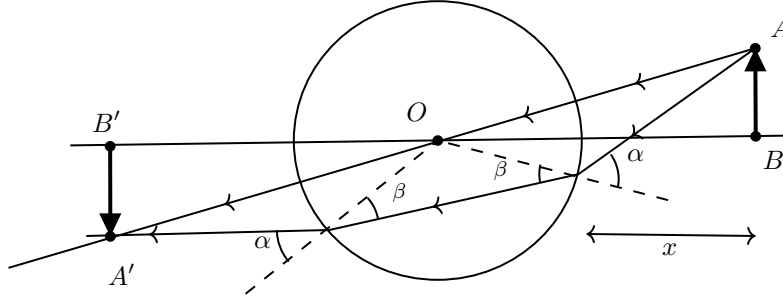
$$\frac{h}{R} = \frac{h'(n-2)}{nR}$$

So the answer is

$$h' = \frac{n}{2-n}h$$

Solution 10. 19

The mirror creates an additional image of the object, so there are two identical objects behind the sphere located symmetrically relative to the mirror. So like any other geometrical optics problem we must draw a diagram that contains everything:



You can go ahead and try to find the solution with using pure geometry, it is up to you. But in this solution we will write more like “cheating” way. So we can think of the sphere as a lens with focus distance $F = \frac{n}{2n-2}R$; this is known from previous problems. Using the lens formula

$$\frac{1}{d} + \frac{1}{f} = \frac{1}{F}$$

So our objects distances are $d_1 = 2R$ and $d_2 = 4R$, an observer will see two different images at distances

$$f_1 = \frac{2nR}{4n-5}$$

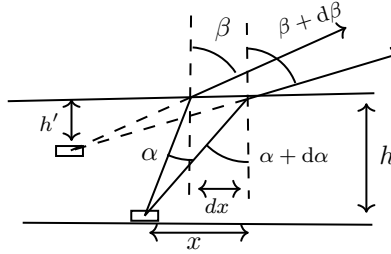
$$f_2 = \frac{4nR}{7n-8}$$

Apparent heights $h' = h \frac{f}{d}$ will be:

$$h'_1 = \frac{n}{4n-5}R \quad h'_2 = \frac{n}{7n-8}R$$

Solution 11. 20

Human eye estimates the distance to the object by focusing several beams in his eye’s “lens”. Key to this problem is that for estimation of distance we have to examine two beams of light. Let’s look at two beams propagating with very close angles α and $\alpha + d\alpha$. After refraction the beams will deviate to angles β and $\beta + d\beta$ respectively. Human brain will think the beams go with straight lines, which result object appear in height h'



We can find the small changes in x and α using basic geometry and derivatives

$$x = h \tan \alpha$$

$$dx = h \sec^2 \alpha d\alpha = h' \tan \beta$$

$$dx' = h \sec^2 \beta d\beta$$

We can see that $dx' = dx$ which yields

$$h' = h \frac{\cos^2 \beta d\alpha}{\cos^2 \alpha d\beta}$$

Now we need to find relation between α and β and its derivatives. In order to do this we can use Snell’s law and its

differentiated version for a refraction

$$n \sin \alpha = \cos \beta$$

$$n \cos \alpha d\alpha = \cos \beta d\beta$$

Solving last three equations yields

$$h' = h \frac{(1 - n^2 \sin^2 \alpha)^{3/2}}{n(1 - \sin^2 \alpha)^{3/2}}$$

Additionally from basic geometry we can see that

$$L = h \tan \alpha + H \tan \beta = \frac{h \sin \alpha}{\sqrt{1 - \sin^2 \alpha}} + \frac{H n \sin \alpha}{\sqrt{1 - n^2 \sin^2 \alpha}}$$

This equation is hard to solve. In order to solve this we will approximate things. From given values we see that $h \ll H$ So we can ignore first term. Then,

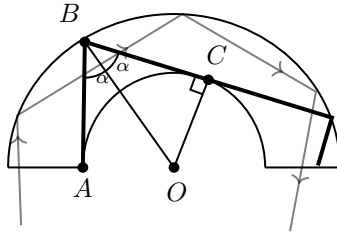
$$\sin^2 \alpha = \frac{L^2}{(H^2 + L^2)n^2}$$

Substituting the angle α into equation of h' and get

$$h' = \frac{n^2 H^3}{((H^2 + L^2)n^2 - L^2)^{3/2}}$$

Solution 12. 21

Rays far from axis O will have large incident angle. So they will emit from other side with no worries. So if we fix the problem with closest ray to axis we can say all the other rays will emerge from other side. The critical condition for this ray occurs when the ray goes tangent to internal boundary point C :



The critical angle α should be

$$n \sin \alpha = 1$$

from Snell's law. Look at the right triangle OCB . We can find the angle from there too

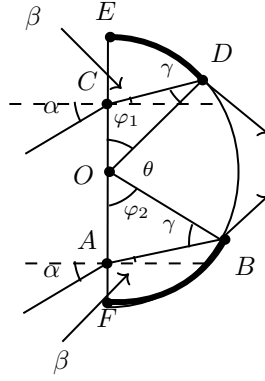
$$\sin \alpha = \frac{r + d}{r}$$

Solving two equations yields

$$d = (n - 1)r = 5.0 \text{ cm}$$

Solution 13. 22

Due to internal refraction of the light the light will not emerge from area ED and FB



Where points B and D are characterized by critical angle γ

$$\gamma = \arcsin\left(\frac{1}{n}\right) \quad (1)$$

From Snell's law

$$\sin \alpha = n \sin \beta$$

For triangles ODC and OBA the sum of internal angles is 180° so:

$$\varphi_1 = 180^\circ - \gamma - (90^\circ + \beta)$$

$$\varphi_2 = 180^\circ - \gamma - (90^\circ - \beta)$$

The angle θ will be

$$\theta = 180^\circ - \varphi_1 - \varphi_2 = 2\gamma$$

The result is surprisingly does not depend on α

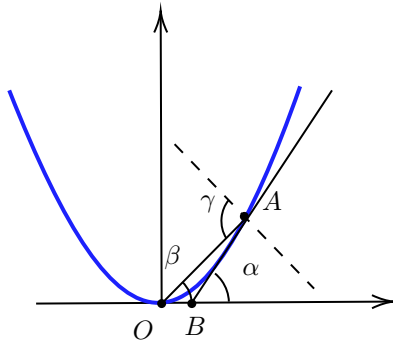
$$N = \frac{R\theta}{\pi R} = \frac{2\gamma}{\pi}$$

Substituting the equation 1 yields:

$$N = \frac{2}{\pi} \arcsin\left(\frac{1}{n}\right) \approx 0.46$$

Solution 14. 23

Let's examine some beam of light going from impurity to the parabola



Observer will not see the impurity if all the rays from it internally reflected. Therefore, the angle γ should be

$$n \sin \gamma \geq 1$$

Let the coordinates of point A be (x, y) . Therefore it can be described as

$$\tan \alpha = \frac{dy}{dx} = 2ax$$

$$\tan \beta = \frac{y}{x} = ax$$

The triangle OAB :

$$\beta + (90^\circ - \gamma) + (180^\circ - \alpha)$$

Substituting into first equation we get $n \cos(\alpha - \beta) \geq 1$. But we know only $\tan \alpha$ and $\tan \beta$. In order to get form of them we need to play with some trigonometric identities. We will use

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta = \cos \alpha \cos \beta (1 + \tan \alpha \tan \beta)$$

$$\sec^2 \varphi = 1 + \tan^2 \varphi$$

So combining all the equations yields

$$n \frac{1 + 2a^2x^2}{\sqrt{1 + 4a^2x^2}\sqrt{1 + a^2x^2}} \geq 1$$

This is product of two things. We want such n so that $nf(x) \geq 1 \forall x$. So when $f(x)$ is minimum we can conclude that the relation holds for other values of x . So we need to take derivative. Substituting $a^2x^2 = z$ and taking derivative we get

$$\frac{df}{dz} = \frac{2z - 1}{\sqrt{1 + 4z}\sqrt{1 + z}} = 0$$

For $z = 1/2$ the equation is true. So plugging in

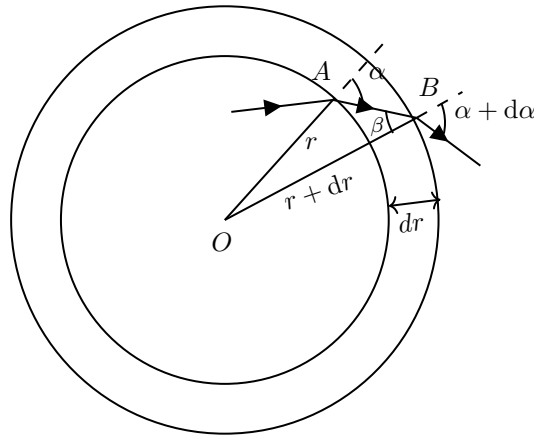
$$f(z)_{\min} = \frac{2\sqrt{2}}{3}$$

Therefore the minimum value of refractive index is

$$n \geq \frac{3}{4}\sqrt{2} \approx 1.06$$

Solution 15. 33

Let's examine a layer with thickness dr



Snell's law of refraction can be written as

$$n(r) \sin \beta = n(r + dr) \sin(\alpha + d\alpha)$$

The law of sines for triangle OAB

$$\frac{r}{\sin \beta} = \frac{r + dr}{\sin(180^\circ - \alpha)}$$

Substituting $\sin \beta$ to first equation we get

$$n(r)r \sin \alpha = n(r + dr)(r + dr) \sin(\alpha + d\alpha)$$

So for any refraction above the surface it's correct that

$$n(r)r \sin \alpha = \text{const.}$$

Maximum height corresponds to $\alpha = 90^\circ$ Then

$$n_0(1 - k \cdot 0)R \cdot 1 = n_0(1 - kH)(R + H) \cdot 1$$

So solving equation for H yields

$$H = \frac{1 - kR}{k} \approx 20.4 \text{ km}$$

Solution 16

(a) We write Snell's Law at the back-surface of the lens (right side in the figure).

$$n \sin \theta = \sin \theta'$$

Using small angle approximation, we get $n\theta = \theta'$. This gives the angle of deflection as $\epsilon = \theta' - \theta = (n - 1)\theta$.

We can also observe from the diagram that $\tan(90^\circ - \theta) = \frac{dx}{dr} \Rightarrow \frac{dx}{dr} = \tan \theta \approx \theta$. Here we have again used the small angle assumption. Hence we get an expression for $\frac{dx}{dr}$ -

$$\boxed{\frac{dx}{dr} = \frac{\epsilon}{n - 1}}.$$

(b) Using $\epsilon = \frac{R}{r}$, we get $\frac{dx}{dr} = \frac{\epsilon}{n - 1} = \frac{R}{(n - 1)r}$.

Integrating, we get

$$x = \int \frac{R}{(n - 1)r} dr = \frac{R}{n - 1} \ln r + C.$$

Using the limit of $r = r_1$ at $x = 0$, we get the $C = -\frac{R}{n - 1} \ln r_1$. Hence we get the equation for the back surface as

$$\boxed{x = \frac{R}{n - 1} \ln \frac{r}{r_1}}.$$

(c) Considering the triangle made by the refracted light ray with the vertical at its point of emergence and the horizontal axis gives $\tan \epsilon = \frac{2}{30} \Rightarrow \epsilon = \tan^{-1} \frac{2}{30} = \frac{R}{2} \Rightarrow R = 2 \tan^{-1} \frac{2}{30} = 0.133$.

This gives $\boxed{R = 0.133 \text{ cm}}$.

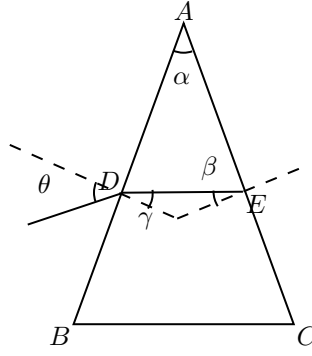
(d) As can be easily seen from the problem statement, we have $R = \frac{4GM}{c^2} \Rightarrow M = \frac{Rc^2}{4G} = 4.49 \times 10^{23} \text{ kg}$.

Note that this mass is quite huge, hinting at the fact that optical deviation due to gravitational systems is very small.

Solution 17. 1

We need to consider several cases

1. First we need to look at light is incident below the normal as shown in the figure below



From Snell's law:

$$\sin \theta = n \sin \gamma$$

Where θ is incident angle. The ray has to reflect from side AC . So total internal refraction angle:

$$n \sin \beta \geq 1 \quad (2)$$

From triangle ADE :

$$\alpha + (90^\circ - \gamma) + (90^\circ - \beta) = 180^\circ$$

From last equation we get that $\beta = \alpha - \gamma$. Plugging it to equation 2 and combining with the first equation and using the trigonometric formula for sine and rearranging the terms in order to solve for θ :

$$\sin^2 \theta + 2 \cos \alpha \sin \theta + (1 - n^2 \sin^2 \alpha) \geq 0$$

Solving the equation for the critical situation(equal sign):

$$\sin \theta = -\cos \alpha \pm \sqrt{n^2 - 1} \sin \alpha$$

Using trigonometric relations

$$\sin \theta = n \sin \left(\alpha - \arcsin \left(\frac{1}{n} \right) \right)$$

The angle θ is in range $(0, \pi/2)$ so:

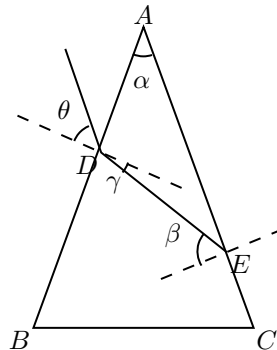
$$0 \leq \sin \theta \leq \min \left\{ 1, n \sin \left(\alpha - \arcsin \left(\frac{1}{n} \right) \right) \right\}$$

For happening of total internal refraction for any angles we should have

$$n \sin \left(\alpha - \arcsin \left(\frac{1}{n} \right) \right) \geq 1$$

$$\alpha \geq 2 \arcsin \left(\frac{1}{n} \right)$$

2. Second we need to look at light is incident above the normal as shown in figure below



Like previous part from triangle ADE we get:

$$\beta = \alpha + \gamma$$

For the internal refraction:

$$n \sin(\alpha + \gamma) \geq 1$$

Using trigonometric identities and solving quadratic equation and again using trigonometric identities yields:

$$\sin \theta = n \sin \left(\arcsin \left(\frac{1}{n} \right) \pm \alpha \right)$$

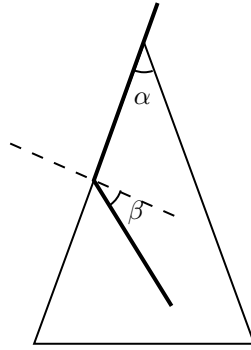
Obviously we want our expression inside of sine to be negative so:

$$\max \left\{ 0, n \sin \left(\arcsin \left(\frac{1}{n} \right) + \alpha \right) \right\} \leq \sin \theta \leq 1$$

And angle α should satisfy

$$\alpha \geq \arcsin \left(\frac{1}{n} \right)$$

Additionally we must ensure that the ray always reaches the side AC



The ray will not reach to the side when refracted ray and side are parallel. So

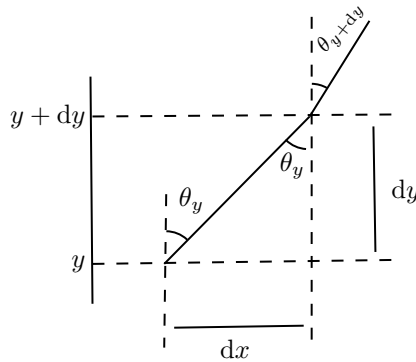
$$\alpha + \beta > 90^\circ$$

So the angle α should satisfy:

$$\arcsin \left(\frac{1}{n} \right) < \alpha < 90^\circ - \arcsin \left(\frac{1}{n} \right)$$

Solution 18. 3

(a) Let's examine a thin layer in which the refractive index is not changing



If we write the Snell's law for this layer

$$n_y \sin(\theta_y) = n_{y+dy} \sin(\theta_{y+dy})$$

We can easily see that

$$n(y) \sin \theta = \text{const.}$$

. From diagram it is obvious that

$$\frac{dx}{dy} = \tan \theta = \frac{\sin \theta}{\sqrt{1 - \sin^2 \theta}}$$

Substituting the $\sin \theta$ from Snell's law

$$\frac{dx}{dy} = \sqrt{\frac{n^2}{n_0^2 \sin^2 \theta_0} - 1}$$

We are given that $n^2 = n_0^2 + \alpha^2 y$. So plugging in and seperating variables

$$\frac{dy}{\sqrt{\frac{n_0^2}{\alpha^2} \cos^2 \theta_0 + y}} = \frac{\alpha}{n_0 \sin \theta_0} dx$$

Integrating with simple substitution $u = \frac{n_0^2}{\alpha^2} \cos^2 \theta_0 + y$ we get

$$2\sqrt{\frac{n_0^2}{\alpha^2} \cos^2 \theta_0 + y} = \frac{\alpha}{n_0 \sin \theta_0} x + \mathcal{C}$$

We can find the integration constant $\mathcal{C}$ from initial value $y(x=0) = 0$. Plugging $y = 0$ and $x = 0$:

$$2\sqrt{\frac{n_0^2}{\alpha^2} \cos^2 \theta_0 + 0} = 0 + \mathcal{C}$$

After solving for y rearranging variables we get that

$$y = \frac{\alpha^2}{4n_0^2 \sin^2 \theta_0} x^2 + \frac{x}{\tan \theta_0}$$

And this is the equation for parabola.

- (b) For maximum x_a it's obvious that if the initial angle θ_0 is $\pi/2$ then the value of x_a can be maximum. Plugging in $\theta_0 = \pi/2$ and solving for x_a yields

$$x_a = \frac{2n_0}{\alpha} \sqrt{y_a}$$

- (c) Let the speed of target v . So its coordinates will change over the time: $(x_0 - vt, y_0)$. So let the coordinates of target at arbitrary moment (x_a, y_a) . Then we have derived before

$$y_a = \frac{\alpha^2}{4n_0^2 \sin^2 \theta_0} x_a^2 + \frac{x_a}{\tan \theta_0}$$

So we must solve for θ_0 . In order to do this we need to use trigonometric identity $\sin \theta_0 = \frac{\tan^2 \theta_0}{1 + \tan^2 \theta_0}$. Plugging it and rearranging the terms for solving $\tan \theta_0$:

$$(4n_0^2 y_a - \alpha^2 x_a^2) \tan^2 \theta_0 - 4n_0^2 x_a^2 \tan \theta_0 - \alpha^2 x_a^2 = 0$$

Solving this quadratic equation with assuming $0 \leq \theta_0 \leq \pi/2$ so $\tan \theta_0 \geq 0$, as a result we care about only positive root:

$$\tan \theta_0 = \frac{2n_0^2 x_a}{4n_0^2 y_a - \alpha^2 x_a^2} + \sqrt{\left(\frac{2n_0^2 x_a}{4n_0^2 y_a - \alpha^2 x_a^2}\right)^2 + \frac{\alpha^2 x_a^2}{4n_0^2 y_a - \alpha^2 x_a^2}}$$

Plugging in values for y_a and x_a :

$$\theta = \arctan \left(\frac{2n_0^2 (x_0 - vt)}{4n_0^2 y_0 - \alpha^2 (x_0 - vt)^2} + \sqrt{\left(\frac{2n_0^2 (x_0 - vt)}{4n_0^2 y_0 - \alpha^2 (x_0 - vt)^2}\right)^2 + \frac{\alpha^2 (x_0 - vt)^2}{4n_0^2 y_0 - \alpha^2 (x_0 - vt)^2}} \right)$$

- (d) It's obvious that minimum value for the speed is 0 because the target is not moving so the bomb can't reach the transmitter. Coming to the upper end of the set the target has to release bomb when it was destroyed by laser so that bomb can reach the transmitter. The time for bomb spends in air is $t_2 = \sqrt{\frac{2y_a}{g}}$. The distance that bomb will go from initial moment will be

$$s = v(t_a + t_2) = v\left(t_a + \sqrt{\frac{2y_a}{g}}\right)$$

We need this distance to be less than maximum initial distance of target $s < x_{\text{amax}}$. We have got the value for maximum distance from part b. Plugging in:

$$v < \frac{\frac{2n_0}{\alpha} \sqrt{y_a}}{t_a + \sqrt{\frac{2y_a}{a}}}$$

So in order to destroy the ship its speed has to have the values:

$$0 \leq v \leq \frac{2n_0(t_a\sqrt{gy_a} - \sqrt{2}y_a)}{\alpha(qt_a^2 - 2y_a)}$$

Solution 19. 4

- (1) For the case $n_1 > 0$ and $n_2 > 0$, we draw Figure 18.

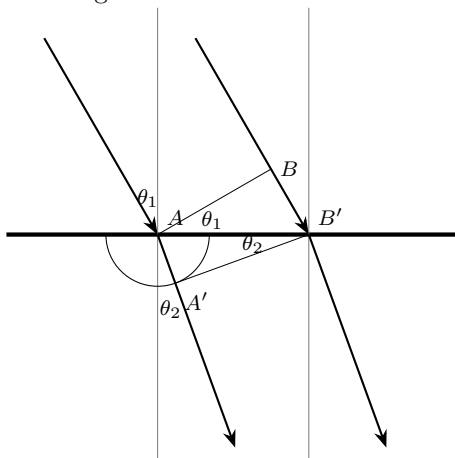


Figure 18: Huygens's principle applied to a typical material.

Let $\overline{AB}$ be a wavefront, along which the wave has the same phase. When the wave at B reaches B' , the front of the wavelet propagating from A is a hemispherical surface of radius r situated within medium 2, whose intersection with the incident plane is a semicircle also of radius r . r is, in turn, determined by

$$n_2 r = n_1 BB'. \quad (3)$$

Let line $A'B'$ be the tangent of the line from B' to the wave at A' ; then $\overline{A'B'}$ is also a wavefront. By the results of geometry, we have

$$\begin{aligned} BB' &= AB' \sin \theta_1, \\ r &= AB' \sin \theta_2, \end{aligned}$$

from which it is straightforward to derive Snell's law:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2. \quad (4)$$

We now treat the case $n_1 > 0$ and $n_2 < 0$. Recall that the optical path length of light travelling in medium 2 is negative.

Alternative 1. As shown in Figure 20, the wavelet propagating from B' is given by the semicircle of radius r situated within medium 2, and the corresponding optical path length is given by $n_2 r < 0$. Therefore, the wavefront including A is given by the tangent from A to this semicircle, whose point of tangency we name A' . The radius of the semicircle is determined by

$$n_1 BB' + n_2 A'B' = 0. \quad (5)$$

By geometry

$$BB' = AB' \sin \theta_1, \quad B'A' = AB' \sin |\theta_2|,$$

from which we may derive the generalised Snell's law:

$$n_1 AB' \sin \theta_1 = -n_2 AB' \sin |\theta_2| = n_2 AB' \sin \theta_2 \quad (6)$$

$$n_1 \sin \theta_1 = n_2 \sin \theta_2. \quad (7)$$

Alternative 2. As shown in Figure 21, $\overline{AB}$ is a wavefront of the incident light. When the wave at B reaches B' , the wavelet propagating from A is a circle of centre A and radius r , corresponding to the optical path length given by $n_2 r < 0$.

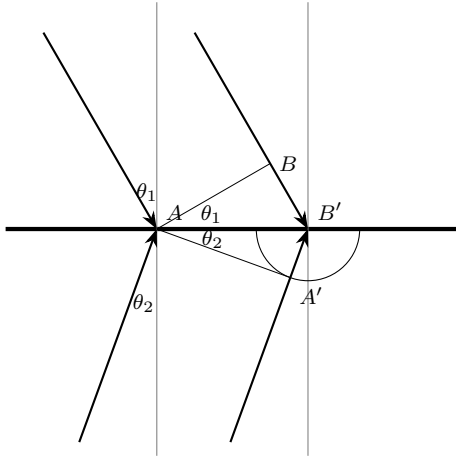


Figure 20: Huygens's principle applied to a NIM.

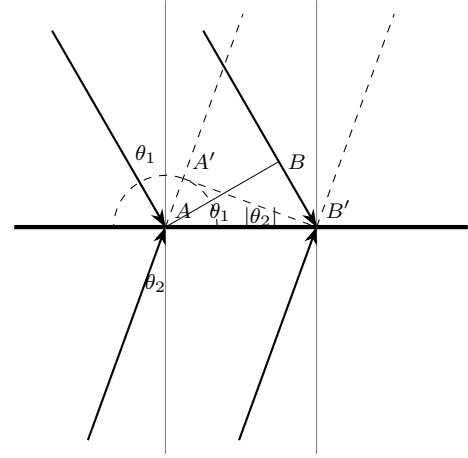


Figure 21: Huygens's principle applied to a NIM—Alternative.

The radius is determined by

$$|n_2| r = n_1 BB'. \quad (3)$$

Since the optical path length is negative, we may as well treat the wavefronts in medium 2 as moving in the opposite direction as compared to the typical case, i.e. away from medium 2 and towards medium 1. Thus the wavelet of interest is the dashed semicircle in Figure 21. If we construct line $B'A'$ tangent to the semicircle at A' , then $\overline{B'A'}$ is a wavefront. By geometry

$$BB' = AB' \sin \theta_1, \quad r = AB' \sin |\theta_2|, \quad (4)$$

so we obtain

$$n_1 \sin \theta_1 = n_2 \sin \theta_2. \quad (5)$$

- (2) We take $s_1 > 0$ when the object is outside the NIM and $s_2 > 0$ when the image is inside the NIM; the reverse is true for $s_1 < 0$ and $s_2 < 0$. We take $\theta_1 > 0$ (incident angle) and $\theta_2 < 0$ (refracted angle), and take all other named angles in Figure 3 as positive. We notice that

$$\theta_1 = \alpha_1 + \beta, \quad -\theta_2 = \alpha_2 - \beta. \quad (6)$$

Under the paraxial approximation, all these angles are small, so we have

$$\sin \theta_1 = \theta_1, \quad \sin \theta_2 = \theta_2$$

and

$$\begin{aligned} \tan \alpha_1 &= \sin \alpha_1 = \alpha_1 = \frac{h}{s_1}, \\ \tan \alpha_2 &= \sin \alpha_2 = \alpha_2 = \frac{h}{s_2}, \\ \tan \beta &= \sin \beta = \beta = \frac{h}{R}, \end{aligned} \tag{7}$$

where h is the distance of the point of incidence M from the optical axis. Snell's law yields

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_1 \left(\frac{h}{s_1} + \frac{h}{R} \right) = -n_2 \left(\frac{h}{s_2} - \frac{h}{R} \right) \tag{8}$$

$$\frac{n_1}{s_1} + \frac{n_2}{s_2} = \frac{n_2 - n_1}{R}, \tag{9}$$

where (9) is the required effective lens equation. Note that since $n_2 < 0$ and $n_1 > 0$, the RHS of (9) is negative.

To find the magnification, we draw Figure 22 and consider object $\overline{Q_1 P_1}$ and its image, $\overline{Q_2 P_2}$.

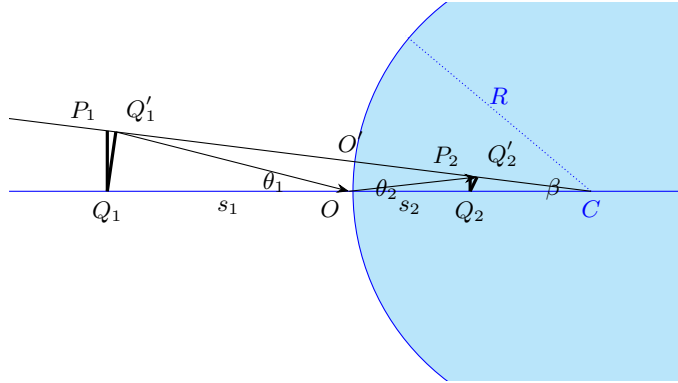


Figure 22: Construction for finding the magnification of the spherical NIM. Location of Q'_1 and Q'_2 not to scale for clarity.

Now suppose we rotate the parts in blue clockwise about C , bringing points Q_1 , O , and Q_2 along with them, such that P lies on the rotated optical axis. Let Q'_1 , O' , and Q'_2 be the transformed positions of the previously-named points respectively. Under the paraxial approximation, since the angle through which the blue parts are rotated by is small, we may write

$$\begin{aligned} Q_1 P_1 &= Q_1 Q'_1, \\ Q_2 P_2 &= Q_2 Q'_2. \end{aligned} \tag{10}$$

Thus by the above and geometry we have

$$Q_1 P_1 = s_1 \theta_1, \tag{11}$$

$$Q_2 P_2 = s_2 \theta_2, \tag{12}$$

which yields our desired magnification,

$$\frac{Q_2 P_2}{Q_1 P_1} = \frac{-s_2 \theta_2}{s_1 \theta_1} = -\frac{s_2 n_1}{s_1 n_2}. \tag{13}$$

- (3) As shown in Figure 23, if the spherical NIM is absent, then the light rays would converge at the focus of the convex lens after passing through it, as shown by the dashed lines.

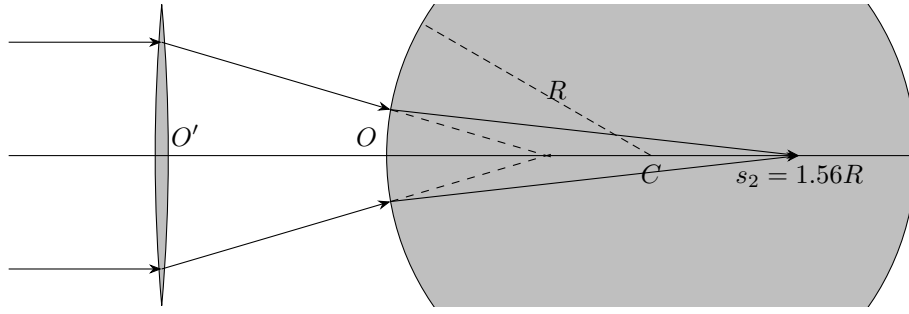


Figure 23: Ray diagram for lens-NIM setup.

When the NIM is introduced, the original point of convergence becomes the virtual object awaiting transformation by the NIM, so the corresponding object distance is given by

$$s_1 = -(f - d). \quad (14)$$

If $n_2 > 0$, the equivalent lens equation is given by (this is in the syllabus of CPhO finals):

$$\frac{n_1}{s_1} + \frac{n_2}{s_2} = \frac{n_2 - n_1}{R}, \quad (15)$$

which is of the same form as (9). Solving for s_2 in both cases yields

$$s_2 = \frac{n_2 s_1 R}{n_2 s_1 - n_1 (s_1 + R)} = \frac{n_2 R (f - d)}{(n_2 - n_1)(f - d) + n_1 R}. \quad (16)$$

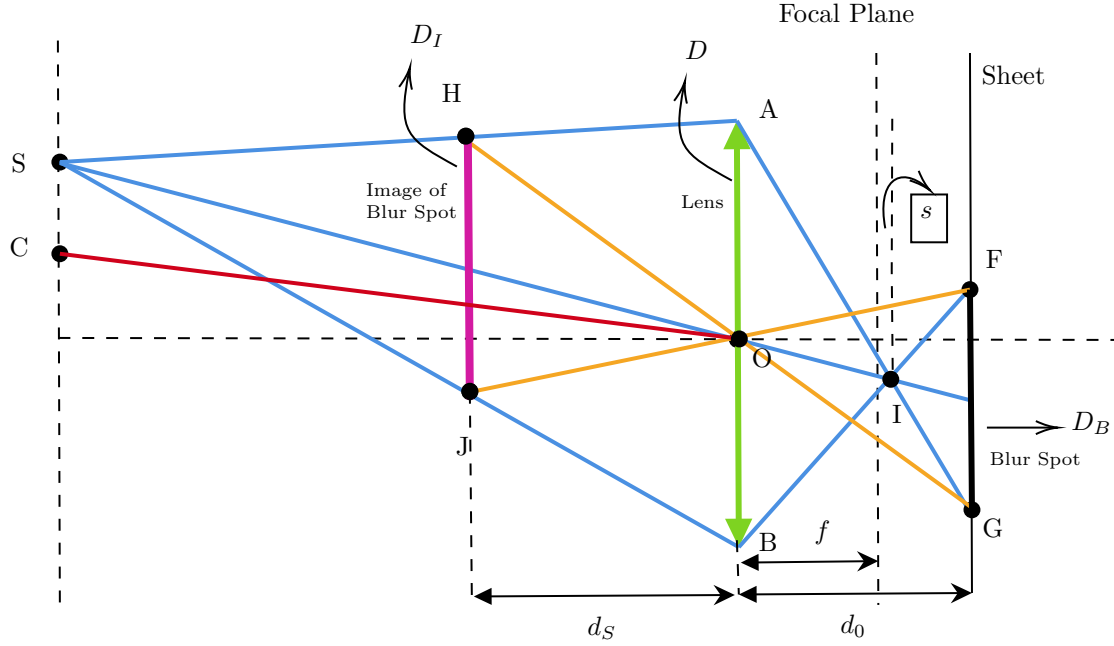
The required results are shown in Table 1. The required diagram is shown in Figure 23.

Case	n_2	d	s_2
1	1.5	$0.35R$	$1.10R$
2	1.5	$0.85R$	$0.74R$
3	-1.5	$0.35R$	$0.92R$
4	-1.5	$0.85R$	$1.56R$

Table 1

Solution 20

In any Geometrical Optics problem, drawing a clear diagram is the first task to complete. Such a diagram should ideally have all the elements of length that could hold significance in the problem. The next job to do should be the analysis of the graph given in the problem i.e. how the regions that appear on the graph come about. Here is a complete diagram -



A quick observation that can be made is the fact that $L \gg f$. Now we should consider the schematic diagram to see what similar things we can conclude. Taking the definition as in the diagram above, we can judge that d_0 , the distance between the lens and the plane where we are observing the **blur spot** (the projection of the image of the source on the screen) will be of the same order of magnitude as f while L is approximately 9 times f .

Now we look into the Physics of the problem. The graph can be considered to have four broad regions - the first plateau at around 3.4, the second one at 4.4, then the minima being very close to 0 and the constant value of 1.95. Just a casual look can explain three of these parts. The brightest part has to be the projection of the image of the point source formed by the lens on the screen (the blur spot). The constancy of intensity there indicates the even distribution of the light from the source onto the blur spot. The minima is the portion shaded by the black annular part of the disc. The further constancy in the value can be easily understood to be due to the light that didn't pass through the lens but fell directly on the screen. Note that this value's significant constancy can help us conclude the fact that the incidence is close to normal on the screen when light falls on it. This was also evident in part from the schematic diagram. We now use the near-normal incidence of light onto the lens to conclude that the luminous flux reaching the paper sheet from the lens per unit area of the lens and per unit solid angle subtended by the paper sheet is constant (This is a quite accurate assumption given that the plateaus are quite straight on the graph).

Now we are left with the region that is at $x < 100$. This region is more bright than the constant part at $x > 420$. There obviously has to be some extra light falling here. The problem statement mentions the non-ideality of the lens that has been used to model the cat's eyes. Evidently, some light will be scattered by the elements of the lens forming the image of the blur spot to this region. Note that in case the lens had been ideal (in terms of the no scattering), this region would be lower in intensity than that at $x > 420$ due to the absorption of some light by the lens.

This scattering will also affect the blur spot. Even though the the blur spot is 10 times as large as the region of $x < 100$, this contribution should be subtracted to ensure accuracy is within limits desired in the problem because otherwise there could have been an error of around 10%. We can assume that such scattering will hardly affect the regions that lie beyond the projection of the lens on the screen because of their inclined direction when seen from the lens elements.

The light that was incident onto the glass lens got compressed into a smaller region on the plane sheet. Hence intensity ratios will be in the ratio of their areas. Define the intensity at the paper sheet to be I_p . The same will hold approximately at the lens surface (as described earlier). Then the intensity at the blur spot will be I_p/r , where r is the ratio of the areas. Now the lens will form an image of the blur spot. Suppose the distance of the image of this blur spot from the lens is d_s . Then using lens formula, we get that

$$d_s = \frac{d_0 f}{d_0 - f}.$$

The diameter of the image of the blur spot (whose diameter is D_B) will be

$$D_I = \frac{D_B d_s}{d_0} = \frac{D_B f}{d_0 - f} \quad [\text{Using similarity of triangles}].$$

We can calculate D_B as

$$D_B = \frac{D(d_0 - f - s)}{f + s} \approx \frac{D(d_0 - f - s)}{f}$$

where s is distance of the image of the point source from the focal plane. Note that $f + s \approx f$ because the object is quite far from the lens compared to f and thus the image will be close to focal plane. Also we didn't use above approximation in the numerator because $d_0 - f$ already made it small enough and we can no longer be sure of its largeness compared to s .

Using Newton's formula for the source of light's image in the lens, we get

$$s = \frac{f^2}{L - f - d_0} \approx \frac{f^2}{L}.$$

Using the equations above, we get that

$$D_B = D \left(\frac{d_0 - f}{f} - \frac{f}{L} \right) \Rightarrow D_I = D \left[1 - \frac{f^2}{L(d_0 - f)} \right].$$

Using $d_0 - f = d_0 f / d_s \approx f^2 / d_s$, we get

$$D_I = \frac{D(L - d_s)}{L}.$$

The above few lines have involved a lot of calculations. Lets see the motivation. First we used the lens equation to get relations between d_s, d_0, s, f, L . Next we used similarity to get relations between the diameters of each of the extended images. We then used the fact that $d_0 \approx f$ (very crude approximation but helps towards having only one unknown). After these eliminations, we get the relation between D_I and D only in terms of L and d_s , which is really neat.

Rearranging the terms of the final expression, we see that when observed from the camera, the angular size of the image of the blur spot $\theta_I = D_I / (L - d_s)$ equals that of the lens aperture $\theta_L = D / L$.

Notice that apart from the approximations related to distance, we have not yet used any data from the problem statement. We haven't yet obtained any real numbers for any of the variables we have used/assumed so far. Let's pause to see how we can use that for the analysis.

First we should try to somehow use ratios of intensities of light at the camera to get some meaningful details about the situation.

We can see that the image of the blur spot as seen in the picture given in the problem is a part of a circle, one which has been cut off by the blackened part of the lens. The light which fell on the lens got concentrated into the image of the source in the lens and then expanded to fill the area on the screen that comprises the blur spot. So we can get the ratio of the areas of the blur spot and the lens using the given intensity diagram. If we write I_{out} to be the intensity in the region $x > 420$ and I_{max} for that of the blur spot, we get

$$I_{out} \cdot \frac{A_{lens}}{A_{blur\ spot}} = I_{max} \Rightarrow I_{out} \cdot \left(\frac{D}{D_B} \right)^2 = I_{max}.$$

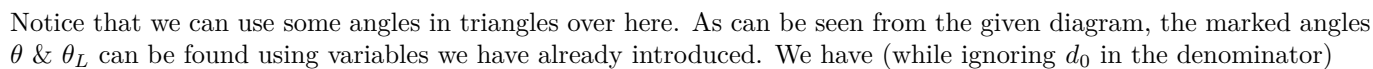
Note that here we used the intensity at the screen to be that at the lens because of d_0 being significantly small compared to L .

Regarding the ratio of intensities, we can use the graph. Note that the light initially incident on the lens was either concentrated or scattered resulting in a loss of the sharpness and a decrease in the intensity of the blur spot. We should be using a slightly higher value for the intensity of the blur spot but (luckily) due to the accuracy bars specified

So we use this data to get that

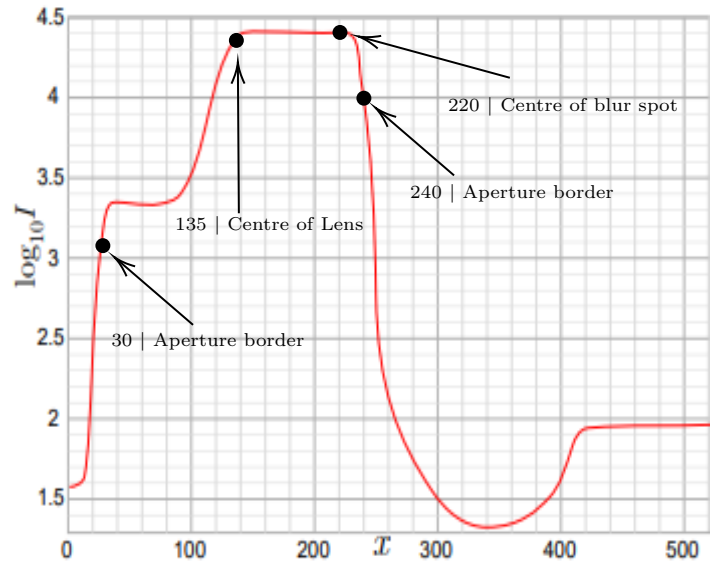
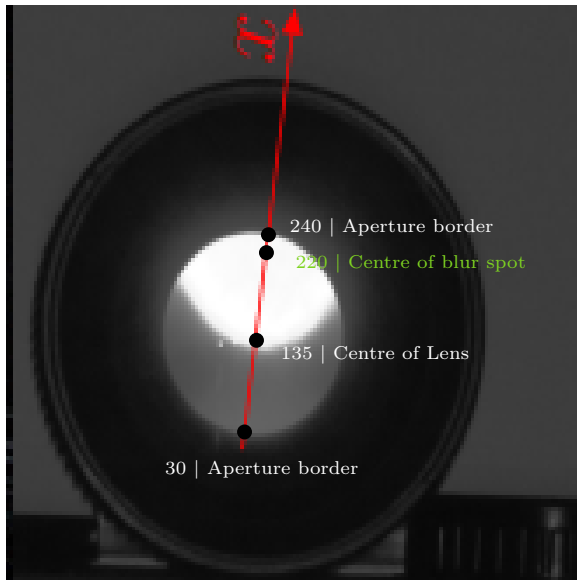
Now we use the ratio we had found earlier in the solution i.e.

This gives $\frac{d_0}{f} \approx 1.07$. Now have a look at the following diagram -



so the ratio is

69



Also notice that the diagram with x marked on it (an edited version of which is given above) can be used to find the ratio of exactly the same angles. The aperture has its borders at $x = 30, x = 240$ pixels and the blur spot has its centre at around 220 pixels (slightly off the border of the aperture, as shown). The centre of the lens is at approximately 135 pixels $\left(\frac{30 + 240}{2}\right)$. Writing $\frac{\theta}{\theta_L}$ as $\frac{d_z}{D'}$ where d_z is the projection of h' in the given image and D' is that of the lens diameter, we get

$$h = \frac{d_z}{D'} \cdot \frac{D(L - d_S)}{d_S}.$$

Putting in the values gives us $h \approx 76$ cm.

Solution 21

Let the impact point of rod will be A and the point which second rod receive the signal B . The signal can reach point B with the least time using one of the two different paths depending on the values of parameters which are given in the problem statement.

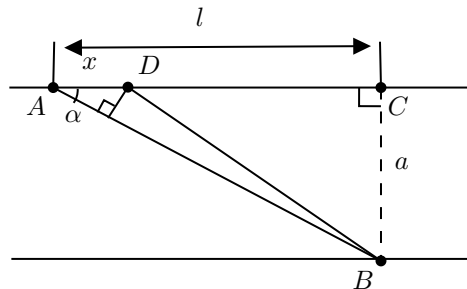


Figure 24

If $c \geq v$, obviously, the sound reach point B faster, along with the straight line AB . The time for that will be

$$T_1 = \frac{\sqrt{a^2 + l^2}}{c}$$

Now let $c < v$. Let's look at the time for reaching point B of two signals with two different paths. One is along the straight line AB , and the other goes a small distance x along the rod, till the point D , then from point D it goes to point B . (see figure 24). Let the angle $\angle CAB = \alpha$. Since the x is small enough we can assume $AB - DB$ is $x \cos \alpha$.

The time difference between two signals will be

$$\Delta t = \frac{AB}{c} - \left(\frac{x}{v} + \frac{DB}{c} \right) = \frac{x \cos \alpha}{c} - \frac{x}{v}.$$

For the condition

$$\cos \alpha = \frac{l}{\sqrt{l^2 + a^2}} < \frac{c}{v}$$

the Δt is negative, so sound will reach much quickly with the line AB , rather than spending some time on the rod. When $\cos \alpha > \frac{c}{v}$, the signal will go faster along the path which first goes some distance in rod then it goes water.

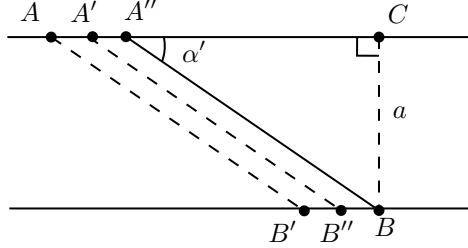


Figure 25

In the last case let the change from rod to water of signal will be in the point A' (see figure 25), and the angle will be $\cos \alpha' = \frac{c}{v}$, because after that if it spends time on the rod that will not be beneficial because of the discussion in the previous part. The propagation time will be

$$T_2 = \frac{l - \frac{a}{\tan \alpha'}}{v} + \frac{a}{c \sin \alpha'} = \frac{l}{v} + a \left(\frac{1}{c \sin \alpha'} - \frac{1}{v \tan \alpha'} \right).$$

Since $\cos \alpha' = \frac{c}{v}$, $\sin \alpha' = \sqrt{1 - \frac{c^2}{v^2}}$, and $\tan \alpha' = \sqrt{\frac{v^2}{c^2} - 1}$. So

$$T_2 = \frac{l}{v} + a \sqrt{\frac{1}{c^2} - \frac{1}{v^2}}.$$

Combining the considered cases, we get the answer for the minimum sound propagation time between points A and B:

$$T = \begin{cases} \frac{\sqrt{a^2 + l^2}}{c} & \text{for } \frac{c}{v} \geq \frac{l}{\sqrt{l^2 + a^2}} \\ \frac{l}{v} + a \sqrt{\frac{1}{c^2} - \frac{1}{v^2}} & \text{for } \frac{c}{v} < \frac{l}{\sqrt{l^2 + a^2}} \end{cases}$$

Note that if the conditions $c < v$ and $\cos \alpha > \frac{c}{v}$ are met simultaneously, the signal can propagate both along the path $AA'B$ and along the paths $AB'B$ and $AA''B''B$. In this case, signals propagating along all three paths will reach point B simultaneously, but their intensities will be different. Signals passing along the $AA'B$ and $AB'B$ paths are refracted at the rod-water interface once and will have approximately the same amplitude, while the signals transmitted along all $AA''B''B$ paths are refracted twice and will be much weaker. Note also that after the first signal arriving at point B, a rather prolonged hum will be heard, since the sound is emitted by all points of the rod along which the shock wave runs, but these signals come later. The situation in this case resembles what is heard during the flight of a supersonic aircraft or a lightning strike: first, the observer hears the strike, and then the rumblings. In conclusion, we note that the problem can be solved in another way, by constructing the so-called "Mach cone".

Source: 2001 Russian Physics Olympiad for grade 9 problem 2

Solution 22

The sound can go with various paths:

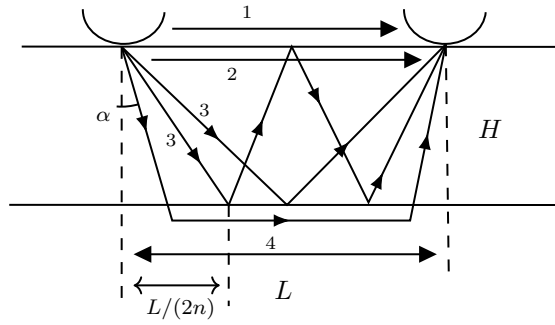


Figure 26

- 1) Directly by air. In this case the time for sound propagation is $t_1 = L/v_3 \approx 9.01$ s
- 2) Directly by water. In this case the time for sound propagation is $t_2 = L/v_2 \approx 2$ s.
- 3) By water, with a number of successive reflections from the boundaries of the sections "water-bottom" and "water-air". Consider n reflections with the bottom and $n - 1$ reflections with the water-air boundary. Then length of such path (we can find using right triangles) is

$$S_3 = 2n \sqrt{\left(\frac{L}{2n}\right)^2 + H^2} = \sqrt{L^2 + 4n^2 H^2},$$

and the time for that

$$T_n = \frac{S_3}{v_1} = \frac{\sqrt{L^2 + 4n^2 H^2}}{v_1}.$$

For $n = 1, 2, \dots$ propagation time is $T_1 = 2.40$ s, $T_2 = 3.33$ s, $\dots$

- 4) On the water and along the bottom – sound passes through the water to the bottom, penetrates into the bottom rocks, moves in them along the "water-bottom" interface, re-enters the water and either reaches the second ship or is reflected from the water surface, after which the cycle "water - bottom - water" is repeated until the sound reaches the second ship. Obviously, the minimum propagation time of sound along a path of this type will be obtained with one "water - bottom - water" cycle. In order for the sound that has reached the bottom to begin to propagate along the interface between the media, it is necessary that the angle of its incidence on the interface is equal to the angle of total internal reflection, which is $\alpha = \arcsin \frac{v_1}{v_2} \approx 0.3398 \text{ rad} \approx 19.5^\circ$. The sound propagation time in one cycle "water - bottom - water" is:

$$t_4 = \frac{L - 2H \tan \alpha}{v_2} + \frac{2H}{v_1 \cos \alpha} \approx 1.92 \text{ s}$$

in two cycle

$$t_5 = \frac{L - 4H \tan \alpha}{v_2} + \frac{4H}{v_1 \cos \alpha} \approx 3.19 \text{ s}$$

and so on.

As a result, comparing the found times, we find that the first sound signal will reach the second ship in 1.92 s along the path "water - bottom - water", the second - after 2 s directly through the water, the third - after 2.4 s through the water with one reflection from the bottom.

Solution 23

Music may sound out of tune if, due to the approach or distance from the viewer of the sounding accordion, the frequency of the sound changes when the same key is pressed more than $a = \sqrt[24]{2}$ times. The change in frequency is associated with the Doppler effect. Before starting to solve the problem, let us consider in more detail the origin of this effect.

Let a sound source emitting waves with a frequency $\nu_0 = 1/T_0$, where T_0 is the oscillation period of the source, moves towards the receiver with a constant velocity v . Imagine that a wave consisting of N whole waves (needs to be fixed

here) is emitted in time $t = NT_0$. If the speed of sound is c , then after time t , when the first wave has passed the distance ct , the last wave has just had time to leave the source, and the source itself has passed the distance vt . This means that the distance between the beginning and the end of the wave after time t will be $(c - v)t$, and at this distance N wavelengths will fit. Hence, the wavelength emitted by a moving source is equal to

$$\lambda = \frac{(c - v)t}{N} = (c - v)T_0$$

and corresponding frequency recorded by receiver is

$$\nu = \frac{c}{\lambda} = \frac{c}{(c - v)T_0} = \nu_0 \frac{c}{c - v}$$

This is the formula for Doppler effect. If the source moves away from the receiver, then in front of the source velocity v in this formula, obviously, there should be a plus sign. Note again that the Doppler effect is associated with a change in the wavelength emitted by a moving source, and not with a change in the speed of sound, which depends only on the properties of the medium.

So, we found out that the approach of the accordion to the viewer when the carousel rotates leads to an increase in the frequency of the sound, and the removal of the accordion - to a decrease. In our case, the change in frequency is maximal when the listener is in the same plane with the carousel, and the accordion player's speed, equal in magnitude to ωR , is directed along the listener-player line (see Fig. 27). The ratio of the increased sound frequency to the nominal ν_0 or the nominal frequency to the reduced frequency should not exceed $a = \sqrt[24]{2}$. Since, obviously, $\omega R \ll c$, where ω is the angular speed of rotation of the carousel, using the approximate formula $(1 + x)^n \approx 1 + nx$, in the first case we get

$$\frac{\nu}{\nu_0} = \frac{c}{c - \omega R} \approx 1 + \frac{\omega R}{c} < a$$

and for the second case

$$\frac{\nu}{\nu_0} = \frac{c + \omega R}{c} \approx 1 + \frac{\omega R}{c} < a$$

which is the same relation. We can find ω from formula above

$$\omega < \frac{c}{R}(a - 1) \approx 2 \text{ rad/s}$$

Note that both the first and second inequalities can be solved directly, without approximating. This gives two frequency values:

$$\begin{aligned} \omega < \omega_1 &= \frac{c}{R} \frac{(a - 1)}{a} \approx 1,9698 \text{ rad/s} - \text{in the first case,} \\ \omega < \omega_2 &= \frac{c}{R}(a - 1) \approx 2,0276 \text{ rad/s} - \text{in the second case.} \end{aligned}$$

However, the frequency ω_1 differs from the frequency ω_2 so little that the frequency shift caused by such a change in angular velocity is much less than a quarter tone and cannot be detected by ear. Therefore, it is physically more correct to combine the cases and give one approximate answer.

Note that the real angular velocity of the carousel rotation, of course, much less than 2 rad/s, since at such an angular velocity the centripetal acceleration of the accordion player would be equal to $\omega^2 R = 20 \text{ m/s}^2 \approx 2g$!

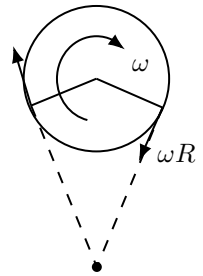


Figure 27

Solution 24

The change in the frequency of the electromagnetic signal reflected from the tornado is due to the fact that the objects reflecting this signal are moving, and the Doppler effect takes place. Unlike problem 46, where this effect was considered for sound waves in air, here we are talking about radio waves, the speed of which c is equal to the speed of light. Let the object reflecting the signal move in the direction of the radar with the maximum speed v . It is clear that this speed is very small compared to c , and the quadratic effects associated with the theory of relativity can be neglected. Then only the Doppler effect, linear with respect to v/c , remains, and by analogy with Problem 46, the

frequency of the signal absorbed by an object in a tornado should be expressed by the formula

$$f' = f_0 \left(1 + \frac{v}{c}\right)$$

and change in the frequency $\Delta f_1 = f' - f_0$ can be derived from the expression above

$$\frac{\Delta f_1}{f_0} = \frac{v}{c}$$

Since the signal from the radar is first absorbed by a moving object, and then re-emitted by it, the relative change in frequency recorded by the radar will be twice as large:

$$\frac{\Delta f_2}{f_0} = 2 \frac{v}{c}$$

In other words

$$v = \frac{c}{2} \cdot \frac{\Delta f_2}{f_0}$$

From the graph given in the condition, it can be seen that the maximum speed of objects in the tornado corresponds to a change in the frequency of the reflected signal $\Delta f_2 \approx 6500\text{Hz}$. Substituting numerical data into the resulting formula, we have:

$$v \approx \frac{3 \cdot 10^8}{2} \frac{6500}{10^{10}} \approx 100\text{m/s}$$

Solution 25

Let's split the ocean into thin horizontal layers so that the speed of sound within one layer can be considered constant. The law of refraction of "sound beams" during the transition from one layer to another layer, as in optics, has the form: $\frac{\sin \varphi_1}{\sin \varphi_2} = \frac{c_1}{c_2}$, where c_1 and c_2 are the speeds of sound in adjacent layers, φ_1 and φ_2 are the angles at which

sound comes out of the first layer and enters the second. Further, for the next pair of adjacent layers: $\frac{\sin \varphi_2}{\sin \varphi_3} = \frac{c_2}{c_3}$,

etc. Multiplying such equalities written for n layers, we obtain: $\frac{\sin \varphi_1}{\sin \varphi_n} = \frac{c_1}{c_n}$.

Let the sound penetrate the ocean to the maximum depth H . Then $\frac{\sin \alpha}{\sin \varphi(H)} = \frac{c_0}{c(H)}$, where $\varphi(H)$ is the angle at which the sound enters the deepest layer. But $\varphi(H) = \pi/2$ (if this were not so, then the sound would go even deeper). Hence, $\sin \alpha = \frac{c_0}{c_0(1 + aH)}$ and

$$H = \frac{1 - \sin \alpha}{a \sin \alpha}$$

Solution 26

To solve the problem, let's make a drawing (see Fig. 28). Let us denote the initial and final bus stops on it by the letters A and B , by the letter C - the point from where the stop pavilion B seemed to the pedestrian $k = 1.5$ times lower than the pavilion A , by the letter D - the point from where the stop pavilion A seemed to the pedestrian at $k = 1.5$ times smaller than pavilion B . Since the apparent size of the pavilion is inversely proportional to the distance to it, the following proportions are valid:

$$\frac{AC}{CD + DB} = \frac{1}{k}, \quad \frac{DB}{AC + CD} = \frac{1}{k}$$

From equations above we get

$$\frac{AC}{AC + CD + DB} = \frac{1}{k + 1} = \frac{AC}{AB}, \quad \frac{DB}{AC + CD + DB} = \frac{1}{k + 1} = \frac{DB}{AB}$$

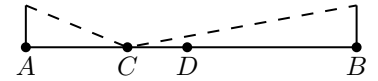


Figure 28

Also $CD = AB - AC - DB$, so

$$\frac{CD}{AB} = 1 - \frac{AC}{AB} - \frac{DB}{AB} = 1 - \frac{1}{k+1} - \frac{1}{k+1} = \frac{k-1}{k+1}$$

And we are given that $CD = L$, so

$$AB = \frac{k+1}{k-1}L = 500 \text{ m}$$

Solution 27

Let's begin with drawing a diagram of what is going on

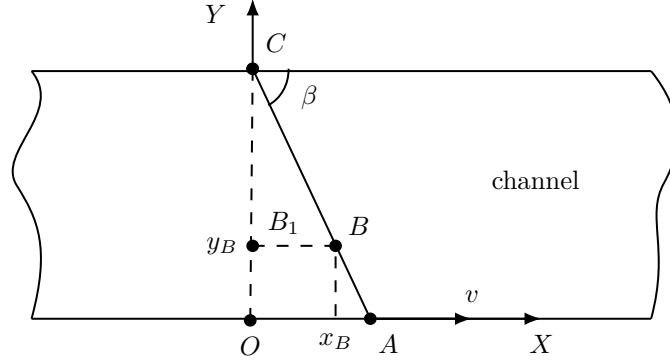


Figure 29

This is the top view of the channel (see Fig. 29) and we have marked the positions of the bus, flare and pillar on it with the letters A , B and C , respectively. Let at time $t = 0$ the bus was at the origin of the XOY coordinate system – point O , and the line OC was perpendicular to the channel banks. Then $OA = vt$. We also denote $OC = L$, $AC = l$, $AB = l_1$, $BC = l_2$.

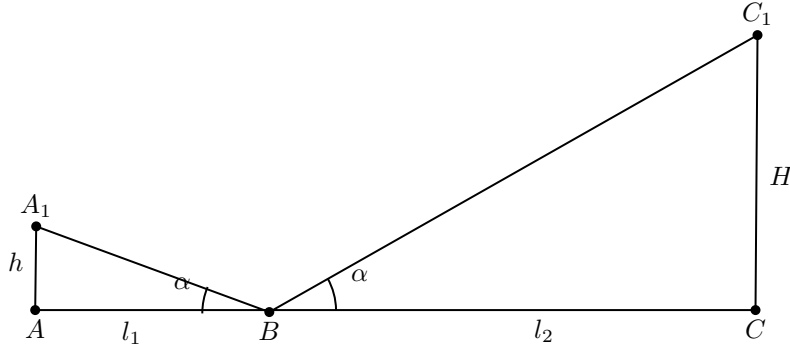


Figure 30

Consider the side view in the plane ACB (see Fig. 30) and denote the location of the passenger's eyes A_1 , and the top of the pillar C_1 . Since the angle of incidence A_1BA is equal to the angle of reflection C_1BC , then

$$\tan \alpha = \frac{h}{l_1} = \frac{H}{l_2}$$

Also,

$$l_2 + l_1 = l = \frac{L}{\sin \beta}$$

Expressing the value of l_2 from the first equation and substituting it into the second, we find l_1 :

$$l_1 = \frac{hL}{(h+H)\sin \beta}$$

Hence the coordinates of the flare:

$$x_B = vt - l_1 \cos \beta = vt - \frac{hL \cot \beta}{h + H} = vt - \frac{hL \cdot \frac{vt}{L}}{h + H} = v \frac{H}{h + H} t,$$

$$y_B = l_1 \sin \beta = \frac{hL}{h + H}$$

It follows from these formulas that the flare does not shift in the direction perpendicular to the shores. Thus, the speed of the flare motion is directed parallel to the channel banks and is equal in magnitude

$$u = v \frac{H}{h + H}$$

Alternative solution

Another way to solve the problem is based on the consideration of similar triangles and allows you to do without trigonometry. Indeed, triangles AA_1B and CC_1B are similar. Therefore

$$\frac{BC}{AB} = \frac{H}{h}$$

It follows that

$$\frac{BC}{AC} = \frac{BC}{AB + BC} = \frac{H}{h + H}$$

Draw through point B in Figure 29, a straight line parallel to the channel banks; it will cross the perpendicular CO at point B_1 . From the similarity of triangles CBB_1 and CAO we obtain

$$\frac{B_1C}{OC} = \frac{BC}{AC} = \frac{H}{h + H}$$

that is, the ratio B_1C/OC is a constant. This means that point B_1 does not change its position over time. Thus, the flare moves along a straight line passing through point B_1 parallel to the channel banks. Let's find its speed. The lengths of the segments B_1B and OA are equal to ut and vt , respectively. The similarity of triangles CBB_1 and CAO implies the ratio:

$$\frac{B_1B}{OA} = \frac{ut}{vt} = \frac{BC}{AC} = \frac{H}{h + H}$$

from which the previous expression for the flare speed is obtained.